

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

You. Rajesh Rishiyur Chandrasekaran

Role. Senior Supply Chain Manager, NACF Supply Chain — Global Capacity Planning team. Job ID 3195330. Bellevue, WA.

Team mission, in their words. Plan and deliver North America Customer Fulfillment capacity to meet customer demand at the desired speed and quality while improving the long-term cost structure of the network.

Their customers. S&OP, PPT, Finance, FC Operations, WW and Operations Engineering, Real Estate, WW Design, Launch, Safety/EHS. Partners in SCOT, Product, Engineering and Construction.

Format. One hour. Expect two to four Leadership Principle stories with hard follow-ups, and one functional block on capacity planning. If this is the hiring manager, the functional block gets longer.

Your position. You have designed and consolidated DC capacity, allocated demand across a 3-DC / 100-store network through peak, and turned measured lane data into planning parameters. You have not built an Amazon FC capacity plan. Say so once, early, then show the mechanism.

Read sections 2, 3, and 4 out loud once — section 3 has "tell me about yourself" in the Amazon shape and "why Amazon / why this team / why now." Section 5 is the main story — Flexport Shipment Visibility: the WBR, automation rate, human-in-the-loop, coverage — lead with it. Section 6 is organized by Leadership Principle: the stories for each LP with hooks, the full STAR blocks with math and solution design, and every one of the 128 questions in Amazon's question bank answered under its LP. Section 7A is the deep model — how a unit flows, what it costs, and the WBR this role would run; read it once, slowly. Section 7 is the functional block; walk Case A on paper once before Friday.

1. What they are actually scoring

The L6 bar, in one table

L6 GUIDELINE SAYS	WHAT THE INTERVIEWER LISTENS FOR	WHERE YOU PROVE IT
Ambiguity. Strategy may not be defined. You pick the stakeholders, set the goals, design the long-term solution.	Did you start from a vague ask and define the problem yourself?	Elementum: the ask was a report; you built lane-level lead-time planning. Flexport: inherited an 18-month plan and re-costed the sequence.
Scope and influence. A team, sometimes an org or country. Influences 1–2 levels up.	Did you move a decision above your level with evidence, not seniority?	Roadmap flip ratified in Bellevue after private math with your manager. CIO/COO weekly pack across 60+ stakeholders.
Execution. Tactical and strategic. Decides where to simplify or extend. Mitigates long-term risk. Trade-offs short vs long.	Can you name the trade-off and the number that decided it?	Partner-first vs operator-first (CTS ~50% vs ~25%). Robots at 30% vs the launch date. Mock tolerance vs go-live.
Complex. Visible risks, roadblocks, constraints. You design an appropriately simple solution.	Was the solution simpler than the problem? Did a gate exist before the crisis?	0.5% tolerance published before mock one. 98% pick-accuracy gate published before the robots arrived.

L6 GUIDELINE SAYS	WHAT THE INTERVIEWER LISTENS FOR	WHERE YOU PROVE IT
Designing or modeling the supply chain, not working in it.	Are your stories about the network's shape and parameters, or about running a shift?	Allocation across 3 DCs + 100 stores. Two buildings into one omni DC. 200 lanes measured. Safety-stock parameters re-set from data.
Process improvement. Drives operational-excellence best practice.	Did you leave a mechanism that ran without you?	WBR grammar at Flexport; QA-authored release gate that survived your leaving; carrier escalation and drift flags.

How the interviewer is trained to probe — Amazon's STAR worksheet

Amazon's interviewer guide gives every interviewer the same worksheet: establish the story in STAR, then **probe** on the assigned Leadership Principle, then **challenge** the candidate's decisions. Results are scored against a legend — \$ cost savings or revenue, # volume/scale, % change, **time** saved, **customer/team impact**, **quality**. Every story in Sections 5 and 6 is written so the probes below are already answered before they are asked, and every Result is written to that legend.

PHASE	WHAT THEY ARE TRAINED TO ASK	HOW YOUR ANSWER PRE-EMPTS IT
Situation / Task	Why is this important? What was the goal? What was the initial scope? What were the challenges? What were the risks if nothing happened?	Open every story with <i>problem</i> → <i>goal</i> → <i>the metric and its baseline</i> → <i>the cost of doing nothing</i> . Fifteen seconds. Then stop describing and start doing.
Action	Were you the key driver? Biggest contribution? Unique value? Most significant obstacles? What did you do versus the team? How did you set priorities / get buy-in? What decisions did you challenge?	Every action sentence starts with "I" and names an artifact — the gate, the table, the rule, the log, the query. Name the obstacle and the decision you overruled. Name what you deliberately did <i>not</i> do.
Results	How did you measure success? Why these results? What percentage change year over year? What trade-offs (quality, cost, time)? "I'm concerned about... tell me more."	Give the metric, grain, baseline, target, actual — in that order. Then the trade-off you paid. Then the mechanism that made it stick.
Challenge	Why this story? Can you give a more recent one? What were the lessons? What would you do differently? How would you implement this at Amazon? How did results compare to the goal you stated?	Every story below ends with "Did the metric move" against the stated goal, a real "differently," and an "At Amazon" sentence that lands in capacity planning.

Three words they must hear in every story: supply, demand, capacity. Each story in Sections 5 and 6 carries a box naming what supply was, what demand was, what capacity was, and which one bound. Say the box out loud when the question is functional; keep it in your head when the question is behavioral — but make sure the story's logic follows it.

The JD, translated into what you say

- **"Take end-to-end ownership of building and manifesting our capacity plan."** Say: a plan is a set of numbers with named owners and triggers, not a deck. You have run that as a WBR.
- **"Balance capacity and demand."** Say: that is allocation at Manhattan and consolidation at Sephora UDC. Demand did not fit the building; you changed where units went, then changed the building.
- **"Define metrics, develop program management mechanisms, invent/simplify processes."** Say: input lever vs output metric, pre-wired reds, COE that changes a mechanism. Flexport Snowflake story.
- **"Craft concise written narratives to influence executive decisions."** Say: the two-column cost-to-serve table that flipped an 18-month roadmap. Offer to write it.

- **"Adherence to lean philosophy at scale."** Say: gates that can stop the line. Andon is not a poster.

Vocabulary you should use correctly, and what to ask about

TERM	USE IT AS
NACF	North America Customer Fulfillment — the FC network that picks, packs and ships customer orders.
Capacity plan	Units of demand the network must absorb, by week, by building type, against what each building can store and push out. Two constraints, not one: storage (cube) and throughput (units per hour through inbound, stow, pick, pack, ship). Say both every time.
S&OP	The weekly plan that reconciles demand forecast, inventory and capacity; the basis of labor planning and steering decisions. Your Elementum forecast-alert story lives here.
OP1 / OP2	Amazon's annual operating-plan cycles (fall and spring). Long-range capacity decisions — new builds, expansions, retrofits — are argued in these. You can say "the OP cycle" without pretending you have sat in one.
SCOT	Supply Chain Optimization Technologies — the science and product org that owns forecasting, buying, placement and planning systems. The interface you would manage: they own the model, you own whether the plan is honest.
PPT	Listed as a customer of the team. Do not guess the acronym. Ask: "what does PPT own relative to Global Capacity Planning?"
Sortable / non-sortable / AR	FC types by item size and automation. Capacity is planned per type; a sortable AR building and a non-sort building are not interchangeable. Say "by building type" rather than "by FC."
Launch / Real Estate / WW Design	The teams that put new capacity on the ground. Your UDC and dual-DC stories are launch stories; speak to them in launch gates and freeze windows.
Speed	Promised delivery speed to the customer. Capacity in the wrong place is a speed miss, not just a cost miss. Your promise-equation work maps straight onto this.

2. Say this about the gap. Once.

UNPROMPTED, INSIDE THE FIRST FIVE MINUTES

"I have not built an Amazon FC capacity plan. I have done the three things underneath one: I allocated demand across three mega-DCs and a hundred stores through peak when the network did not fit the plan; I consolidated two warehouses into one omni facility with robotics and ran the launch gates; and I replaced assumed planning parameters with measured ones across two hundred lanes and released the inventory the bad assumptions were holding. I know the scale here is different. The mechanism is not."

Then stop. Do not list what else you have not done. If they probe the gap later, the answer is in section 8 ("Why you over an internal transfer").

3. "Tell me about yourself" — Amazon format, 60 seconds

Present → past → future. One sentence each for present and future; three numbered chapters for past, one number each. Under 160 words. No adjectives about yourself. Say the three words — **supply, demand, capacity** — once, in the last line, where they land as a claim about the job.

SAY THIS, THEN STOP — ABOUT 60 SECONDS

Present. "I'm the technical program manager for retail fulfillment and inventory at Sephora — I own how inventory is defined, placed and promised across stores, DC and dotcom on S/4HANA, and I just consolidated two warehouses into one robotic omni DC: processing efficiency up fifteen percent, zero customer incidents through peak.

Past. Before that, three chapters, one job. Manhattan Associates: allocation and promising across three mega-DCs and a hundred stores — 99% fill through Cyber Monday, about eight million protected. Elementum: replaced brochure lead times with measured distributions on two hundred lanes — a million and a half of safety stock released at 99% accuracy. Flexport: took shipment visibility from human-powered track-and-trace to an automated product — seven to sixty-eight customers, cost per container from sixty-six cents to three, the automation rate measured every week in a review I built.

Future. I have not built an Amazon FC capacity plan. What I have done four times is plan supply, demand and capacity as one equation, deliver it through gates, and keep it honest in a weekly review. I want to do that at network scale, in the one place where it is the operating system."

If they want the long version, add one sentence per chapter from the table in Section 11 — for Flexport, the roadmap flip (\$5M avoided, value in six months). Do not add adjectives; add numbers. If they interrupt, the last two sentences are the only ones that must land.

Why Amazon. Why this team. Why now.

The generic version — "customer obsession, scale, Leadership Principles" — scores zero because every candidate says it. This version is three things Amazon runs on, each matched to a time you did the same thing with a number attached. Three different stories — forecast, weekly review, one inventory number — so no two reasons sound alike. Plain words: no "WBR," no "p99," no "cube and hours" in the spoken version. The omni DC and the WMS migration are delivery stories; keep them for "deliver results," not for "why Amazon."

WHAT AMAZON RUNS ON	WHERE YOU HAVE ALREADY DONE IT	WHY IT LANDS
The whole network is planned off a forecast and measured lead times. Planners are expected to trust the model, not argue with it.	Elementum. 200 lanes planned on the carrier's advertised transit time; you replaced it with what actually happened on each lane (the slowest one-in-a-hundred shipment). ~\$1.5M of inventory released, lead-time accuracy 99%. Shortage alerts went from 70% to 85% accurate and became the first alerts planners acted on in the weekly S&OP.	You have done the job of making a planning number trustworthy enough that people stopped overriding it. That is the daily job here.
The weekly business review: outputs over inputs, every red explained before the meeting, an owner and a date on every action.	Flexport Shipment Visibility. Leadership asked for a weekly review of container tracking; nobody owned it. You built it from a blank sheet, ran it with the VP every week for a year, then automated the data under it. Over that year: 7 → 68 customers, cost per container \$0.66 → \$0.03, share tracked with no human touch ~60% → 97%.	You have run Amazon's core mechanism, by choice, somewhere it was not the default. Here it is.
One inventory number the customer promise rests on, kept honest across hundreds of buildings and many systems every day.	Sephora S/4HANA. Five teams, five definitions of "available." You wrote one definition per stock state, made returns real stock, reconciled SAP against the warehouse system every night, and cut over with a numeric rollback that never fired. Zero oversell incidents, legacy system retired.	Amazon's promise is the same single number at a thousand times the scale. You know what it takes to keep it true.

WHY AMAZON — THE 60-SECOND PITCH, SPOKEN. SAY THIS, THEN STOP.

"Three reasons.

First, the forecast. At Elementum, two hundred shipping lanes were planned on the carrier's promised transit time. I replaced that with what actually happened on each lane. That freed up about a million and a half dollars of inventory, and the shortage alerts built on it were the first ones planners actually acted on. Amazon plans the whole network off a forecast. I've done the work of making people trust one.

Second, the weekly review. At Flexport, leadership asked for a weekly review of shipment tracking and nobody owned it. I built it and ran it with the VP every week for a year; cost per container went from sixty-six cents to three. That weekly review is how Amazon runs. I've already lived that way.

Third, one version of the truth. At Sephora I own how inventory is counted and promised across stores, the warehouse and the website. Five teams had five definitions of 'available.' I got it to one, checked every night, and we stopped overselling. Amazon's promise to the customer rests on that same one number, at a thousand times the scale.

Why now: the work at Sephora is live. This is the biggest version of the same problem."

If they split the question. "*Why this team?*" — "The charter owns both halves of the trade: meet demand at the promised speed, and lower the long-term cost. Most roles own one half and lobby against the other. This one owns the dial." "*Why leave Sephora?*" — "The inventory model is live and is the standing reference; the migration is done. If I stay I run the same play again. I would rather run it once where the network is the product."

If they probe a reason, go to the story: reason one → S3 and S4 (under Think Big and Bias for Action); reason two → MAIN (Section 5); reason three → S10 (under Ownership). **If asked "why Amazon" after you have already told the MAIN story:** reasons one and three only.

Do not say: "most customer-centric company," "I love the Leadership Principles," "Day 1," or anything about compensation or Bellevue here. **Keep it under a minute.** Three reasons, one number each, stop.

4. The sixty-minute clock

MINUTE	WHAT IS HAPPENING	YOUR JOB
0–3	Intros. They describe the team.	Listen for the word they use for their pain: "peak," "utilization," "launch," "forecast," "cost per unit." That word picks your lead story.
3–7	"Tell me about yourself." Often followed by "why Amazon."	Section 3, Amazon format. Land the gap sentence. Keep "why Amazon" to the 60-second pitch: three reasons, one number each, stop.
6–40	Two to four LP questions. Each gets four or five follow-ups.	One story per question, 90 seconds, then stop and let them drill. Answer follow-ups in one or two sentences. Say "I" for decisions. Name the data pull.
40–52	Functional block: build a plan, handle a gap, define metrics.	Sections 7A and 7. Clarify out loud for 60 seconds before you solve. Say both currencies — cube and hours. Do a math checkpoint. End with owners and triggers.
52–58	Your questions.	Two from section 9. Not four.
58–60	Close.	Section 10.

Amazon interviewers write notes the whole time and score after. Pauses are fine. A tight 90 seconds followed by "happy to go deeper on any part" beats a four-minute story every time.

5. The main story — Flexport Shipment Visibility

Lead with this one. It is the story that has every element the worksheet scores and every word in the job description: a customer problem, a **supply / demand / capacity** mechanism, a weekly business review you built from nothing and then automated, **automation rate**, **human-in-the-loop**, **coverage**, cost per unit falling while volume grew, and a public hold on a date. It is also the story the interviewer can probe deepest, because every number has a definition and every definition has a source: the E-Team takeaways (Feb 1, 2022), the BRD you authored (TSR-852, Zendesk data in Snowflake), the engineering org's LSA WBR you borrowed the format from, and your own metric nugget.

MAIN — Shipment Visibility: replacing human-powered track-and-trace with an automated product, and the WBR that ran it (Flexport, 2022 · domain story)

LPs: Customer Obsession · Ownership · Invent and Simplify · Dive Deep · Insist on the Highest Standards · Frugality · Deliver Results · Success and Scale Bring Broad Responsibility. **AIQB:** CO 5, 7 · OW 3 · IS 5 · DD 5 · HS 1, 6 · FR 7 · DR 3, 7 · HB 5 · ET 5 · TB 5 · LC 7 · SS 1, 7.

Problem	Flexport had launched Shipment Visibility (SV): tracking containers Flexport did <i>not</i> forward ("4PL" shipments) so a client could see all their freight in one app. First service pulled out of the monolith into a microservice — and human-powered : when the Visibility Service failed to initiate tracking automatically for a submitted container, it opened a Zendesk ticket; a Data Services operator found the container on the carrier or terminal site and typed the milestones in. Seven active customers, ~2,000 shipments/week, 400 monthly active users, cost-to-serve \$0.66 per container , system accuracy ~60%, no product analytics team, no WBR. The E-Team (Feb 2022) set Goal 1: lower cost-to-serve by replacing human track-and-trace with a highly automated product ; Goal 2 (public tracking site as a lead funnel) explicitly second. Their action item: "Visibility team to publish its own WBR — priority POD-ETA accuracy and interaction elimination."
Goal	Monthly active customers (MAC) 7 → 68; MAU 400 → 3×; weekly shipment volume 2,000 → 2× — with cost-to-serve down and accuracy up . Growth absorbed by automation, not by Data Services headcount.
Key metric	Cost-to-serve per container (\$0.66). Automation rate — containers tracking-initiated with no human touch ÷ containers submitted [~60%]. Coverage — % containers tracked (automated) and % tracked (with human inputs). Tickets per active container (the human-in-the-loop). POD-ETA accuracy . Timeliness — % 4PL milestones within SLA vs 3PL, and the delta. System accuracy vs source of truth (60%).
Did it move?	Yes . 68 customers (~10×), two months early. Shipment volume 5×. System accuracy 60% → 90%. Cost-to-serve \$0.66 → \$0.03 (~95%). Ticket response 72% → 90% at 5× volume. WBR assembly 30 h/week → 6 h/week once Zendesk landed in Snowflake (~\$103,680/yr).

SAY THIS — THE FULL VERSION, ABOUT 2½ MINUTES; THEN STOP

Situation. "In 2022 Flexport had just launched Shipment Visibility — tracking containers Flexport did not forward, so a client could see all their freight in one place. It was the first service pulled out of the monolith, and it was human-powered: when the visibility service could not start tracking a container on its own, it opened a Zendesk ticket, and a Data Services operator found the container on the carrier's or terminal's site and typed the milestones in by hand. Seven active customers, about two thousand shipments a week, sixty-six cents a container to serve, system accuracy around sixty percent, and no product analytics team. The executive team's first goal was explicit — lower the cost to serve by replacing human track-and-trace with automation — and their action item was 'the visibility team publishes its own WBR; priority is POD-ETA accuracy and interaction elimination.' Nobody owned that.

Task. I took it. Grow from seven to sixty-eight customers, three times the monthly active users, double the weekly volume — with cost to serve going *down* and accuracy going *up*. Which meant the growth had to be absorbed by automation, not by hiring operators.

Action. Four things. First, I made the ticket the sensor. Every Zendesk ticket was one container the machine could not track. So *tickets per active container* became the input metric that measured the human-in-the-loop, and *automation rate* — containers tracking-initiated with no human touch over containers submitted — became the number the engineering backlog was ranked on: tickets times cost per ticket, carrier by carrier, terminal by terminal, trans-Pacific eastbound and trans-Atlantic westbound lanes first because that is where the volume was. Second, I built the WBR by hand before the pipeline existed — four systems into Excel every Monday, about two hours — using the engineering org's own WBR conventions: output metrics, input metrics, a roll-up column saying which output each input moves, five weeks and the months, red at five percent unfavorable, call-outs with an owner and a date. Running it by hand for a quarter debugged the definitions before anyone coded them. Then I wrote the BRD to land Zendesk in Snowflake joined to container numbers, priced the status quo at a hundred and thirty thousand dollars a year of our own labor, and automated the review. Third, I decided where the human loop applied: client tracking opened tickets, because a paying customer needs a consistent experience and manual updates fill the automation gaps; the public tracking site did not, because the bar was lower and the volume would have buried Data Services. Fourth, I gated onboarding: six graduation criteria a wave had to clear at current load — data quality at or above ninety against source, ETA accuracy at or above eighty, verification SLA, engagement, stability, CSAT — and when wave three missed on ETA accuracy I held it, publicly, with the source that caused it and the date we would re-test.

Result. Sixty-eight customers, two months early. Shipment volume up five times. System accuracy from sixty to ninety percent. Cost to serve from sixty-six cents to three cents a container — ninety-five percent — and ticket response from seventy-two to ninety percent at five times the volume. The WBR itself went from thirty hours a week of assembly to six, about a hundred thousand dollars a year. What I am proudest of is *how* the human loop shrank: by measurement. Every operator ticket was a row in the backlog until the machine took it."

THE 90-SECOND CUT — WHEN THEY ONLY WANT THE HEADLINE

"Flexport's shipment-visibility product was human-powered: when automation could not start tracking a container, a Zendesk ticket went to an operator who typed the milestones in. Sixty-six cents a container, seven customers, sixty percent accuracy, no WBR. I owned the WBR and the ramp. I made the ticket the sensor — tickets per active container measured the human-in-the-loop, automation rate ranked the engineering backlog by tickets times cost, carrier by carrier. I ran the review by hand for a quarter to debug the definitions, then wrote the BRD that put Zendesk in Snowflake and automated it. I gated onboarding on six criteria at current load and held a wave publicly when it missed. Sixty-eight customers two months early, volume five times, accuracy sixty to ninety, cost to serve sixty-six cents to three. The human loop shrank by measurement, not by decree."

Supply · Demand · Capacity. Demand: tracking requests — containers clients submit, growing 5× as waves boarded. **Supply:** the data — carrier APIs and EDI, terminal and port feeds, later CargoSmart — the raw material; a missing or wrong feed is the defect. **Capacity:** the system's ability to turn a submitted container into trustworthy milestones = automated capacity (the Visibility Service, per carrier × terminal × milestone) + human capacity (Data Services operators, metered by tickets). **What bound:** capacity at quality. Waves metered demand to measured capacity; the backlog grew automated capacity where tickets said the human loop was most expensive; coverage-with-human minus coverage-automated was the human loop's share, and it fell every week. This is a capacity plan with two currencies — automated and manual — the way an FC plan has cube and hours.

What was innovative. The usual plan was a headcount curve for Data Services and a dashboard someday. I did five things differently. (1) *The ticket as the automation-gap sensor* — a Zendesk ticket is a container the machine failed; tickets per active container made the human-in-the-loop a measured, shrinking input metric instead of a department. (2) *The WBR before the pipeline* — I ran the review by hand on the engineering org's conventions so the definitions were debugged on paper before they were coded, then wrote the BRD and priced the status quo in our own labor so the data-infra team prioritized it themselves. (3) *Where the human loop applies is a product decision* — tickets for client tracking, none for public tracking, source toggles per calling product. (4) *An ROI-ranked automation backlog* — tickets × cost per ticket by carrier × terminal × milestone, TPEB and TAWB first, "easy to get the data" only as a tiebreaker. (5) *Graduation gates* — admission to the platform earned at current load, so cost per container fell as the base grew instead of rising.

Human-in-the-loop — how it was designed, not just tolerated

Trigger	Visibility Service cannot initiate tracking for a submitted container (no source matched, source rejected the container number, or first milestone not received within [24 h]). Later: milestone gap > [48 h] after an expected event; ETA confidence below threshold.
What happens	One Zendesk ticket, tagged with container number, client, carrier, terminal, lane, failure reason. Data Services operator looks the container up at the source, enters milestones; manual entries are tagged <code>source = manual</code> so they count in <i>coverage with human inputs</i> and never in <i>coverage automated</i> .
SLA	Ticket closed within [24 h] for client tracking; the "4PL % within SLA" line in the WBR is the customer-facing read on whether the human loop is keeping up.
Where it does <i>not</i> apply	Public tracking site (lower bar, free, SMB funnel): no tickets — a public request that automation cannot serve returns "not covered" rather than consuming an operator. Source toggles per calling product (SV client app, FF App, public) decide which sources and whether a human loop is allowed.
Exit criterion	A carrier or terminal leaves the human loop when its automation rate $\geq$ [95%] for four consecutive weeks; it re-enters if it drops below [90%] (the kill switch per source class).
What the loop produces	Every ticket is a labeled failure: the ticket log is the engineering backlog, ranked weekly by tickets × cost per ticket. The human loop's output is not only the milestone — it is the row that retires itself.

The math

DEFINITIONS (grain: container × week; the "unit" is the container)

Submitted (S)	= containers clients asked us to track this week (4PL = containers not forwarded by Flexport)
Auto-initiated (A)	= containers the Visibility Service began tracking with no human touch
Automation rate	= $A \div S$ [~60%] → 97%
Tickets generated (T)	= $S - A$ (one Zendesk ticket per failed initiation = the human-in-the-loop, counted)
Active containers (N)	= containers in transit at week end $\approx S \times [5]$ weeks of transit
Tickets / active ctr	= $T \div N$
Coverage (automated)	= containers with $\geq [6 \text{ of } 8]$ milestones sourced automatically $\div N$

Coverage (w/ human) = containers with $\geq$ [6 of 8] milestones from any source $\div$ N
 Human-loop share = Coverage (w/ human) - Coverage (automated) = the gap humans close
 POD-ETA accuracy = containers with $|\text{ETA predicted at ATD} + [7]\text{d} - \text{actual POD}| \leq [1 \text{ day}]$
 $\div$ containers arrived
 Timeliness 4PL / 3PL = milestones posted within [24 h] of the event $\div$ milestones
 Delta = 4PL - 3PL (3PL = Flexport-forwarded, the control group)
 Interactions eliminated = Σ (auto-tracked containers $\times$ [8] milestones each) - manual updates
 System accuracy = milestones matching the source of truth $\div$ milestones audited
 60% $\rightarrow$ 90%
 Cost-to-serve (CTS) = (Data Services hours $\times$ loaded rate + platform/data cost)
 $\div$ containers tracked

COST-TO-SERVE DECOMPOSITION (bracketed inputs illustrative, chosen to land on the real endpoints)

Baseline, 7 customers: S = 2,000/wk ; automation 60% $\rightarrow$ T = 800 tickets/wk
 cost per ticket = [12 min] $\times$ [\$75/h] = \$1.50 ; human CTS = $800 \times 1.50 \div 2,000 = \0.60
 platform/data = [\$120/wk] $\div$ 2,000 = \$0.06 $\rightarrow$ CTS = \$0.66 ✓
 End, 68 customers: S = 10,000/wk (5 $\times$) ; automation 97% $\rightarrow$ T = 300 tickets/wk
 cost per ticket falls to [\$0.80] (pre-filled carrier page, lookup tooling)
 human CTS = $300 \times 0.80 \div 10,000 = \0.024 ; platform/data = [\$150/wk] $\div$ 10,000 = \$0.015
 $\rightarrow$ CTS $\approx$ \$0.04 $\rightarrow$ \$0.03 with [bulk source pricing] ✓
 Decomposition of the -95%: volume alone (fixed cost diluted 5 $\times$) would give $\approx$ \$0.13 ;
 the rest is automation rate 60 $\rightarrow$ 97 (tickets -62% in absolute terms while volume $\times$ 5)
 Sensitivity: 1 point of automation rate at 10,000/wk = 100 tickets/wk $\approx$ \$80/wk $\approx$ \$4K/yr
 - the backlog was ranked on exactly this number

WHAT TO AUTOMATE NEXT (ROI stack by carrier $\times$ terminal $\times$ milestone; TPEB and TAWB lanes first)

Value_c = tickets_c/wk $\times$ cost per ticket $\times$ 52 + interactions_c/wk $\times$ [\$0.10] $\times$ 52
 Rank = Value_c $\div$ engineering weeks_c ; "easy to get the data" breaks ties only
 Worked: carrier X 140 tickets/wk $\rightarrow$ \$10.9K/yr + 140 $\times$ 5 $\times$ 8 $\times$ \$0.10 $\times$ 52 = \$29K
 $\rightarrow$ \$40K $\div$ 3 eng-wk = \$13K per eng-wk
 terminal Y 60 tickets/wk $\rightarrow$ \$4.7K/yr + \$12.5K = \$17K
 $\rightarrow$ \$17K $\div$ 0.5 eng-wk = \$34K per eng-wk
 $\rightarrow$ terminal Y first, even though carrier X generates more tickets

WAVE GATE (all six must pass at CURRENT load before a wave boards)

data quality $\geq$ 90% vs source $\cdot$ POD-ETA accuracy $\geq$ 80% $\cdot$ verification SLA $\cdot$ engagement floor
 $\cdot$ stability $\cdot$ CSAT
 Wave-3 hold: ETA accuracy 76% vs 80% ; one source class = [70%] of misses
 $\rightarrow$ fix + kill switch $\rightarrow$ re-test 1 week $\rightarrow$ boarded

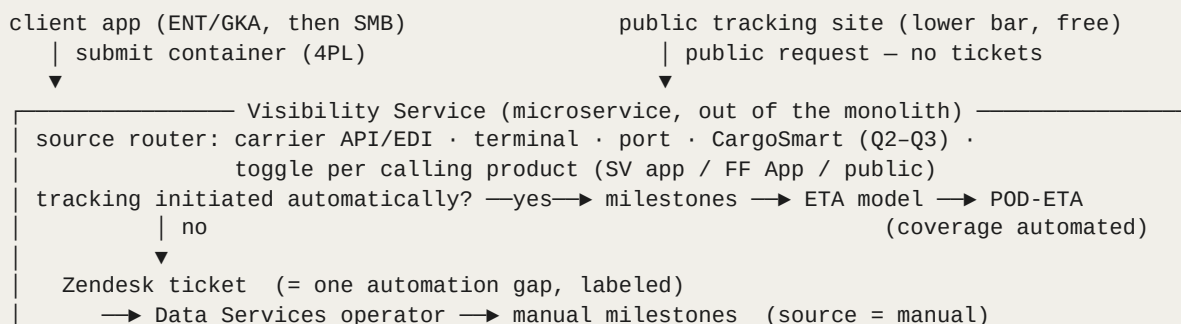
THE WBR'S OWN COST (BRD TSR-852 - Zendesk $\rightarrow$ Snowflake, join ticket_id $\leftrightarrow$ container_number in Crux)

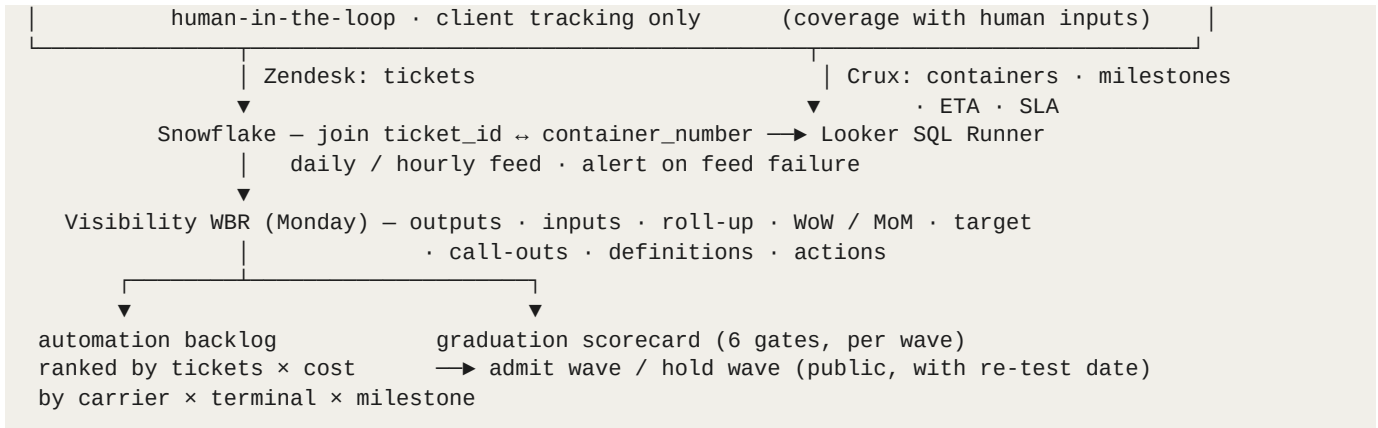
Before: 3 people $\times$ 5 h/wk $\times$ 2 WBRs = 30 h/wk $\times$ \$90/h $\times$ 4 wk = \$10,800/mo $\approx$ \$129,600/yr ;
 1.5-2 h per weekly snapshot, by hand, me as the single point of failure
 After: 1 h/wk per person per WBR = 6 h/wk $\times$ \$90 $\times$ 48 wk = \$25,920/yr
 Saved: $\approx$ \$103,680/yr (SPOF removed; feed daily/hourly, alert on failure - BR-3, BR-4)

GREEN / RED (borrowed from the LSA WBR): relative variance (B - A) $\div$ A ;

favorable $>$ 5% green, unfavorable $>$ 5% red ; five most-recent weeks + all months ;
 Target and Vs-Target columns ; every call-out carries an owner and a date

Solution design / flow





The Visibility WBR — the sheet itself

Same skeleton as the engineering org's LSA WBR: line, metric, owner, roll-up (which output line the input moves), five most-recent weeks, WoW, target, vs-target, then call-outs and definitions. Weeks below are an **ILLUSTRATIVE trajectory** from the last quarter of the ramp; the endpoints (\$0.03, 90%, 68, 97%) are yours. Green/red is relative variance > 5%.

#	METRIC	ROLL-UP	WK 45	WK 46	WK 47	WK 48	WK 49	WOW	TARGET	VS TGT
Output metrics										
1	\$, cost-to-serve per container tracked	—	0.11	0.09	0.07	0.05	0.03	−40%	0.05	−0.02
2	%, POD-ETA accuracy (±1 day at ATD+7)	—	74	76	79	81	83	+2.5%	80	+3
3	%, 4PL milestones within SLA (24 h)	—	71	74	78	82	85	+3.7%	90	−5
4	%, 3PL milestones within SLA (24 h)	—	88	88	89	89	90	+1.1%	90	0
5	Δ pts, 4PL − 3PL timeliness	—	−17	−14	−11	−7	−5	+2	0	−5
6	#, monthly active customers (MAC)	—	41	48	55	61	68	+11%	68	0
7	#, shipments tracked / week ('000)	—	6.2	7.1	8.0	9.1	10.0	+10%	—	—
8	%, system accuracy (audit vs source of truth)	—	82	85	87	89	90	+1.1%	90	0
Input metrics										
9	%, automation rate (tracking initiated without a human)	1, 3	88	91	93	95	97	+2.1%	95	+2
10	%, coverage — containers tracked (automated)	1, 2	79	82	85	88	90	+2.3%	90	0
11	%, coverage — containers tracked (with human inputs)	3, 5	95	95	96	96	97	+1.0%	97	0
12	#, tickets generated (Zendesk)	1, 3	744	639	560	455	300	−34%	500	−200
13	#, tickets per 100 active containers	1, 3	2.4	1.8	1.4	1.0	0.6	−40%	1.0	−0.4
14	#, interactions eliminated ('000, cumulative)	1	42	50	58	67	76	+13%	—	—
15	#, carriers + terminals fully automated (cum.)	9, 10	31	33	36	38	41	+8%	40	+1
16	#, blocker / critical bugs open (Jira)	2, 8	6	4	4	2	1	−50%	2	−1
17	%, Snowflake feed fresh (≤ 24 h) on review day	all	100	100	86	100	100	+16%	100	0

#	METRIC	ROLL- UP	WK 45	WK 46	WK 47	WK 48	WK 49	WOW	TARGET	VS TGT
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Call-outs (owner · date)

Wk 47 — terminal Y automated (0.5 eng-wk): tickets -12% WoW; line 13 crosses 1.5. · Wk 47 — line 17 red: Zendesk feed failed Sunday 02:00, alert fired, re-run before review (BR-4 worked). · Wk 48 — **wave 3 held**: line 2 at 76% vs 80% gate in the wave cohort; source class Z = 70% of misses; kill switch built; re-test Wk 49 — boarded. · Standing: any carrier below 90% automation two weeks running re-enters the human loop and the backlog.

Definitions

Lines 1–17 as in "The math" above. Line 3/4 SLA = milestone posted within 24 h of the physical event. Line 10 vs 11: the difference is the human loop's share. Line 9 is the *leading* indicator of lines 1 and 3; line 13 is the *cost* of the gap; line 15 is the backlog burning down.

How the review ran

Monday, 30 minutes. VP, the core visibility team, and the EMs of the dependent teams (Data Services, data infra, FF App). Every red arrives pre-decomposed by carrier × terminal × milestone, with the owner and the date in the call-out line — a red without a decomposition is sent back. The action-items log (owner · created · target · update) is the first slide; a Jira dashboard of blocker/critical issues is tied to the input lines so a bug has a metric, not just a priority. **Rules:** no new metric without a frozen definition and a roll-up; no metric change during a quarter; a line that is green four weeks running is challenged for removal (the LSA WBR's own question: "if it is 100%, is it done?").

Outputs: cost-to-serve per container; POD-ETA accuracy; 4PL vs 3PL timeliness and delta; MAC, MAU, volume; system accuracy. **Inputs:** automation rate; coverage automated vs with human inputs; tickets and tickets per active container; interactions eliminated; carriers/terminals automated; blocker bugs; feed freshness. **Red rules:** automation rate down two weeks running → the carrier is named and goes to the top of the backlog; 4PL–3PL delta widening → Data Services SLA conversation that day; a feed-freshness red → the review runs on last week's numbers with a banner, never on silently stale data.

SITUATION PROBES

Why was this important? Cost of doing nothing? Visibility was free to the client by design — the E-Team decided to give it away to win the rest of the client's freight. A free product with a sixty-six-cent human cost per container scales into a loss the day it succeeds. Sixty customers at pilot economics was a support organization, not a product.

Who set the goals? The E-Team set Goal 1 (cost-to-serve) and the direction; sales set the 68 and the date. I added the two conditions — cost down, accuracy up — and owned the WBR nobody had been assigned.

What was the initial scope you were handed? "Report progress." I made it "own the metric tree, the cadence, and the backlog ranking." The E-Team action said "publish a WBR"; I treated the WBR as the operating system of the ramp, not a report on it.

Who was the customer? Importers and exporters — enterprise and global key accounts first, SMB later through the public site. The internal customer was Data Services, whose work I was trying to shrink; I told them that on day one.

ACTION PROBES

Why make the ticket the metric? Because it already existed, it was already one-to-one with an automation failure, and it carried the labels — carrier, terminal, lane, reason. Inventing a new "automation failure" event would have taken a quarter; the ticket was free and honest.

Why run the WBR by hand first? That is two hours a week of a program manager. Because the expensive part of a WBR is the definitions, not the pipeline. A quarter of running it by hand changed four of the seven definitions — "active container," for one — before anyone had coded them. Then the BRD had a fixed target and the build was cheap. Automating a review you have not run yet automates your mistakes.

Why not simply hire Data Services operators to the volume? The volume was going to grow five times.

Operators scale linearly with containers; automation scales with carriers. There are a few hundred carriers and terminals and millions of containers. The only cost curve that bends is the one indexed to carriers.

Why tickets for client tracking and none for public tracking? A paying client needs a consistent experience — manual updates fill the automation gaps. A public request is a lead, not a commitment; routing it to an operator would have consumed the human loop on people who were not customers yet, and the public site's job was to tell us where demand and coverage were, not to be perfect.

Why TPEB and TAWB first? Volume. The ROI stack is tickets $\times$ cost; the trans-Pacific eastbound and trans-Atlantic westbound lanes carried most of the containers, so a carrier fixed there retired more tickets per engineering week. "Easy to get the data" was only the tiebreaker, because easy data on an empty lane retires nothing.

Why not buy Project44's data? The E-Team's call, which I agreed with: visibility was a competitive advantage, and buying the data would have bought the gap rather than closed it. We measured ourselves against them instead and invested in our own sources, Data Services included, until automation took over.

What did you do versus the team? I defined the metric tree and the roll-ups, ran the Monday review, wrote the BRD and priced it, decided where the human loop applied, ranked the backlog weekly, and made the wave-three hold. Engineering built the source integrations and the kill switch; data infra built the Snowflake feed; Data Services ran the loop.

What decision did you challenge? Hiring support ahead of the ramp — and the assumption that the public site was a distraction. I argued the public site was a cheap volume source for finding where coverage was weak, as long as it created no tickets. The E-Team kept Goal 1 first, and the site launched behind it.

RESULTS PROBES

How did you compute cost-to-serve? Data Services hours on visibility tickets at a loaded rate, plus the platform and data-source cost attributable to visibility, divided by containers tracked that month. Sixty-six cents at seven customers; three cents at sixty-eight. The method was agreed with finance before the first number was published.

Isn't the 95% mostly denominator — five times the containers? Partly, and I say so. Fixed cost diluted five times would have taken sixty-six cents to roughly thirteen. The rest is the automation rate going from about sixty to ninety-seven percent: tickets fell in absolute terms while volume quintupled. The WBR shows both lines so nobody has to take my word for the split.

How did you measure "system accuracy" — 60 to 90? A weekly audit sample of milestones compared to the source of truth — the carrier or terminal record — share matching within tolerance. Sixty percent when I started; ninety at the end. Not self-reported by the system.

How did you measure coverage, and why two coverage lines? Containers with at least six of eight expected milestones present, over active containers — once counting only automated sources, once counting any source. The gap between the two lines is the human loop. One line would have hidden the thing I was trying to shrink.

Compared to goal? 68: met, two months early. MAU 3 $\times$: met. Volume 2 $\times$: beaten, 5 $\times$. Cost down: -95%. Accuracy: 60 $\rightarrow$ 90. Response: 72 $\rightarrow$ 90 at 5 $\times$ volume. The WBR's own cost: -\$103K a year.

Trade-offs? One week on wave three and a hard conversation with sales. The public site launched later than its advocates wanted. Data Services shrank by attrition into other roles, which I had told them would happen.

CHALLENGE

What would you do differently? Build the kill switch per source class before wave one — I knew bad sources existed and let the first miss force it. And land Zendesk in Snowflake in month one instead of month four; I ran the review by hand a quarter longer than the definitions needed.

At Amazon. A capacity plan has a human-in-the-loop too: every manual override of the model's output — a planner moving volume, a site manager rejecting a forecast — is a ticket. I would run the same tree: *automation rate of the plan* (share of building-weeks planned without override), *overrides per building per week* as the human-loop cost, *coverage* of the network by a measured ceiling instead of a nameplate, and a backlog ranked by overrides $\times$ cost, so the model earns the buildings back one at a time. The WBR conventions are the same ones this role already runs.

6. Leadership Principles — stories, STAR answers, probes, math and design

Organized the way the hour is organized: by Leadership Principle, in Amazon's order. The interviewer will not say the LP's name; they will say "tell me about a time." Hear the LP, go to its block. Each block has four things. **(1)** One line on what the LP means and what the interviewer's worksheet tells them to listen for. **(2)** The stories that fit it, with the hook sentence to open with and where the full block lives. **(3)** The full block of the story (or two) that is home here — **Problem** → **Goal** → **Key metric** → **STAR** → **supply / demand / capacity** → **what was innovative** → **the math** → **solution design / flow** → **how it translated to a WBR** → **the worksheet probes**. **(4)** Every question in Amazon's question bank for that LP, each answered in STAR with a **Math** line, the two probes most likely to follow with one-line answers, and the **WBR / design** translation. Questions marked (*Manager*) in the bank are answered with matrix leadership — say "matrix" out loud; do not invent a headcount.

Two kinds of stories. *Domain stories* — where you designed the supply/demand/capacity logic: MAIN (Shipment Visibility, Section 5), S10 (S/4HANA ATS and reconciliation), S3/S4/S5/S16 (Elementum), S7/S8/S15 (Flexport), S2/S11 (Manhattan). *Program-management stories* — where you ran a large delivery: S1 (UDC commingle) and S9 (RP JDA). Lead with domain stories for capacity, inventory, data and metrics questions; use S1 and S9 for delivery, launch, deadline and obstacle questions. **Lead with these four** unless their pain word says otherwise: MAIN (visibility, automation rate, WBR), S2 (peak allocation), S3 (200 lanes), S10 (ATS and reconciliation). Every story carries a "The math" block, a "Solution design / flow" diagram where there is a system, and "How it translated to a WBR." In the math blocks, numbers in [brackets] are illustrative inputs chosen so the arithmetic lands on your real result; replace them with your actual inputs before Friday. The bolded result numbers are yours.

The one-page map — which story, which hook

LP	LEAD / BACKUP	THE HOOK SENTENCE
Customer Obsession	MAIN / S11	"The customer was paying 66 cents a container for a human to type milestones. I made the automation rate the metric and the human loop the designed exception." / "Silent corrections are loans against trust. The customer heard before the fix, and re-ranked the fix."
Ownership	S10 / MAIN	"Nobody owned what "available" meant. I wrote the definitions, took them to steering with the failure mode attached, and owned the cutover." / "The E-Team wrote "Visibility team to publish its own WBR." Nobody was assigned. I built it from a borrowed sheet that week and ran it for a year."
Invent and Simplify	S8 / S16	"New metrics without killing old ones is complexity wearing a badge. I froze seven definitions and killed the spreadsheet." / "A thousand routes, one gate. The gate replaced a thousand onboarding conversations."
Are Right, A Lot	S2 / S10	"70% of the information is enough when waiting is a customer miss and every lever is a two-way door." / "I forbade promising against in-transit units. I was wrong, the data said so, and the fix went through the same signed policy."
Learn and Be Curious	S14 / S9	"I taught myself the stack on weekends and shipped a triage agent that cut cost per ticket from \$25 to 40 cents." / "I had never run a WMS migration. I did not fake it. I built the gate and used the specialists."
Hire and Develop the Best	S12 / S13	"I gave the skeptic the gate. Development that depends on me staying is not development." / "I staffed two go-lives on one team by naming the owner of each gate and protecting their week."

LP	LEAD / BACKUP	THE HOOK SENTENCE
Insist on the Highest Standards	MAIN / S1	"I held wave three at 76% against an 80% gate in front of the VP. Standards without a stop-the-line are posters." / "I published the 98% gate before the robots arrived, and throttled them to 30% when they hit 94%."
Think Big	S3 / S7	"The ask was a report. The product was lane-level planning as a system." / "Think Big without sequencing is a slogan. The cheaper half first paid for the harder half."
Bias for Action	S2 / S1	"Timestamped decision at 4 AM, then a post-mortem that made silence an alarm." / "Throttling the robots was a two-way door we had built. I turned it the same day."
Frugality	MAIN / S7	"Cost-to-serve down 95% while the book grew ten times. Headcount would have scaled the waste." / "Two columns of cost-to-serve beat eighteen months of roadmap. \$5M avoided, value in six months."
Earn Trust	S11 / S10	"Silent corrections are loans against trust." / "Sixty stakeholders, one weekly pack, metrics frozen so nobody could re-litigate a red by changing the chart."
Dive Deep	S5 / MAIN	"I pull until the unit is a shipment, an order, a lane or a building — not a national average." / "Every red decomposed to carrier and ticket reason code before the review, with an owner and a date."
Have Backbone; Disagree and Commit	S7 / S11	"Backbone is the private table. Theater is the public ambush." / "I disagreed with my manager about disclosure in the morning and committed to his repair sequence by noon."
Deliver Results	S9 / S10	"Results with a rollback are not results. The gates are why it counted." / "343-task runbook, rollback as a number, never fired, legacy system off, zero customer incidents."
Strive to Be Earth's Best Employer	S13 / S12	"Heroics are a process miss. I re-cut the plan so nobody worked a 24-hour day." / "I gave the skeptic the gate and the credit."
Success and Scale Bring Broad Responsibility	MAIN / S1	"We scaled ten times and ticket response improved. The cost of scale was paid in gates, not in someone else's backlog." / "A gate I published before the robots arrived protected the floor team from a narrative."

The follow-up discipline

- **"What did you do, specifically?"** One sentence that starts with "I." Name the artifact: the gate, the table, the log, the SQL.
- **"Why not [the obvious alternative]?"** Name the cost of the alternative in the same currency as the result.
- **"How did you measure?"** Metric, grain, baseline, target. Never "we saw improvement."
- **"What would you do differently?"** A real change you have since made. Not "nothing" and not "communicate more."
- **"Walk me through the data pull."** Tables joined, grain, filters. If you did not pull it yourself, say who did and what you checked.

Customer Obsession

Start with the customer and work backwards; earn and keep trust; pay attention to competitors but obsess over customers. The interviewer listens for: who the customer was, how you learned what they needed, what you gave up for them.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
MAIN	"The customer was paying 66 cents a container for a human to type milestones. I made the automation rate the metric and the human loop the designed exception." MAIN — Shipment Visibility (Flexport)	Section 5
S11	"Silent corrections are loans against trust. The customer heard before the fix, and re-ranked the fix." S11 — The \$2.5M discrepancy I refused to fix quietly	below
S3	"I treated the brochure lead time as a lie the customer was paying for in inventory." S3 — Two hundred lanes: plan lead time at the empirical p99, release the inventory the brochure was holding	under Think Big
S1	"I will not ship a customer-facing miss to protect a robotics narrative." S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived	under Insist on the Highest Standards

■ S11 — The \$2.5M discrepancy I refused to fix quietly (Manhattan Associates)

LPs: Earn Trust · Customer Obsession · Have Backbone. **AIQB:** Earn Trust 6 · Customer Obsession 1, 8 · Have Backbone 1.

Problem	An OMS-to-WMS handshake defect had mis-stated a couple hundred customer orders — roughly \$2.5M of order value — and the manager's instinct was to correct the data quietly.
Goal	Protect the customer's real priorities and the account, with zero undisclosed corrections.
Key metric	Orders corrected to the customer's priority; disclosure before fix (yes/no); account outcome.
Did it move?	Yes. Full disclosure; the customer prioritized the orders that mattered to their business; the class was bulk-patched; we later won a larger initiative with them.

Situation. "On an OMS-to-WMS handshake we found a discrepancy across a couple hundred orders, roughly two and a half million dollars of order value. **Task.** The manager's instinct was to correct it quietly and move on. I thought the customer had to decide the order of repair, because they knew which orders mattered. **Action.** I argued for full transparency and won it: we showed the customer the list the same day, let them prioritize, and bulk-patched the defect class rather than the instances — so the fix was structural, not cosmetic. It was uncomfortable; I was the junior voice in that room. **Result.** They protected the orders that mattered to their business, the class never recurred, and we later won a larger initiative with them. Silent corrections are loans against trust, and the interest rate is brutal. Same instinct at Flexport later: a truncated week printed 99.4% automation and I discarded it, because a VP steering on three data points is a decision-quality problem."

Supply · Demand · Capacity. **Demand:** the customer's orders, mis-stated. **Supply:** the truth about them. **Capacity:** the account's trust — the capacity to be believed next time. **What bound:** the willingness to show a bad number.

The math

Affected orders = orders where OMS allocation $\neq$ WMS pick confirmation for the same order line (handshake defect class)

count $\approx$ [200] orders ; value = Σ order value $\approx$ \$2.5M
Repair order = customer-ranked (their business priority), not system-ranked (order age)
Fix scope = the defect CLASS (handshake mapping), verified by zero recurrences over [4] weeks
– not the 200 instances

How it translated to a WBR

Output: OMS $\leftrightarrow$ WMS handshake mismatches per 10K lines (target 0, published to the customer weekly). **Input:** interface error queue age. **Red rule:** any customer-impacting data defect is disclosed with the list before the fix, every time.

Why not fix it and tell them after? Same outcome. Not the same outcome: they would have learned we fix things without telling them. Also not the same repair — they re-ordered the fix by business priority, which we could not have known.

What did the manager say? That it would create noise. I agreed it would, and said the noise now was cheaper than the noise later. He let me make the call and own it.

Why bulk-patch the class, not the instances? Two hundred orders were the symptom. The handshake was the cause. Fixing instances would have made the next two hundred our fault twice.

At Amazon. If the capacity plan is wrong, the WBR hears it from me first with the decomposition attached, before anyone finds it in a speed miss. Reds that arrive pre-decomposed are the version of this story that scales.

The question bank — Customer Obsession (9 questions, in the bank's order)

Format: question $\rightarrow$ **story** and STAR answer $\rightarrow$ **Math** $\rightarrow$ **Probes** $\rightarrow$ **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

CO-1. A difficult interaction with a customer. How did you deal with it? Outcome? Differently?

S11. Situation. At Manhattan Associates a reconciliation showed about two hundred of a retail customer's orders mis-stated between OMS and WMS — roughly \$2.5M of inventory the customer believed was somewhere it was not. My manager preferred to fix it quietly and report the corrected state. **Task.** Decide, that morning, whether the customer heard about it before or after the fix. **Action.** I pulled the complete affected list and the cause class myself, argued for disclosure first, took the list to the customer the same day, and asked them to re-rank the repair by their priorities — stores first, dotcom second. We then bulk-patched the handshake defect class, not the two hundred instances. **Result.** The customer protected what mattered to them, the class never recurred over the following quarter, and we won a larger initiative later. The anger lasted a meeting; the trust lasted the account.

Math. Affected orders = lines where OMS allocation $\neq$ WMS pick confirmation (same order line) ; exposure = Σ units $\times$ cost $\approx$ \$2.5M ; fix scope = defect class, verified by zero recurrences over [4] weeks.

Probe "What did the customer say?" — Anger first, then "thank you for telling us" once they saw they could re-prioritize.

Probe "Would you do this at \$25K?" — Yes; the amount changes the meeting, not the rule.

WBR / design: customer-visible defect metric published weekly; disclosure-before-fix as a standing rule.

CO-2. Went above and beyond for a customer. Why? Response? Outcome?

S3. Situation. An Elementum planner asked for a lead-time variability report. Safety stock on two hundred lanes was sized against carrier-brochure transit times plus a pad "about a week." **Task.** Deliver the report — and, I decided, fix the number the report was about. **Action.** I built per-lane empirical distributions from shipment scans, replaced the planning lead time with the measured p99, and released safety stock only where the measured spread was tighter than the pad implied; where it was wider I *added* stock first, so planners saw the system give before it took. I added a drift flag so the parameter stayed true. I pitched leadership with mockups before building and delivered the report on time as good faith. **Result.** Planned lead time held 99% of the time; 580K units and $\sim$ \$1.5M of safety stock released; the drift flag still fires. The planner got a parameter, not a PDF.

Math. Per lane, sort n transit times; $LT_{plan} = \text{empirical } p99 = t([0.99n])$. Lane A: n 400, mean 21 d, σ 4 d, p99 34 d; a normal fit gives $21 + 2.33 \times 4 = 30.3$ d and would have held only 96.5%

(14 of 400 arrivals beyond it); brochure 18 d + pad $\approx$ 25 d held 93%. Safety stock = $d \times (LT_{plan} - LT_{mean}) + z \times \sigma_d \times \sqrt{LT_{mean}}$: Lane A (d 500/day, σ_d 150) = $500 \times 13 + 2.33 \times 150 \times \sqrt{21} = 6,500 + 1,601 \approx 8,100$ units, vs $\approx 12,900$ before (ERP z-formula with an assumed σ_{LT} of 8 d, plus a 7-day manual pad) $\rightarrow$ 4,800 units released. Release only where σ_{LT} measured $<$ σ_{LT} assumed; lanes where it was wider (Lane B: σ 11 d) took stock in. Net over 200 lanes: $\sim$ 675K out, $\sim$ 95K in = 580K units $\times$ [\$2.59] $\approx$ \$1.5M working capital ($\sim$ \$300K/yr carrying cost at 20%). Full derivation: S3 block under Think Big.

Probe "Who authorized the bigger scope?" — I pitched leadership with mockups before building; the ask was mine to make. **Probe** "What did it cost?" — My time, and engineering time to productize the drift flag.

WBR / design: lane LT distribution pipeline $\rightarrow$ planning parameter $\rightarrow$ drift flag $\rightarrow$ S&OP review line.

CO-3. Anticipated a customer need with a solution they didn't know they needed. How did you know? Response?

S4. Situation. Planners asked for shortage alerts. The SAP views under them — requirements (MD04) and stock (MB52) — disagreed daily by 8–12%, and the first prototype's alerts were dismissed unread. **Task.** Ship alerts planners would act on in S&OP, not another email. **Action.** I watched them dismiss, and built what they needed instead of what they asked for: a daily MD04 $\leftrightarrow$ MB52 reconciliation held at zero for three weeks before any alert fired; a thirty-second reason line on every alert ("coverage falls to 1.6 weeks in week 7 because PO 4471 slipped 9 days"); and a public weekly false-alarm burn-down where the top dismissal category was fixed. **Result.** Live in three months; forecast accuracy 70 $\rightarrow$ 85%; $\sim$ \$2M of safety stock released under a two-signal rule; the alert entered the weekly S&OP without a mandate. Adoption was the metric: % of alerts actioned, weekly.

Math. Recon drift = $\sum_{plant} |MD04 - MB52| \div \sum MB52$ (daily) ; alert fires only if coverage $<$ LT weeks + safety weeks AND drift = green ; false-alarm rate = dismissed $\div$ fired, by cause.

Probe "How did you measure adoption?" — % of alerts actioned in S&OP, weekly. **Probe** "What did they push back on?" — The three-week wait; the drift number falling daily kept them.

WBR / design: recon drift (input) $\rightarrow$ alert with reason line $\rightarrow$ % actioned (input) $\rightarrow$ stockouts without prior alert (output).

CO-4. Asked for customer feedback and used it to drive improvement. Response?

S4 (and S16). Situation. The first alerts were being dismissed; I had the dismissals but not the reasons. **Task.** Turn "noise" into a backlog. **Action.** Every dismissed alert required a reason category — data, logic, timing, ignored. The dismiss rate went on the wall; the top category was fixed in a public weekly forum, with the planner who flagged it named next to the fix. The first top category was timing — right alert, a week early — so we lag-matched the forecast. At Elementum's EDI program (S16) I did the same with routing-guide overrides: a reason code on every override, so skeptics' feedback became the lane-fix backlog. **Result.** False-alarm rate fell week over week; alerts adopted into S&OP; S16 completeness 60 $\rightarrow$ 95% with overrides legitimized rather than policed.

Math. False-alarm rate_w = dismissed_w $\div$ fired_w, split by cause ; weekly burn-down target: top cause $\sim$ 50% in two weeks ; override reason-code coverage = coded $\div$ overrides = 100%.

Probe "What was the top false-alarm category?" — Timing: right alert, fired a week early; we lag-matched the forecast.

Probe "How did you close the loop?" — The fix and the planner who flagged it were named in the forum.

WBR / design: false-alarm rate and override reason-code mix as input metrics with a weekly burn-down.

CO-5. Evaluated the customer experience of your product or service. What did you do? Result?

MAIN. Situation. Shipment Visibility's customer experience was its data: a late milestone or a wrong ETA is the whole product failing. Sixty percent system accuracy at seven customers, and sixty more waiting. **Task.** Evaluate the experience as a number, per wave, before each wave boarded. **Action.** I made data quality $\geq$ 90% against the source of truth and POD-ETA accuracy $\geq$ 80% two of six graduation criteria, measured on the current load; added tickets per customer as the experience metric that customers actually feel; put 4PL-vs-3PL timeliness delta on the WBR so the visibility client's experience was compared to the forwarded client's every Monday; and held wave three publicly when ETA accuracy printed 76%. **Result.** Accuracy 60 $\rightarrow$ 90%; ticket response 72 $\rightarrow$ 90% at 5 $\times$ volume; 68 customers two months early; the 4PL–3PL gap closed from double digits to single.

Math. POD-ETA accuracy = containers with $|ETA_{pred} - ATA| \leq 1$ d $\div$ arrived ; system accuracy = audited milestones matching source $\div$ audited ; Δ timeliness = 4PL % within 24 h – 3PL % within 24 h $\rightarrow$ 0.

Probe "How did you measure ETA accuracy?" — Predicted vs actual milestone within a tolerance window, per shipment, per data source. **Probe** "What did customers notice?" — Fewer tickets; that is the experience metric that moved.

WBR / design: graduation scorecard by wave; tickets per customer as the experience output.

CO-6. A customer came to you for something that wouldn't address their need. How did you approach it? Result?

S5. Situation. An Elementum sponsor in Austria had promised a carrier-routing engine as milestone one; freight was late and "carriers" was the thesis. **Task.** Deliver auditable freight savings — which is what he needed, not what he had asked for. **Action.** I pulled three years of shipment history myself and found we could not measure lateness: 55% true data quality at the planning grain (reported as 40%), carriers under five spellings, brochure transit times stored as actuals. I showed him three real lanes where the engine would have been wrong on current data — privately, one page — and re-sequenced: measurement to 99% by defect class first, engine second, savings method signed with finance per shipment before any claim. **Result.** ~\$8M across 25 locations, every dollar traceable to a shipment. He got the outcome; the thing he asked for came one milestone later.

Math. $P(\text{recommendation on bad input}) \approx 1 - 0.82^3 \approx 45\%$ at 55% completeness ; savings = $\sum_{\text{shipments}} (\text{baseline lane rate} - \text{executed rate}) \times \text{units}$, lanes < 99% quality held out of the claim.

Probe "How did you keep him on side?" — I kept his savings target and his locations; I only moved the first milestone.

Probe "Was he right about anything?" — Yes: some carriers were bad; we could prove which ones only after the data was fixed.

WBR / design: measurement scorecard as milestone one; savings ledger per shipment as the output.

CO-7. Pushed back or said no to a customer request. What did you say or do?

MAIN (wave three); S15 as backup. Situation. Sales — my internal customer — wanted the onboarding pool boarded on their timeline. Wave three's cohort printed POD-ETA accuracy at 76% against an 80% gate. **Task.** Say no without the no becoming a fight about sales versus product. **Action.** I said no publicly, as a date: the criterion, the number, the source class causing 70% of the misses, the fix (a kill switch per source class), and the re-test date one week out. Then I had engineering build the switch. At Flexport's FF App (S15) the same shape: operators asked for their screens in the MVP; the scored scope table said no for two quarters, in public. **Result.** Wave three boarded a week later; ticket response rose to 90% at 5× volume; nobody argued a gate after they had watched one work. The no was a date, not a refusal.

Math. Gate: $\text{ETA accuracy} \geq 80\%$ on the wave cohort at current load ; miss decomposition: source class $Z = [70\%]$ of misses $\rightarrow$ fix $Z \rightarrow$ projected accuracy $76 + 0.7 \times (80 - 76) \div \dots \rightarrow$ re-test ; cost of boarding anyway $\approx \Delta \text{tickets} \times \$ \text{ per ticket} \times \text{cohort weeks}$.

Probe "How did Sales react?" — Badly for a week; nobody argued a gate after they saw it work. **Probe** "When would you have said yes?" — At 80%, or with a signed exception naming who eats the tickets.

WBR / design: gate scorecard public to the requester; every "no" carries the criterion and the re-test date.

CO-8. Balanced the needs of the customer with the needs of the business. What did you do? Result?

S1. Situation. Sephora's omni-DC consolidation launched robotics into a window that ended at peak. The business wanted bots at full volume on the plan date — the ROI the program was sold on. Customers wanted accurate orders. Week two, pick accuracy printed 94% against a 98% zone gate I had published before the bots arrived. **Task.** Serve both without pretending the tension away. **Action.** I throttled bots to the ~30% of volume in zones that had cleared 98%, kept the certified manual path fully staffed as paid insurance, sat the zone-logic debug with the vendor, and restored full volume in week three as zones cleared. The three weeks of robotics ROI were priced against one week of a six-point accuracy miss at peak — an order of magnitude apart. **Result.** Zero customer-facing incidents through peak; +15% processing efficiency on the year; launched on date. The business paid three weeks; the customer paid nothing.

Math. Insurance = $3 \text{ wk} \times [\text{incremental manual hrs/wk}] \times [\text{loaded rate}]$; miss cost = $[\text{peak units/wk}] \times 6\% \times [\text{re-ship} + \text{concession} + \text{store OOS per mis-ship}]$; throttle share = $\sum \text{volume in zones} \geq 98\% \div \text{total} \approx 30\%$.

Probe "How did you price the three weeks?" — Incremental manual hours vs one week of a 6-point accuracy miss at peak; an order of magnitude apart. **Probe** "Who signed?" — The gate was signed before the bots arrived; the throttle needed no new signature.

WBR / design: bot share and zone accuracy as inputs; efficiency and incidents as the two outputs that must both be green.

CO-9. (Manager) A change you implemented in your team or organization to meet customer needs. Result?

S10. Situation. Sephora's inventory organization had five meanings of "available" — stores, DC, dotcom, finance, returns — and the symptoms were oversells, returned units shipped to second customers, and credit issued ahead of the physical unit. **Task.** Change what the organization meant by "available," across sixty-plus stakeholders in a matrix, so the customer promise was true. **Action.** One signed definition per stock state with the failure mode attached — the customer harmed if that state is promised — before configuration; a returns plant so nothing is promiseable until disposition; RTV as pick-reserve; and a nightly ATS rebuilt from the WMS supply basis, reconciled to SAP with a bucketed out-of-balance and an owner per bucket. One weekly pack, metrics frozen. **Result.** Zero promises against non-sellable stock in production, zero go-live incidents, legacy system retired. The customer saw fewer cancellations and no second-hand "new" units.

Math. $ATS = \text{supply basis} - \text{pack reserve} - \text{ecomm pick reserve} - \text{ecomm pack reserve} - \text{SNB} + \text{mech reserve} + \text{L\&F}$; $OOB = \text{SAP supply} - (\text{basis} + \text{SNB})$; $|OOB| \leq 0.5\%$; worked: 10,000 → 7,400 ATS, OOB = 0.

Probe "What did customers see?" — Fewer cancelled orders and no returned units shipped to a second customer.

Probe "How is it still running?" — The nightly OOB pack with owners; a bucket repeating three nights becomes a defect.

WBR / design: $ATS = \text{basis} - \text{reserves} - \text{SNB} + \text{mech} + \text{L\&F}$; OOB vs SAP supply; oversells as output, buckets as inputs.

Ownership

Think long term; act on behalf of the whole company; never say "that's not my job." They listen for: a problem nobody owned, a decision outside your lane, and what you did when the easy move was to hand it off.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S10	"Nobody owned what "available" meant. I wrote the definitions, took them to steering with the failure mode attached, and owned the cutover." S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback	below
MAIN	"The E-Team wrote "Visibility team to publish its own WBR." Nobody was assigned. I built it from a borrowed sheet that week and ran it for a year." MAIN — Shipment Visibility (Flexport)	Section 5
S3	"Nobody owned the seam between brochure lead times and the plan. I took the seam, not the ticket." S3 — Two hundred lanes: plan lead time at the empirical p99, release the inventory the brochure was holding	under Think Big
S8	"I was the single point of failure in the review. I wrote the BRD to remove myself." S8 — Four systems into one pipeline: the weekly review is only as good as the data under it	under Invent and Simplify

S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback (Sephora S/4HANA ↔ ISM)

LPs: Ownership · Invent and Simplify · Earn Trust · Dive Deep · Deliver Results · Are Right, A Lot (being wrong).

AIQB: Ownership 1, 3 · Invent and Simplify 1, 3 · Earn Trust 1, 6 · Are Right, A Lot 4, 5 · Deliver Results 2.

Problem	S/4HANA was becoming the inventory system of record for stores, retail DC and dotcom with a warehouse execution system beside it. "Available" meant something different in every system because each definition protected a different number. Customer returns, RTV stock, quality-hold and lost-and-found were being promised to customers because somewhere they counted as available.
Goal	One signed definition per stock state, with one owner and one promising rule; returns handled as first-class inventory; cutover with zero customer-facing incidents and a rollback that is a number, not a paragraph; the legacy inventory system (MCS) retired.
Key metric	Units promised against non-sellable states (returns, QI, blocked, in-transit) → zero; nightly out-of-balance (OOB) between SAP supply and WMS supply within tolerance; customer-facing incidents at go-live; interface sync and p95 latency vs rollback triggers; MCS retired.
Did it move?	Yes. Non-sellable promises → zero, verified in production. Nightly OOB between SAP and WMS inside tolerance with every exception bucketed and owned. Zero customer-facing incidents on staged go-lives. 343-task runbook; rollback triggers never fired. MCS retired. One rule I wrote was wrong and was corrected through the same signed policy.

Situation. "Sephora was standing up S/4HANA as the inventory system of record for stores, the retail DC and dotcom, with a warehouse execution system beside it. The problem was not the software. 'Available' meant something different in every system, because each definition protected a different number — fill rate for stores, labor for the DC, the promise for dotcom, the credit memo for finance. Customer returns, return-to-vendor stock, quality-hold and lost-and-found were being promised to customers because somewhere they counted as available.

Task. One definition per stock state, one owner, one rule about who may promise against it — signed by the business before anyone configured anything. Returns as first-class inventory. A cutover with zero customer-facing incidents. And retire the legacy inventory system.

Action. I redesigned the inventory operating model. Every stock state — available-to-sell, reserved, in-transit, quality-inspection, blocked, lost-and-found — got one definition, one owner, and one promising rule, and I took the whole set to steering with the failure mode attached to each: here is the customer who gets hurt if this state is promised. Returns got a plant of their own. Return-to-vendor stock-transport orders were modeled as pick-reserve so credit could never run ahead of the physical unit: receive, grade, then restock, scrap or RTV — nothing promiseable until disposition. The daily proof was the available-to-sell computation. S/4HANA was the aggregated book of record; the Inventory Supply Management layer computed supply and demand across store, retail DC and dotcom; and every night it rebuilt ATS from the WMS supply basis — minus pack reserve, minus e-comm pick and pack reserves, minus shipped-not-billed, plus mech reserve and lost-and-found added back — then reconciled it to SAP supply and published the out-of-balance by SKU and DC with a tolerance and a bucket for every unit outside it. If the OOB was not explained by six in the morning, it carried an owner. I ran it as a program: RAID log, a weekly CIO/COO pack across sixty-plus stakeholders on frozen metrics so a red stayed red, and a 343-task weekend cutover runbook with rollback written as a number before the weekend — interface sync under 99% for four hours, or p95 latency over twice budget for one hour. One rule I wrote was wrong: I forbade every channel from promising against units on a truck. It starved dotcom of sellable units while in-transit inventory rose. The data showed the miss; the fix went through the same signed policy — a per-lane in-transit promise window from measured lane data.

Result. Could a unit in quality-hold or on a returns pallet still be promised to a customer? No — verified in production. Zero customer-facing incidents on the staged go-lives. The rollback never fired. MCS came out only after the new definition held in production. And the in-transit correction recovered the sellable units I had hidden, through policy, not a hotfix."

Supply · Demand · Capacity. **Supply:** inventory by state — the point is that only some states are supply. **Demand:** three channels promising against the same pool. **Capacity:** returns are inbound volume the plan did not forecast; quality-hold is storage you cannot sell; in-transit is supply you may or may not count. **What bound:** definitions. A capacity plan that counts non-sellable states as available overstates usable capacity, and the promise lies. Giving each state a definition and an owner is the same move as giving each building type its own ceiling.

What was innovative. Treating "available" as a policy document signed by the business before configuration, with the failure mode written next to each state; modeling RTV as pick-reserve so finance could not out-run the floor; and writing rollback as two numbers so a tired bridge at 2 AM could call it without a meeting. Also: correcting my own wrong rule through the policy, not around it.

The math — available-to-sell (ATS) and the end-of-day reconciliation

This is the computation the program lived on. S/4HANA is the **aggregated book of record** (plant / storage-location stock in MM-IM); the **Inventory Supply Management (ISM)** layer computes supply and demand across **store, retail DC and dotcom** and publishes ATS by channel; the WMS holds the physical and the reserves. Every night the three had to agree, and the number that proved it was the **out-of-balance (OOB)**.

STEP 1 — SUPPLY BASIS (from WMS, by SKU × DC, EOD snapshot)
Supply basis = physical on-hand in sellable locations

(excludes QI, blocked, damaged, returns-plant stock)

STEP 2 – AVAILABLE TO SELL (ISM computes by SKU × DC, then splits by channel policy)

ATS = Supply basis

- Pack reserve store/retail orders waved, picked, at pack (committed)
- Ecomm pick reserve dotcom orders allocated, not yet picked
- Ecomm pack reserve dotcom orders picked, at pack
- Shipped-not-billed left the dock, goods issue not yet posted in SAP
(gone physically, still in SAP stock)
- + Mech reserve units parked in automation buffers – physically
present and sellable; WMS shows "reserved"
- + Lost & found found, not yet re-binned – present, sellable once binned

STEP 3 – RECONCILE TO SAP (the OOB)

SAP supply = SAP unrestricted (MB52 / MARD) – SAP reservations, same SKU × plant

WMS supply = Supply basis + Shipped-not-billed

(add SNB back: SAP still holds it until GI posts)

OOB (units) = SAP supply – WMS supply ; OOB (\$) = OOB × standard cost

Tolerance = $|OOB| \leq 0.5\%$ of SAP supply per SKU × DC

AND $\sum |OOB \$| \leq [\text{materiality}]$ per DC per day

Buckets = timing (in-flight IDocs) · SNB > 24 h · adjustment posted one side
· receipt posted one side · L&F ageing

WORKED EXAMPLE – one SKU, one DC, one night (illustrative units)

Supply basis (WMS physical, sellable)	10,000	
- Pack reserve	-1,200	
- Ecomm pick reserve	-800	
- Ecomm pack reserve	-500	
- Shipped-not-billed	-300	
+ Mech reserve	+150	
+ Lost & found	+50	
= ATS	7,400	→ split by channel policy (dotcom floor, store allocation)

SAP unrestricted	10,420
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- SAP reservations (STO / delivery)	-120
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= SAP supply	10,300
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WMS supply = 10,000 + 300 (SNB)	10,300
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OOB = 10,300 – 10,300 = 0 ✓

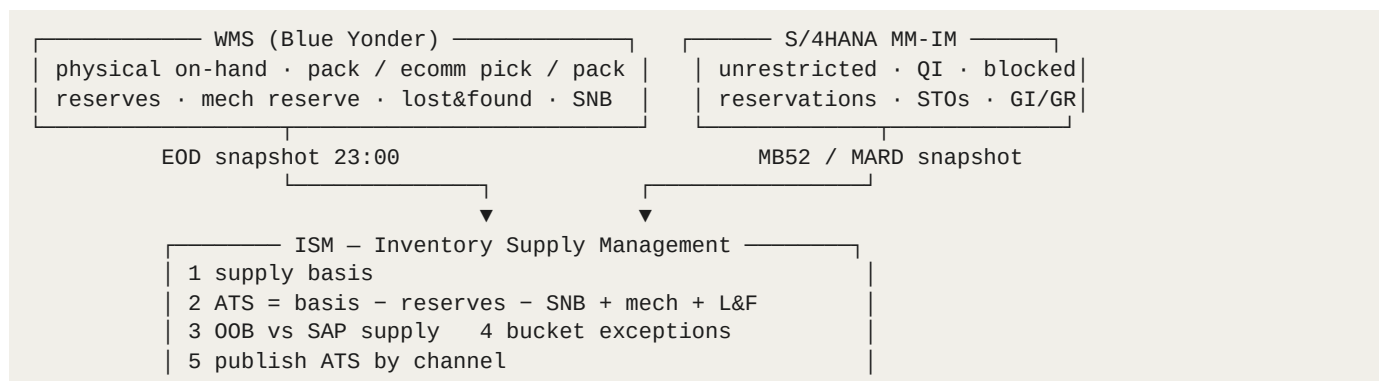
(if SAP showed 10,420 after reservations → OOB = +120

→ classify: 90 timing IDoc, 30 adjustment posted one side)

WHY EACH TERM IS THERE

- reserves subtracted: already promised to someone – selling them twice is the oversell
- SNB subtracted from ATS (it is gone) but added back for the SAP comparison
(SAP has not posted the goods issue yet)
- mech + L&F added back: physically here and sellable – WMS "reserved" ≠ "promised"
- returns, QI, blocked NOT in supply basis at all – nothing is promiseable until
disposition (the returns plant)

Solution design / flow — EOD reconciliation and ATS publication



dotcom OMS (ATS)
promise ≤ ATS

store replenishment
allocation ≤ ATS

DC / finance
OOB pack + exceptions queue
fix by 06:00 or carry with owner

STOCK-STATE MODEL (signed before configuration)

state	owner	promiseable?	example
available	ISM	yes (by channel)	stowed, sellable location
reserved	WMS	no (already is)	pack / pick reserves
in-transit	logistics	inside lane window	STO on a truck (my wrong rule: "never" → fixed to per-lane window)
quality-inspect	QA	no	returns awaiting grade
blocked	QA/finance	no	damaged, recall
lost & found	DC ops	after re-bin	found on floor, not in a bin
returns plant	returns	no until disposition	receive → grade → restock / scrap / RTV (RTV STO = pick-reserve)

How it translated to a WBR

Outputs: oversells (orders promised > physical) — target 0; OOB \$ by DC by day within tolerance; ATS publication by 06:00. **Inputs:** OOB by bucket (timing / SNB ageing / one-sided adjustments / L&F ageing); SNB > 24 h units; L&F > 48 h units; IDoc failure count; in-transit units outside the lane window. **Red rule:** an OOB bucket that repeats three nights running becomes a defect with an owner, not a nightly fix. **Cutover rollback triggers:** interface sync < 99% for 4 h, or p95 latency > 2× budget for 1 h — never fired.

SITUATION PROBES

Why important? Cost of nothing? Every oversell was a cancelled order and a customer who did not trust the site; every return promised before grading was a damaged unit shipped to a second customer. Finance was issuing credit on units the floor had not touched.

What was the initial scope? Configure S/4HANA to match the legacy definitions. I changed it to: define first, configure second, because the legacy definitions were the problem.

ACTION PROBES

Why a signed policy before configuration? Most oversell bugs are vocabulary bugs. A config change nobody signed becomes next quarter's argument. Once the sentence was signed, the design arguments got small — the plant and the STOs were just the implementation of a sentence everyone had agreed to.

Why a separate returns plant? That's heavy. Because a returned unit is not the same supply as a new one until someone has graded it, and the only way to make the system enforce that was to make it physically and logically elsewhere. The heaviness is the point: nothing in that plant is promiseable until it leaves.

Why pick-reserve for RTV? So finance's credit memo could not run ahead of the physical unit. The reservation ties the money to the pick. Before, credit was issued on paper and the unit was sometimes still on a store shelf.

Why rollback as a number? A rollback that is a paragraph gets debated at 2 AM by tired people. A number gets called by whoever is on the bridge. The triggers existed so the weekend could start.

How did you keep sixty stakeholders from re-litigating reds? Frozen metrics in the weekly pack. A red could not be made green by changing the chart; it could only be made green by moving the number.

What did you do versus the team? Wrote the stock-state definitions and the failure modes; designed the returns plant and the pick-reserve RTV model; owned the RAID, the pack, the runbook, and the rollback triggers; made the MCS retirement call. SAP and the integrator configured.

RESULTS PROBES

Walk me through the ATS computation. Supply basis is WMS physical on-hand in sellable locations. Subtract what is already promised — pack reserve for stores, e-comm pick reserve and pack reserve for dotcom — and subtract shipped-not-billed because it has physically left. Add back mech reserve, units parked in automation buffers that WMS flags as reserved but nobody has promised, and lost-and-found, units physically present that will be

sellable once re-binned. That is ATS, split by channel policy. Then reconcile: SAP supply is unrestricted minus SAP-side reservations; WMS supply is basis plus shipped-not-billed, because SAP still carries it until the goods issue posts. The difference is the OOB.

Why add back shipped-not-billed for the SAP comparison but subtract it from ATS? Two different questions. ATS asks "what can I promise" — SNB is gone, so it is out. The OOB asks "do the two books agree" — SAP has not posted the goods issue, so SAP still holds it; to compare like with like I add it back on the WMS side. If SNB ages past 24 hours it stops being a timing difference and becomes a defect with an owner.

What did the OOB usually come from? Timing — IDocs in flight at snapshot — was the biggest bucket and self-cleared by morning. Then one-sided adjustments, SNB ageing, and lost-and-found not re-binned. The nightly pack showed the buckets; a bucket that repeated three nights became a defect, not a nightly fix.

How did you verify "non-sellable promises → zero"? Promise log joined to stock state at promise time, daily in hypercare. The count of promises against QI, blocked, returns-plant or lost-and-found stock: zero.

How did you know the in-transit rule was wrong? Sellable units by channel fell while units in transit by lane rose. Two lines diverging is the signal. Fix: a per-lane in-transit promise window from measured lane transit — the 200-lane idea applied to promising instead of planning.

What did the wrong rule cost? Weeks of dotcom running with fewer sellable units than it physically had. I will not quote a dollar I did not measure; I can say the recovery was visible in sellable units the week the window went live.

Compared to goal? Definitions signed: yes. Returns first-class: yes. Zero incidents: yes. Rollback never fired: yes. MCS off: yes. One rule wrong and corrected: owned.

CHALLENGE

Differently? Set the in-transit window from lane data on day one instead of starting from "never." Conservatism has a price, paid in the same currency as risk. And sample the OMS-to-WMS handshake daily in hypercare from day one.

At Amazon. Returns are inbound volume the plan did not forecast; quality-hold is cube you cannot sell; in-transit is supply you may count only inside a measured window. I would want the capacity plan's "available" to be a signed definition by stock state — and I would want to know which states today's utilization number includes.

The question bank — Ownership (6 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

OW-1. Took on something significant outside your area. Why important? Outcome?

S10. Situation. I was the fulfillment TPM. "What does available mean" sat between stores, finance, dotcom and the DC, and belonged to nobody; every oversell and mis-shipped return traced to it. **Task.** Take the seam, not the ticket. **Action.** I wrote the stock-state model — state, owner, promiseable?, example — ran the workshops where each group named the customer harmed by its own definition, took the set to steering for signature before configuration, designed the returns plant and RTV-as-pick-reserve, and specified the nightly ATS/OOB reconciliation that proved SAP and WMS agreed. I ran it as a program: RAID log, weekly CIO/COO pack, 343-task cutover runbook with rollback written as two numbers. **Result.** Zero non-sellable promises, legacy system off, OOB nightly within tolerance with a bucket owner for anything outside it. Nobody had asked me to own the definition; the definition was the whole problem.

Math. Rollback triggers: interface sync < 99% for 4 h OR p95 latency > 2× budget for 1 h ; OOB buckets: timing · SNB > 24 h · one-sided adjustment · one-sided receipt · L&F ageing ; a bucket repeating 3 nights = defect with owner.

Probe "Who should have owned it?" — Everyone said someone else; that was the signal. **Probe** "Did anyone object to you taking it?" — Finance, briefly; the signature process gave them a veto, which is why they accepted it.

WBR / design: stock-state model with named owners — the seam now has an owner per state.

OW-2. Didn't think you'd meet a commitment. How did you identify and communicate the risk? Differently?

S8 (and MAIN). *Situation.* Mid-build of the Snowflake pipeline for the Monday review, I learned Zendesk ticket text carries customer content and needed a data-security review. The cutover date was dead the moment I knew. *Task.* Identify the risk, size it, and say so before anyone found out from the slip. *Action.* I confirmed the classification with security that morning, got the review's lead time (two weeks), re-sequenced the build so non-sensitive sources landed first, and told my manager the same day with the new date, the reason, and the fact that the parallel run kept Monday alive on the manual path. *Result.* Two weeks late, zero business impact, and a kickoff-checklist item I have used on every data program since: "whose approval does this data need, and what is its lead time." The pipeline still landed at 60 → 90% accuracy and \$103,680/yr saved.

Math. Slip cost = 2 wk × 30 h/wk × \$90 = \$5,400 of manual assembly – small, because the manual path was still running ; the unpriced cost was credibility, repaired by owning it the same day.

Probe "How did your manager react?" — Fine, because the business impact was zero and the date was already re-planned. **Probe** "What if the parallel run had not existed?" — Then a missed Monday, and a worse conversation.

WBR / design: approval lead times as RAID rows; parallel run as the safety net.

OW-3. An initiative that benefited the company but wasn't any group's responsibility.

S3; MAIN as second example. *Situation.* The seam between carrier-brochure lead times and the inventory plan: carriers owned the brochure, planners owned the plan, nobody owned whether the number was true. *Task.* Own the truth of the parameter. *Action.* I measured two hundred lanes from ship and arrival scans, replaced the parameter with the empirical p99, built the drift flag so it re-measured itself, and got finance to sign the release rule. At Flexport (MAIN) the same move: the E-Team's action said "visibility team publishes a WBR," assigned to no one; I took it and made it the ramp's operating system. *Result.* ~\$1.5M released at 99% accuracy; the flag still fires. At Flexport, a WBR with seventeen lines, run by hand then automated, and cost-to-serve \$0.66 → \$0.03.

Math. Drift flag: rolling-90-day p99 > planned LT + [2 d] OR $\sigma_{90} > 1.25 \times \sigma_{\text{baseline}}$; MAIN: tickets/active container = (S - A) ÷ N as the owned input metric.

Probe "How did you get permission?" — Delivered the report I was asked for on time and pitched the parameter change with a three-lane proof. **Probe** "Who owns it now?" — Planning, with the drift flag as their trigger.

WBR / design: drift flags open as a planner-owned input.

OW-4. Sacrificed short-term gain for long-term value. Outcome? Differently?

S4 (and S1). *Situation.* Planners wanted alerts immediately; the data under them did not reconcile. *Task.* Choose between a visible win now and alerts that would be trusted for years. *Action.* I spent three weeks holding the MD04 ↔ MB52 reconciliation at zero before a single alert fired, and published the drift number daily so the sacrifice looked like progress. At Sephora (S1) the same trade: three weeks of robotics ROI spent on a 30% throttle to keep zero customer incidents through peak. *Result.* Alerts adopted into S&OP un-mandated; forecast accuracy 70 → 85%; ~\$2M released. S1: zero incidents, +15% efficiency. Differently on S4: show the drift falling from day one, not day five.

Math. Cost of waiting = 3 wk × [stockouts a trusted alert would have prevented] ; cost of not waiting = alerts ignored forever (adoption → 0) – unbounded ; drift target = 0 for [5] consecutive days before go-live.

Probe "How did you know the long-term value was real?" — Alerts ignored cost more than alerts delayed; I had watched the first prototype get dismissed. **Probe** "Who absorbed the short-term pain?" — Planners, and I told them why in the first meeting.

WBR / design: recon drift published daily — the sacrifice made visible as a metric.

OW-5. Transitioned a project you owned to a new owner; and a time you got involved in one you had transitioned.

S12 (transition) and S3 (re-involvement). *Situation.* I had built a release gate at Sephora; I was rolling off. Separately, a quarter after handing Elementum's lane system to planning, a drift flag fired and sat untouched for two weeks. *Task.* Transfer ownership so it survives me; and later, step back in without taking it back. *Action.* Transition: written criteria, an exception path with reason codes, and two release cycles of the QA lead defending the gate with me silent in the room. Re-involvement: I did not decide the lane — I made the flag's *ageing* a red on the planner's own line, so an ignored flag became visible to their manager. *Result.* The gate outlived my engagement and was copied by neighboring teams; the ageing red made ignored flags impossible. I got involved because the mechanism was at risk, not the lane.

Math. Durability test = gate still in use [N] quarters after roll-off ; flag ageing red: days open > [10] → red on planner line ; the decision itself stayed with planning.

Probe "How did you keep from taking it back?" — I changed the rule, not the decision. **Probe** "Signs a transition worked?" — Decisions made without you that you would have made.

WBR / design: owner named on the artifact; ageing of open flags as the health check.

OW-6. Saw a peer struggling and stepped in. Why? Actions?

S13. Situation. Sephora's engineering lead was carrying two go-lives on one team and every night call. Not my half of the program; my go-live chain. **Task.** Take load off him without taking the program from him. **Action.** I built the two-pass war room — a day pass that triages, a night pass in another time zone that executes under written decision rights for reversible actions — one RAID log, one 45-minute daily, nobody on both passes. I took the day-pass triage myself for the first two weeks. **Result.** Issue age fell, both go-lives landed on date, ~\$1M per DC protected, no 24-hour days. He kept the program; the clock absorbed the load.

Math. Issue age = hours open → resolved ; two passes: work moves 16–20 h/day on 8-h shifts ; people on both passes = 0 by rule, checked from the log ; cost of delay ≈ \$200K/month/DC.

Probe "Did he want help?" — Not at first; the RAID log showing his name on 70% of rows made the case without a conversation about him. **Probe** "What did it cost you?" — Two weeks of early mornings.

WBR / design: rows per owner as a load input; people on both passes as the red.

Invent and Simplify

Expect innovation from yourself and the team; always find ways to simplify; be externally aware. They listen for: the usual approach, what you did instead, and what got removed.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S8	"New metrics without killing old ones is complexity wearing a badge. I froze seven definitions and killed the spreadsheet." S8 — Four systems into one pipeline: the weekly review is only as good as the data under it	below
S16	"A thousand routes, one gate. The gate replaced a thousand onboarding conversations." S16 — A thousand routes, one gate	below
MAIN	"Automation rate as the one number: every ticket is a failed automation, so the cost line and the quality line became the same line." MAIN — Shipment Visibility (Flexport)	Section 5
S10	"Most oversell bugs are vocabulary bugs. One definition, one owner, one rule per stock state." S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback	under Ownership

S8 — Four systems into one pipeline: the weekly review is only as good as the data under it
(Flexport)

LPs: Invent and Simplify · Dive Deep · Frugality · Ownership. **AIQB:** Invent and Simplify 1 · Dive Deep 5 · Frugality 5, 6 · Ownership 2.

Problem	The Visibility WBR and the ops review both ran on four systems stitched together in Excel — Zendesk tickets, the platform's container records (Crux), CRM and ops. Each weekly snapshot took 1.5–2 hours of a person's hands to pull and join on container number; across three people and two reviews that was ~30 hours a week, ~60% agreement with source systems, one person (me) as the single point of failure, and a VP steering on numbers that disagreed with their sources.
Goal	A review whose data is boring: accuracy ≥ 90%, zero manual assembly, no single point of failure — without a new team or a new tool budget.
Key metric	Metric accuracy vs source systems (60%); manual hours per week (~30: 3 people × 5 h × 2 reviews); cost of status quo (\$129,600/yr, priced in the BRD at the infrastructure team's \$90/h rate); parallel-run delta (<1%).
Did it move?	Yes. Accuracy 60% → 90%. Manual hours ~30 → 6 (review and exception handling only): \$129,600 → \$25,920/yr, \$103,680/yr saved . Parallel run within 1% for consecutive weeks before cutover. First automated catch: a 4% ticket-sync drop the manual era never saw.

Situation. "Flexport's visibility review ran on four systems stitched together in Excel — Zendesk tickets, the platform's container records, CRM and ops. Every weekly snapshot was two hours of someone's hands joining tickets to container numbers; across three people and two reviews it was thirty hours a week, about sixty percent agreement with the source systems, and a VP making decisions on it. I was the single point of failure, and I knew the numbers were soft.

Task. Make the data under the review boring: ninety percent accuracy or better, no manual assembly, no dependence on me — and do it without a new team.

Action. I wrote the BRD myself — Zendesk ticket data into Snowflake, joined to the platform's container numbers, loaded daily with an hourly option, an alert to the owner on any failed load — and froze seven metric definitions exactly as the review already used them: coverage automated, coverage with human inputs, tickets generated, tickets per active container, 4PL and 3PL percent within SLA, and the delta between them. No new metrics, no renamed ones, because changing the vocabulary during a migration lets everyone relitigate every red. I priced doing nothing at a hundred and thirty thousand dollars a year — thirty hours a week at the infrastructure team's own ninety-dollar rate — and the target state at twenty-six, so the request competed on their numbers, not mine. I gated cutover on a parallel run: manual versus pipeline within one percent for consecutive weeks, published. Four systems became one Snowflake pipeline. I hit one wall I did not plan for — ticket data carries customer content, and the security review cost two weeks. I owned it upward the day I found it and re-sequenced; the parallel run meant Monday never missed.

Result. Accuracy from sixty to ninety percent. Thirty manual hours a week to six — about a hundred and four thousand dollars a year back — and the review no longer stopped when I was out. The first automated catch was a four percent ticket-sync drop the manual era had never seen, because the manual era could not see it. The weekly cadence only works if the data underneath it is boring."

Supply · Demand · Capacity. **Demand:** decisions the review had to support every Monday. **Supply:** data from four systems. **Capacity:** the review's capacity to see — bounded by thirty manual hours a week and one person who knew the join. **What bound:** the review itself. A WBR is a capacity system for decisions; this story is about expanding its throughput and removing its single point of failure.

What was innovative. Most "automate the dashboard" projects redesign the metrics at the same time and lose the room. I froze the vocabulary as a contract, made the status quo compete on the infrastructure team's own cost, and treated cutover like a WMS go-live — a numeric parallel-run gate. The pipeline was the easy part. The query layer was deliberately boring too: Looker's SQL Runner on the Snowflake tables, so an analyst could re-derive any of the seven numbers without a ticket to engineering.

The math

Accuracy	= metric-weeks within tolerance of source-system recompute ÷ metric-weeks 60% → 90%
Status-quo cost	= 3 people × 5 h/wk × 2 reviews × \$90/h × 4 wk = \$10,800/mo ≈ \$129,600/yr
Target-state cost	= 6 h/wk (review + exceptions) × \$90/h × 4 wk × 12 = \$25,920/yr
Savings	= \$129,600 - \$25,920 = \$103,680/yr
Per snapshot	= 1.5–2 h hand-join (Zendesk ↔ container no.) → 0 h (daily load, hourly opt.)
Tickets / active	= tickets opened in week ÷ containers with ≥1 milestone in week (one of 7)
Coverage (auto)	= containers with milestones and no ticket ÷ containers submitted
Coverage (w/ human)	= containers with milestones (any path) ÷ containers submitted
Parallel-run gate	= manual - pipeline ÷ manual ≤ 1% for each of 7 metrics, [3] consecutive weeks
First catch	= ticket-sync completeness fell 4% (records in source - records in pipeline) ÷ source

Solution design / flow

```
Zendesk · platform(Crux) · CRM · ops → Snowflake (daily/hourly) → 7 FROZEN definitions (Looker)
                                     |
                                     |→ Monday review
                                     |→ parallel run vs Excel (≤ 1%) → cutover
                                     |→ Excel
                                     |→ retired
security review (customer content in tickets) → now a kickoff-checklist item with lead time
failed load → alert to pipeline owner (BRD BR-4); metric shown flagged that week, never stale
```

How it translated to a WBR

Outputs: review metric accuracy; decisions reversed due to data error (target 0). **Inputs:** pipeline freshness; source completeness by system; definition-change requests pending. **Red rule:** any metric whose source completeness drops > 2% is shown with a flag that week, not silently.

SITUATION PROBES

Why important? A VP was steering on numbers that were forty percent wrong somewhere. One Monday the wrong forty percent would be the one that mattered.

Cost of nothing? Thirty hours a week across three people, priced at \$129,600 a year; the latent cost of a bad decision, unpriced; and a review that stopped the week I was out.

ACTION PROBES

Why freeze the definitions? Some were bad. Some were. I wrote the bad ones down for a later change with its own approval. Changing definitions mid-migration means nobody can tell a pipeline bug from a definition change, and every red becomes an argument about the chart instead of the business.

Why price the status quo in their rates? Because the infrastructure team had to build it, and "it saves three people a few hours" is not a priority for them. "It costs you \$129,600 a year of your own people's time and the fix costs \$25,920" is.

Why a 1% parallel run — why not just switch? Because the review was live. The parallel run is the mock cutover: I would not go live on a WMS without one and I would not go live on the VP's numbers without one.

What went wrong and what did you do? I did not plan the data-security review; ticket text carries customer content. Two weeks. I told my manager the day I found it, with the new date and the reason, and the manual path kept Monday alive. Every data program since has "whose approval does this data need, and what is its lead time" on the kickoff checklist.

What did you do versus the team? BRD, frozen definitions, status-quo pricing, parallel-run gate, the escalation. Infrastructure built the pipeline.

RESULTS PROBES

How did you measure 60% and 90%? Each of the seven metrics, weekly, recomputed from source systems and compared to the review value. Share of metric-weeks within tolerance.

The 4% catch — why does it matter? Ticket sync had been silently dropping records, which made response rates look better than they were. The manual process could not detect a drop it was built on top of. The first week of the pipeline, it flagged. That is the difference between a dashboard and a mechanism.

Compared to goal? 90%: met. Zero manual assembly: met (six hours a week of review and exception handling remain, by design). No SPOF: met. Two weeks late: owned.

CHALLENGE

Differently? Security review on the kickoff checklist. I have never missed it since.

At Amazon. Before I change a single metric in a capacity WBR I would freeze the existing definitions, run the new source in parallel against the old for a published number of weeks, and only then cut over. New metrics come after, through their own door.

S16 — A thousand routes, one gate (Elementum EDI)

LPs: Insist on the Highest Standards · Deliver Results · Dive Deep. **AIQB:** Highest Standards 1, 3 · Deliver Results 3 · Dive Deep 5.

Problem ~1,000 supplier and carrier routes to onboard onto X12 EDI with SAP PO and ASN integration; shipment completeness at 60%, so planners could not see what was coming.

Goal All routes live with completeness $\geq 95\%$; no cohort graduates on bad data.

Key metric Valid ASN rate per cohort (gate $\geq 95\%$); shipment completeness (60% baseline).

Did it move? **Yes.** Completeness 60% $\rightarrow$ 95%. Every cohort graduated at $\geq 95\%$ valid ASNs. Routing-guide overrides carried reason codes.

Situation. "About a thousand supplier and carrier routes had to go onto X12 EDI with SAP PO and ASN integration; shipment completeness sat at sixty percent, so the plan was blind to forty percent of what was in motion. **Task.** All routes live, completeness to ninety-five, and no cohort allowed through on bad data. **Action.** Cohorts graduated only at ninety-five percent or better valid ASNs — one gate, published, no exceptions by relationship. And a detail that mattered: the routing-guide skeptics were often right about specific lanes, so I made overrides legitimate — every override carried a reason code, which turned sabotage into data and gave me the lane list to fix. **Result.** Shipment completeness from sixty to ninety-five percent. Every cohort graduated at the bar. The reason codes became the backlog for the routing guide's next version."

Supply · Demand · Capacity. **Supply:** inbound shipments — visible only if the ASN is valid. **Demand:** planners' need to see supply in motion. **Capacity:** inbound planning capacity is bounded by visibility — you cannot schedule a dock for a shipment you cannot see. **What bound:** data completeness.

The math

Valid ASN rate (cohort)	= ASNs passing X12 validation + PO match $\div$ shipments ; gate $\geq 95\%$ to graduate
Shipment completeness	= shipments with a valid ASN before arrival $\div$ shipments 60% $\rightarrow$ 95%
Override reason codes	= overrides with code $\div$ overrides = 100% ; top codes $\rightarrow$ routing-guide backlog
Residual	= 5% edge formats, documented with owner (cost to fix > visibility gained)

Solution design / flow

```
supplier / carrier  $\rightarrow$  X12 EDI (856 ASN, 850/855 PO)  $\rightarrow$  SAP PO  $\rightarrow$  validation (format · PO · qty)
├─ pass  $\rightarrow$  ASN accepted
└─ fail  $\rightarrow$  supplier scorecard
    └─ cohort gate ( $\geq 95\%$ )

routing guide  $\rightarrow$  override + reason code  $\rightarrow$  lane fix backlog
```

How it translated to a WBR

Outputs: shipment completeness; receiving dock surprises (shipments with no ASN). **Inputs:** valid ASN rate by cohort; override reason-code mix. **Red rule:** a cohort below 95% does not graduate regardless of the relationship.

Why 95% and not 100%? The last five percent were edge formats that cost more to fix than the visibility they bought; I documented them as a known residual with an owner instead of pretending the gate was 100%.

Why let people override the routing guide? Because they were going to anyway. A reason code made the override visible and often correct; forbidding it would have made it invisible and always suspicious.

At Amazon. Inbound capacity planning depends on ASN accuracy and appointment adherence; I would gate vendor cohorts on measured ASN validity and treat every dock-appointment override as a reason-coded data point about where the plan is wrong.

The question bank — Invent and Simplify (7 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

IS-1. A complex problem solved with a simple solution. What made it complex? How do you know it worked?

S10. Situation. Five systems and five meanings of "available"; returns and quality-hold leaking into promises; finance crediting ahead of the floor. Complex because every group's definition was locally right. **Task.** One answer everyone could compute. **Action.** One sentence per stock state — definition, owner, promiseable? — signed before configuration; and one nightly computation: $ATS = \text{supply basis} - \text{reserves} - \text{shipped-not-billed} + \text{mech reserve} + \text{lost-and-found}$, reconciled to SAP supply with a bucketed OOB. The returns plant made "not promiseable until disposition" a system fact rather than a rule to remember. **Result.** I know it worked because the promise log joined to stock state shows zero promises against non-sellable states, the OOB sits inside 0.5% nightly, and the arguments got small — they were about implementation of a sentence everyone had signed.

Math. Worked night: $10,000 - 1,200 - 800 - 500 - 300 + 150 + 50 = 7,400$ ATS ; SAP $10,420 - 120 = 10,300$; WMS $10,000 + 300 = 10,300$; OOB = 0.

Probe "What did you remove?" — Every channel-specific exception; the plant and STOs implemented one sentence.

Probe "What was hardest to make simple?" — Shipped-not-billed: one fact, two treatments.

WBR / design: stock-state table + ATS/OOB flow (S10 block, under Ownership).

IS-2. The most innovative thing you've done and why. One or two more examples — is it a pattern?

S3, then the pattern. Situation. Plans ran on judgment calls — brochure lead times with a pad "by feel." **Task.** Replace the judgment with a rule that fires on its own. **Action.** Planning lead time at the empirical p99 per lane, with a drift flag that reopens the parameter when the lane changes — the plan measures itself. The pattern repeats: MAIN, onboarding admitted on measured ability to serve, and the human-in-the-loop measured as tickets per active container until automation retires it; S10, rollback written as two numbers so a tired bridge can call it at 2 AM; S2, promise-at-risk ranking — dollars × urgency — instead of FIFO. **Result.** ~\$1.5M at 99%; \$0.66 → \$0.03 per container; a rollback that never fired; 99% fill through peak. The innovation is one move made four times: a measured rule where a meeting used to be.

Math. p99 per lane vs normal fit ($21 + 2.33\sigma = 30.3$ d vs empirical 34 d) ; automation rate = $A \div S$; rollback = $\text{sync} < 99\% / 4 \text{ h}$; promise-at-risk = $\$ \times 1 \div \max(\text{hours to cutoff}, 0.5)$.

Probe "Why is p99 innovative — it's a percentile?" — The innovation is the system: per lane, empirical, with a release rule and a drift detector. The percentile is the easy part. **Probe** "What innovation failed?" — "Never promise in-transit" — a rule without a measurement.

WBR / design: rules with numbers and triggers — the WBR grammar itself.

IS-3. Made something simpler for customers. What drove it? Impact?

S4. Situation. Planners received "coverage alert" emails and dismissed them unread; acting on one meant twenty minutes of digging. **Task.** Make the alert do the digging. **Action.** A thirty-second reason line on every alert — the input whose change moved coverage most: "coverage falls to 1.6 weeks in week 7 because PO 4471 slipped 9 days." Behind it, the reconciliation that made the number trustworthy, and a dismiss-reason loop that kept simplifying what the line said. **Result.** Alerts actioned in S&OP without a mandate; false alarms down; forecast accuracy 70 → 85%; ~\$2M released. The simplification was for the customer's thirty seconds, not for the system.

Math. Reason line = $\text{argmax over inputs of } |\Delta \text{coverage attributable to } \Delta \text{input}|$; alert fires only when coverage < LT weeks + safety weeks AND recon green that day.

Probe "How did you compute the reason?" — The input whose delta moved coverage most since yesterday. **Probe** "What did it replace?" — A planner opening four SAP transactions.

WBR / design: every red in a WBR carries its reason line — which input moved, whose lever.

IS-4. Drove new thinking and innovation out of your team. A specific innovation?

S16 (and S4). *Situation.* A thousand supplier and carrier routes on an EDI program; suppliers overrode the routing guide and the instinct was to forbid it. *Task.* Get the team to invent the alternative rather than police the rule. *Action.* I asked the team why suppliers overrode instead of how to stop them. Their answer became the design: reason-coded overrides — legitimate, logged, and mined weekly into the routing-guide backlog. The skeptics were often right about specific lanes. On S4, the reason-line logic was likewise the team's design to my requirement. *Result.* Completeness 60 → 95%; every cohort graduated at ≥ 95% valid ASNs; override reason codes at 100% coverage. The innovation came from a question, not a mandate.

Math. Valid ASN rate (cohort) = ASNs passing X12 validation + PO match ÷ shipments ; gate ≥ 95% ; override reason-code coverage = coded ÷ overrides = 100% ; top codes → lane-fix backlog.

Probe "What did you do versus the team?" — I asked the question and protected the time; they designed the codes.

Probe "How did you make it safe to propose?" — The first idea that failed was mine, in public.

WBR / design: override reason-code mix as the input feeding the routing-guide backlog.

IS-5. A problem the usual approach wouldn't address. How did you select an alternative? Alternatives considered? Result?

MAIN. *Situation.* The usual approach to growing a support-heavy product is to scale the support team with customers. Visibility was free to the client, so every operator hour was pure cost. *Task.* Grow 10× with cost per container falling. *Action.* Alternatives considered: hire Data Services ahead of the curve (cost curve up), slow sales (revenue), or change what grows — gate admission on measured ability to serve at current load, and index automation to carriers rather than containers by ranking the backlog on tickets × cost per carrier × terminal × milestone. I chose the third because it is the only one where cost falls with scale: operators scale with containers; automation scales with a few hundred carriers. *Result.* \$0.66 → \$0.03 per container at ~10× customers and 5× volume; automation rate ~60 → 97%; Data Services shrank by attrition, as I had told them it would.

Math. CTS = (tickets × \$ per ticket + platform cost) ÷ containers ; operators ∝ containers (linear) vs automation ∝ carriers (bounded) ; 1 pt of automation at 10K/wk ≈ \$4K/yr.

Probe "Why not just slow sales?" — Same outcome for the customer, no improvement in the platform. **Probe** "What did the gates force?" — Bad-source fixes and a kill switch — the actual cost reducers.

WBR / design: graduation scorecard by wave.

IS-6. A novel idea or decision with big business impact. What was novel?

S7. *Situation.* Flexport roadmaps were argued in features and dates; an eighteen-month operator-first sequence had been socialized. *Task.* Decide the sequence on a number the business was paid on. *Action.* I priced both sequences in cost-to-serve — identical rows, footnoted assumptions, only the order differing — and took the table to my manager privately. Novel because nobody had put a roadmap in the P&L's own currency, and because the two columns were structurally identical so the argument could only be about sequence. *Result.* Partner-first ratified at the Bellevue onsite: -66% engineering effort to first value, value in six months instead of eighteen, ~\$5M cost avoided. The function kept the two-column table as its way of comparing options.

Math. Cost avoided ≈ $\sum_{\text{months}} (\text{CTS}_{\text{baseline}} - \text{CTS}_{\text{partner-first}}) \times \text{volume} + (E - 0.34E) \times \text{loaded eng rate} \approx \5M ; ratio held under ±20% effort / ±15% volume.

Probe "Would anyone else have done it?" — Someone would have argued for partner-first; nobody had priced it. **Probe** "Pattern?" — Same as S3 and S5: convert the argument into a measured currency.

WBR / design: two-column CTS table; re-run with actuals.

IS-7. (Manager) Enabled your team or a team member to implement a significant change. Problem? Success measure? Result?

S12. *Situation.* Six-week releases at Sephora and a QA lead everyone managed around because she said no. *Task.* Enable her to implement the change, with the success measure set before the work. *Action.* Success measure first: release cadence to four weeks with critical defects per release not up. Then I gave her the gate — definition of done

with numeric entry criteria, authority to hold a release, an exception path with reason codes — and coached her to defend it as numbers in a room of engineers. Her artifact, her name. *Result.* Four-week cadence in six weeks; critical defects per release -50%; the gate copied by two neighboring teams and still in use after I rolled off. Matrix: I owned the enablement, not the headcount.

Math. Cadence 6 → 4 wk (+50% releases) ; critical defects per release, trailing 3: [4.0] → [2.0] = -50% ; gate = test pass %, open criticals = 0, perf within budget.

Probe "What did you remove from her way?" — Engineering's habit of routing around QA; I made the gate the only door. **Probe** "What if defects had risen?" — The gate would have shown which criterion was missing; fix the criterion.

WBR / design: defects per release and gate pass rate on one scorecard with her name.

Are Right, A Lot

Strong judgment and good instincts; seek diverse perspectives; work to disconfirm your beliefs. They listen for: how you decided with partial data, what you were wrong about, and how you found out.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S2	"70% of the information is enough when waiting is a customer miss and every lever is a two-way door." S2 — Cyber Monday, 4 AM: rebalancing allocation across three DCs with 70% of the information	below
S10	"I forbade promising against in-transit units. I was wrong, the data said so, and the fix went through the same signed policy." S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback	under Ownership
S7	"No benchmark existed, so I built one both sides could attack: two columns, same rows, footnoted assumptions." S7 — Flipping a socialized eighteen-month roadmap with a two-column cost-to-serve table	under Have Backbone; Disagree and Commit
S5	"I arrived certain it was carriers. The data said we could not measure lateness. I changed the question." S5 — Austria: I arrived certain it was carriers; the data said we could not measure lateness	under Dive Deep

S2 — Cyber Monday, 4 AM: rebalancing allocation across three DCs with 70% of the information (Manhattan Associates)

LPs: Bias for Action · Are Right, A Lot · Dive Deep · Customer Obsession. **AIQB:** Bias for Action 2, 4, 5 · Are Right, A Lot 1, 3 · Dive Deep 2 · Customer Obsession 8.

Problem	Peak day, 4 AM: backorder lines piling up across a 3-DC / 100-store network. Millions of dollars of orders would miss cutoff. Root cause unknown; full diagnostic an hour away.
Goal	Hold fill rate at 99% through the peak day — which meant stopping backorder growth inside the hour.
Key metric	Fill rate by hour (target 99%); backorder lines; orders at risk of missing promised cutoff (in \$).
Did it move?	Yes. Fill rate held at 99% through peak; ~\$8M of at-risk sales shipped to promise; structural stockouts ~20% over the engagement once the triage logic became standing policy.

Situation. "I owned order allocation and promising across three mega-DCs and about a hundred stores. The node we sourced from determined the lead time the customer saw — choose the node, and you have chosen the promise. Cyber Monday, 4 AM, a VP on the phone: backorder lines climbing, millions at risk, and the full diagnostic was an hour out.

Task. One number: hold fill rate at 99% through the day. That meant stopping the backorder growth inside the hour, with about seventy percent of the information I wanted.

Action. I split what had to be decided now from what could be learned in parallel. On the bridge I ranked every open order by *promise-at-risk* — dollar value times hours to its cutoff — instead of by age, and rebalanced allocation across the three DCs to the nodes that had hours, overriding the engine's assignments manually in that priority order. Every decision timestamped in one log, so the diagnostic team could later separate my effects from the fault's. I kept two actions explicitly off the table: no blanket promise downgrade, and no emergency inbound — both one-way doors at 4 AM. The trace an hour later showed the allocation scheduler had died quietly at the daylight-saving change. Inventory was fine. DC capacity was fine. The engine had simply stopped assigning.

Result. Fill rate held at 99% through the peak. Roughly eight million dollars of orders that would have missed cutoff shipped to promise. The promise-at-risk ranking became the standing allocation priority, and structural stockouts were down twenty percent over the engagement. Afterward I wrote a scheduler heartbeat and a DST checklist into peak readiness. I never want a second 4 AM."

Supply · Demand · Capacity. **Demand:** peak-day orders by promise tier and cutoff. **Supply:** inventory by DC and store — adequate, as it turned out. **Capacity:** two kinds — the physical pick/pack hours by node, which were fine, and the *planning-system* capacity to assign orders to nodes, which had gone to zero. **What bound:** a false capacity signal. The lesson for a capacity planner: when the plan says "capacity is short," first prove the shortage is physical.

What was innovative. The standard move at 4 AM is FIFO — clear the oldest backorders first — and a blanket promise downgrade to stop the bleeding. I ranked by dollar-weighted hours-to-cutoff so the scarce decision bandwidth protected the most promise per minute, and I classified every available action as a one-way or two-way door before taking it. The timestamped log turned a firefight into a dataset.

The math

Promise-at-risk score (order)	= order value (\$) × urgency, urgency = $1 \div \max(\text{hours to cutoff}, 0.5)$
\$400 order, 2 h to cutoff	→ 200 ; \$900 order, 9 h to cutoff → 100 → the \$400 order ships first
Node feasibility check	= units assigned to node before cutoff ≤ node hours before cutoff × pick/pack rate
node B: 3 h × [4,000 units/h]	= 12,000 units of room → assign in score order until full, then next node
Cost of waiting one hour	≈ pool doubles every ~40 min at 4 AM arrival rate → pool(5 AM) ≈ 2.8 × pool(4 AM)
Sales protected	= $\sum$ order value of orders in the 4 AM pool that shipped before cutoff ≈ \$8M
Fill rate (hourly)	= lines shipped before cutoff ÷ lines ordered → held ≥ 99%
Structural stockouts	= SKU-locations with zero ATS at order time ÷ SKU-locations ordered → -20% over engagement

Solution design / flow

```
order → OMS promise (node? lead time?) → allocation scheduler → DC pick/pack → ship
      ▲                                     | (died at DST change:
      |                                     ▼ queue grew, nothing assigned)
promise-at-risk ranking (manual bridge, 4 AM) → node assignment by score,
one timestamped log → reconciled against feasibility-checked
                        scheduler restart (no double allocation)
```

AFTER: scheduler heartbeat (alarm if no run in 15 min) · DST/peak-regime checklist
· ranking rule = standing policy

How it translated to a WBR

Outputs: fill rate by hour vs 99%; orders past cutoff (\$). **Inputs:** allocation queue depth by node by hour; scheduler last-run age; promise-at-risk \$ in pool; node hours vs units assigned. **Red rule:** queue depth rising two consecutive hours with scheduler age > 15 min pages the bridge before anyone sees a backorder.

SITUATION PROBES

Why was this important? Peak day is a disproportionate share of the year's revenue, and a missed cutoff on peak is a customer who does not come back. The VP's phrase was "millions by noon."

What if you had waited for the diagnostic? An hour of growing backorders at peak order rates. We modeled it after the fact: the at-risk pool roughly doubled every forty minutes at that hour's order arrival rate.

What did you know at 4 AM, and what didn't you? Knew: backorder lines, inventory by node, DC hours scheduled. Didn't know: why the engine was not assigning. I decided I did not need the "why" to move orders safely by hand, as long as every move was logged and reversible.

ACTION PROBES

What did you do versus the ops team? I took the bridge, set the ranking rule, made the node assignments in priority order, and kept the log. Ops executed the picks; the engineers ran the trace. The ranking rule was mine.

Why act at 70%? What would have made you wait? Waiting was a customer miss with a dollar sign; acting was a two-way door — manual assignments could be re-run when the engine came back. I would have waited if the action were a one-way door: a promise downgrade or an emergency PO. Those I kept off the table.

Why promise-at-risk and not FIFO? FIFO protects the oldest order; it does not protect the most promise per minute of my attention. With an hour of hand assignment, I wanted every minute to clear the most dollar-hours of risk. FIFO would have spent the hour on low-value orders with late cutoffs.

How did you decide which DC got which order? Hours available at the node against the order's cutoff, then inventory at the node, then distance to the customer — in that order. If a node had the units but not the hours before cutoff, it did not get the order.

What was the risk of the manual rebalance? Double allocation when the engine resumed. The timestamped log was the mitigation: when the scheduler came back, the engineers reconciled against the log before re-enabling auto-assignment.

What decision did you challenge? The first instinct on the call was to downgrade all two-day promises to standard. I refused — that is a one-way door that touches every customer to fix a problem that might be local. It was local.

RESULTS PROBES

How did you compute \$8M? Orders that were in the backorder pool at 4 AM and shipped before their cutoff after the rebalance, at order value. Conservative: it does not count orders that would have joined the pool over the next hour.

How did you measure fill rate? Lines shipped before cutoff ÷ lines ordered, by hour, against the 99% line. Hourly, because a daily average would have hidden a 4 AM hole.

How did the -20% in stockouts come about? That's not from one night. Correct. The ranking rule became the standing allocation priority — scarce units went to the orders with the highest promise-at-risk rather than first-come. Over the engagement, structural stockouts fell twenty percent against the prior-year baseline. The night was the proof of concept.

Compared to goal? Goal was 99% through the day. Met. We did not claw back the backorders that had already missed before 4 AM; I counted those as a miss and said so.

CHALLENGE

What would you do differently? The heartbeat and the DST check existed by the following week. The deeper fix: I now treat every peak as a different operating regime with its own readiness checklist, because the failure modes at peak are not the failure modes at average.

Why this story for Bias for Action and not something more recent? Because it is the cleanest case where speed had a price and I can show the math on both sides. The Sephora robot throttle (S1) is more recent and is the same muscle: a two-way door turned the same day.

At Amazon. When a region shows a capacity shortfall, I would first prove it is physical — hours or cube — before spending a lever. And every lever in the plan would be labeled one-way or two-way before peak, so the bridge at 4 AM already knows which ones it may pull without a meeting.

The question bank — Are Right, A Lot (7 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

AR-1. Didn't have enough data to make the right decision. What path did you take? Did it turn out correct?

S2. Situation. Cyber Monday, 4 AM, Manhattan Associates: backorders climbing across three mega-DCs and a hundred stores; root cause at least an hour away; the pool doubling roughly every forty minutes. **Task.** Decide with about 70% of the information or wait with 100% and miss cutoff. **Action.** I separated what had to be decided now from what could be learned in parallel: ranked open orders by promise-at-risk — order dollars $\times 1 \div$ hours to cutoff — hand-assigned them to nodes with hours, feasibility-checked each node, logged every move with a timestamp, and kept the one-way doors (promise downgrade, emergency inbound) off the table. The trace came back: the allocation scheduler had died at the daylight-saving change. Capacity and inventory were fine. **Result.** Fill rate held at 99%; ~\$8M of orders shipped to promise; the ranking became the standing allocation priority. Correct — and reversible if it had not been.

Math. Score = $\$ \times 1 \div \max(h \text{ to cutoff}, 0.5)$: $\$400 / 2 \text{ h} \rightarrow 200$ beats $\$900 / 9 \text{ h} \rightarrow 100$; node feasibility: units assigned $\leq$ hours before cutoff $\times$ pick/pack rate ; pool(5 AM) $\approx 2.8 \times$ pool(4 AM).

Probe "What would have made you wait?" — If the only available actions were one-way doors. **Probe** "How did you know inventory was fine?" — Stock by node was on the dashboard; the queue was growing with stock sitting still, which pointed at the engine.

WBR / design: scheduler heartbeat as an input metric; "queue rising + stock flat" as a pre-wired red; every lever labeled one-way/two-way before peak.

AR-2. A strategic decision without clear data or benchmarks. Alternatives, trade-offs, risk mitigation?

S7. Situation. No benchmark existed for partner-first versus operator-first on Flexport's forwarding platform; the inherited plan was eighteen months, operator-first, socialized. **Task.** Make a strategic sequencing call without a comparable. **Action.** I built the benchmark: both sequences costed in cost-to-serve with identical rows and footnoted assumptions. Alternatives: operator-first (18 months, ~25% CTS cut), partner-first (6 months, ~50%), both in parallel (no engineers). Trade-off: operators waited a year for screens. Risk mitigation: engineering pressure-tested the table for two days before anyone senior saw it; I protected a dated operator milestone so the losers of the sequence had a date. **Result.** ~66% engineering effort to first value, ~\$5M avoided, value in six months; the ratio held under the pressure test even as several absolutes moved.

Math. Columns: months to value 18 vs 6 ; effort E vs 0.34E ; CTS cut ~25% vs ~50% ; sensitivity $\pm 20\%$ effort, $\pm 15\%$ volume $\rightarrow$ ranking unchanged.

Probe "How confident in 50 vs 25?" — Confident in the ratio, less in the absolutes; I said so. **Probe** "What if the pressure test had broken it?" — Then I would have committed to operator-first with the table as the reason.

WBR / design: the two-column CTS table is the artifact; it re-runs with actuals when partner adoption trails plan two months running.

AR-3. A difficult decision with input from many sources. How did you arrive at it? Was it correct?

S10. Situation. "Available" meant five things to five groups — stores wanted fill rate, DC wanted labor, dotcom wanted the promise, finance wanted the credit memo, returns wanted throughput. **Task.** One decision that every group would compute the same way. **Action.** I ran stock-state workshops where each group had to name the customer harmed if its definition won, wrote one definition per state with that failure mode attached, and took the set to steering for signature before configuration. Returns got a plant; RTV became pick-reserve so finance could not credit ahead of the floor. **Result.** Correct on every state but one: my in-transit rule was too conservative and was corrected through the same signed policy from lane data. The proof was the promise log joined to stock state — zero promises against non-sellable states — and a nightly OOB inside tolerance.

Math. Stock-state table: state · owner · promiseable? · example (7 states) ; OOB = SAP supply - WMS supply, $|OOB| \leq 0.5\%$ per SKU × DC ; in-transit window = per-lane measured transit, not "never."

Probe "Who lost?" — Finance lost the ability to credit ahead of the physical unit; they agreed once they saw the duplicate-credit exposure. **Probe** "How did you know it was correct?" — Promise log joined to stock state: zero promises against non-sellable states.

WBR / design: stock-state model (owner, promiseable?) + nightly ATS/OOB reconciliation; oversells as the output metric, OOB buckets as inputs.

AR-4. A bad decision. Impact? What did you learn and apply?

S10 (in-transit). Situation. Designing the promising rules at Sephora, I forbade every channel from promising against units on a truck. It felt safe. **Task.** Recognize and reverse my own rule when the data disagreed. **Action.** Dotcom's lead showed two lines diverging — sellable units by channel falling while in-transit inventory rose. I asked for the charts, confirmed the starvation, and took the correction to the same policy forum: replace "never" with a per-lane in-transit promise window derived from measured transit, the method I already had from Elementum. **Result.** Sellable units recovered the week the window went live; the policy process proved it could correct its author. Applied since: every default gets a measured number on day one, and conservatism is priced in the same currency as risk.

Math. Hidden units = $\sum \text{lanes in-transit units} \times P(\text{arrive within promise window})$; window_lane = p90 transit from lane data ; paired chart: sellable by channel vs in-transit by lane.

Probe "How long did it take to notice?" — Weeks; the signal was two metrics diverging, which is now a standing chart.

Probe "Cost?" — I will not quote a dollar I did not measure; sellable units recovered the week the window went live.

WBR / design: "sellable by channel vs in-transit by lane" as a paired chart; in-transit units outside the lane window as an input metric.

AR-5. An error in judgment. Impact? Learned? Applied?

S9 (RF guns). Situation. Dual-DC WMS migration at Sephora. I scored the RF-gun conversion as low effort — "just a script." **Task.** Own the misjudgment and protect the date. **Action.** The script failed on newer hardware; COVID night shifts killed the recovery window; training sign-off, a go-live exit criterion, slipped 1.5 weeks. I told leadership the same day, in the steering pack, with the new date and whose miss it was — mine, because I had scored it. I gave the track lead the recovery plan to run. **Result.** The program landed on date because training was off the critical path; recon 0.3% vs 0.5% tolerance; ~3% under \$1.8M. Applied: any single-point dependency in front of an exit criterion gets a what-if plan regardless of estimate. It has caught two since.

Math. Training slip 1.5 wk, float on that path ≥ 2 wk → program date held ; rule: single-point dependency AND exit criterion downstream → what-if plan, red in RAID regardless of estimate.

Probe "Why did you under-score it?" — Familiarity: I had seen it done before and did not test it on our hardware. **Probe** "How did you communicate?" — Same day, with the new date and the reason, in the steering pack.

WBR / design: "single-point dependencies without a what-if" as a RAID input metric, red regardless of estimate.

AR-6. Discovered your idea was not the best course of action. How did you find out? What was? Who provided it?

S5. Situation. I arrived in Austria certain the problem was carriers and the answer a routing engine. **Task.** Find out whether my idea was right before building it. **Action.** I pulled three years of shipment history myself and found we could not measure lateness — 55% true data quality at the planning grain. The data provided the better course: measurement first, 99% by defect class; the finance partner provided the savings method, per shipment, signed before any claim. I

told the sponsor privately with three lanes where his engine would have been wrong. *Result.* ~\$8M across 25 locations, auditable per shipment; a routing engine delivered *after* the data could carry it. Learned: audit the measurement before the thesis — now week one of every new domain.

Math. True quality = fields present AND lane key at grain AND carrier canonical AND transit from scan ; $0.82^3 \approx 0.55$; reports said 40% (field-present only).

Probe "Who did you have to tell?" — The sponsor who had promised the engine; one page, privately, with three real lanes where the engine would have been wrong. **Probe** "Would you have found it later anyway?" — Yes, as wrong recommendations claimed as savings. Much more expensive.

WBR / design: data quality by defect class as a gating input; savings ledger per shipment; any lane under 99% held out of claimed savings.

AR-7. Brought different perspectives together to solve a problem. Which perspectives? How did you seek them? Outcome? Learnings?

S10. *Situation.* Stores, DC ops, dotcom, finance, returns, SAP and the WMS integrator — each with a different "available." *Task.* Get one definition without a committee that never signs. *Action.* Workshops with one rule: name the customer harmed if your definition wins. Finance (credit ahead of the unit) and returns (ungraded units promised) converted themselves. Signature by a date; silence counted as consent in writing. *Result.* One signed definition per state; zero non-sellable promises in production; zero go-live incidents; legacy system retired. Learning: get the failure mode on the table before the design and the arguments get small. Differently: lane data for the in-transit state from day one.

Math. Each state → owner → promiseable? → failure mode (customer harmed) ; sign-by date ; outputs: oversells = 0 ; OOB nightly $\leq 0.5\%$.

Probe "Who was hardest?" — Finance, on credit timing; the duplicate-credit exposure converted them. **Probe** "How did you keep it from being a committee?" — Signature by a date; silence counted as consent in writing.

WBR / design: the stock-state table (state · owner · promiseable · example) is the design; OOB and oversells are the review.

Learn and Be Curious

Never done learning; curious about new possibilities and act to explore them. They listen for: something you did not know, how you learned it, and what changed because you did.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S14	"I taught myself the stack on weekends and shipped a triage agent that cut cost per ticket from \$25 to 40 cents." S14 — Teaching myself a stack: the LLM triage agent	below
S9	"I had never run a WMS migration. I did not fake it. I built the gate and used the specialists." S9 — Dual-DC WMS migration in eight months, tolerance published before mock one	under Earn Trust
MAIN	"I learned how carrier and terminal data actually arrive — EDI, scraped sites, CargoSmart — before I ranked what to automate." MAIN — Shipment Visibility (Flexport)	Section 5

S14 — Teaching myself a stack: the LLM triage agent (independent build)

LPs: Learn and Be Curious · Frugality · Insist on the Highest Standards. **AIQB:** Learn and Be Curious 2, 5, 7 · Frugality 5 · Highest Standards 2.

Problem	Incident triage cost ~\$25 per incident in analyst time; I did not know the stack to automate it.
Goal	Automate triage at a fraction of the cost with accuracy held and every conclusion cited to a source.
Key metric	Cost per incident; triage accuracy vs the evaluation set; citation coverage.
Did it move?	Yes. \$25 → \$0.40 per incident (−98%); accuracy held against the pre-written evaluation set; 100% citation coverage.

Situation. "Incident triage cost about twenty-five dollars an incident in analyst time. I wanted to automate it and did not know the stack. **Task.** Build it, learn the stack, and hold accuracy — not ship a demo. **Action.** Python, FastAPI, tool use. The discipline I brought from supply chain: I wrote the evaluation set before the agent — a hundred-plus real incidents with known-correct triage — and every prompt or model change was gated on the known past still passing. Every output had to cite its source or it did not ship. **Result.** Cost per incident from twenty-five dollars to forty cents — a ninety-eight percent cut — accuracy held against the evaluation set, one hundred percent citation coverage. Same operating system as everything else I run: instrument first, then let the scorecard teach."

Use only if asked about learning something new or about AI. It is a side build, not a program you owned at a company. Say that in the first sentence.

The math

Cost per incident	= (LLM tokens × price + compute + retrieval) ÷ incidents ≈ \$0.40 vs analyst ≈ 20 min × [\$75/h] = \$25 → −98%
Accuracy	= correct triage ÷ evaluation set (written BEFORE the agent, [100+] real incidents) — held
Citation coverage	= outputs with ≥ 1 source citation ÷ outputs = 100%
Gate	= no prompt/model change ships unless the evaluation set still passes at ≥ baseline

How did you learn the stack? By building against a scorecard. Documentation plus a failing evaluation set is the fastest teacher I know.

Why the evaluation set first? Because otherwise "it works" means "it worked on the three I tried." The set is the gate; the agent is the candidate.

At Amazon. Any model or tool I bring to capacity planning gets a backtest against last year's weeks before it touches the plan. The evaluation set is the mock cutover.

The question bank — Learn and Be Curious (7 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

LC-1. Realized you needed deeper expertise to do your job well. What did you do? Outcome? Differently?

S9. *Situation.* A dual-DC WMS migration at Sephora with \$20M of ROI behind it, eight months, nothing documented — and I had never run one. *Task.* Learn enough, fast enough, to make the go/no-go call myself. *Action.* I made discovery a gate with a written exit — every flow documented — which forced the specialists to teach the program what the building did while I learned to read a reconciliation report. I published the 0.5% tolerance before mock one so a 7% first result was data I could interrogate, not a failure I had to defend. I sat the integration sessions and asked the "what breaks if" question on every interface. *Result.* 0.3% recon at go-live, on time, ~3% under \$1.8M. Differently: learn the RF-device layer too — I left one corner to assumption and it cost 1.5 weeks of training sign-off.

Math. Recon error = $|WMS - ERP| \div ERP$ by SKU-location ; mock 1 = 7% → cause buckets (mapping, timing) → mock 3 = 0.3% vs 0.5% tolerance published before mock 1.

Probe "What did you read?" — Our own flows, the integrator's cutover playbook, and every mock's reconciliation line by line. **Probe** "How did you know you knew enough?" — When I could classify a reconciliation miss by cause without asking.

WBR / design: discovery exit criteria as the learning mechanism.

LC-2. Work outside your comfort area. How did you identify what to learn? How did you build expertise? Did you meet the goal?

S14. *Situation.* I built an LLM incident-triage agent in a stack I did not know — retrieval, tool use, evaluation. A side build; I say so. *Task.* Identify what to learn and build real expertise, not a demo. *Action.* I wrote the evaluation set first — a hundred-plus real incidents with correct triage — so the gap between what I built and what passed told me what to learn next. Every failing case was a lesson with a name. I set a gate: no prompt or model change ships unless the set still passes at or above baseline. *Result.* Cost per incident \$25 → \$0.40 (~98%), accuracy held against the set, 100% of outputs with citations. Met the goal, and the method — evaluation set before the build — is how I now scope any analytics the plan depends on.

Math. Cost/incident = $(\text{tokens} \times \text{price} + \text{compute} + \text{retrieval}) \div \text{incidents} \approx \0.40 vs analyst 20 min × [\$75/h] = \$25 ; accuracy = correct triage ÷ evaluation set, written BEFORE the agent.

Probe "Why that project?" — Because it would fail visibly if I was wrong. **Probe** "What transferred to work?" — Evaluation set first is a mock cutover; same discipline.

WBR / design: evaluation set as the gate for every change — backtest before the plan.

LC-3. Didn't know what to do next on a hard problem. How do you learn what you don't know? Options? Decision? Outcome?

S5. *Situation.* Four weeks into Austria the data said my thesis was wrong and I did not yet know what the right program was. *Task.* Learn what I did not know without stalling the team. *Action.* I learn by pulling the rows myself and by asking the people who key them — planners — what each field meant. Options: clean the data on the way into the engine (hides the problem), pause the program (hides it differently), or make measurement the deliverable. I chose measurement because the other two kept the problem invisible. *Result.* Data quality 99% by defect class; ~\$8M of freight savings auditable per shipment; the engine delivered on a foundation that could carry it.

Math. Defect classes: rate present · lane key at correct grain · carrier canonical · transit from scan ; weekly scorecard by class and location until all ≥ 99%.

Probe "How long were you lost?" — About a week between "carriers aren't it" and "measurement is it." **Probe** "Who helped?" — A finance partner who insisted on a signed savings method; that defined the deliverable.

WBR / design: defect-class scorecard.

LC-4. Something you've worked on to improve your work effectiveness. Resources? Impact?

Written narratives over decks (S7, S10). *Situation.* After the Flexport roadmap flip I noticed the two-column table did the work and the slides did not. *Task.* Change how I argue. *Action.* Every steering pack since is a one-page written argument with the table on top — the Flexport visibility WBR call-outs and the Sephora S/4HANA weekly pack are written this way. Resources: Amazon's public writing on narratives, my manager's feedback on the first three, and the discipline of freezing definitions so the writing is about the business, not the chart. *Result.* Shorter meetings; decisions that stayed decided; a WBR where every red carries its own decomposition and owner in a sentence.

Math. Call-out line = metric · delta · decomposition (carrier × terminal × milestone) · owner · date — one sentence, every red.

Probe "What was hard?" — Writing "we were wrong" lines; now a standing section. **Probe** "Example of a decision that stayed decided?" — The stock-state policy: signed once, never re-litigated.

WBR / design: weekly pack as a written narrative with frozen metrics.

LC-5. Explored a new or unexpected area of an existing space. Why hadn't it been explored? Why did you proceed? Results?

S3. *Situation.* Lead-time variability was a known space; nobody had looked at the tail per lane because the brochure was free and the pad was invisible. *Task.* Explore it when the ask was a report. *Action.* I proceeded because the planner's question implied the plan was running on an unmeasured number. I built per-lane empirical distributions, compared the p99 to the brochure and to a normal fit, and found the normal fit understated the tail by about four days on skewed lanes. *Result.* Planning lead time at the p99 per lane, 99% held; 580K units and ~\$1.5M released; a drift flag that keeps exploring on my behalf.

Math. Normal p99 = $21 + 2.33 \times 4 = 30.3$ d vs empirical p99 = 34 d ; brochure 18 d + pad ≈ 25 d held 93% ; $P(t > \text{mean}) \approx 0.40$ on a right-skewed lane.

Probe "What surprised you?" — Most lanes released stock; I expected most to need more. **Probe** "What else was unexplored there?" — Seasonal regimes per lane; the drift flag catches them now.

WBR / design: lane LT distribution pipeline.

LC-6. Someone on your team challenged you to think differently. How did you respond? Outcome?

S10. *Situation.* Dotcom's lead challenged my "never promise against in-transit" rule: "you've hidden sellable units from us." *Task.* Treat the challenge as data, not as a defense of my rule. *Action.* I asked for two charts — sellable by channel, in-transit by lane — and they diverged. I brought it to the policy forum myself and replaced "never" with a per-lane window from measured transit. *Result.* Sellable units recovered; the policy proved it could correct its author; the paired chart is now standing in the WBR. The person who challenged me got the fix with her name on it.

Math. Units hidden = $\sum \text{lanes in-transit} \times P(\text{arrive inside window})$; window = per-lane p90 transit ; input metric: in-transit units outside the lane window.

Probe "How did you respond in the moment?" — "Show me the two lines." Not a defense. **Probe** "What did it change in you?" — Defaults get a number on day one.

WBR / design: divergence chart as a standing input.

LC-7. Used external trends to improve your company's products or services. How did you keep up? How did you decide to apply? Result?

MAIN, with Amazon as the external example. *Situation.* I needed a model for admitting customers to a platform that could not yet serve them all well. *Task.* Find an external pattern and apply only its shape. *Action.* I studied how Seller Fulfilled Prime turned an adoption problem into eligibility criteria with teeth, and applied the shape — six published graduation criteria a wave must clear at current load — to Flexport's onboarding. I also borrowed the engineering org's WBR conventions rather than inventing a format. I keep up through shareholder letters, carrier and port data standards,

and the planners and operators who live in the systems. *Result.* 68 customers two months early at \$0.03 per container; a WBR that read like every other one in the building, so nobody had to learn it.

Math. Gate = six criteria at current load ; adopted format: output/input, roll-up, 5 wk + months, WoW/MoM, target, vs-target, red at > 5% unfavorable relative variance.

Probe "How did you decide it applied?" — Same structure: a program whose cost rises with uncontrolled admission.

Probe "What did you not copy?" — Penalties; gates were enough.

WBR / design: graduation criteria — the external pattern as a scorecard.

Hire and Develop the Best

Raise the bar with every hire and promotion; recognize exceptional talent; develop leaders. They listen for: a specific person, what you saw, what you gave them, and what they did with it.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S12	"I gave the skeptic the gate. Development that depends on me staying is not development." S12 — Giving the skeptic the gate	below
S13	"I staffed two go-lives on one team by naming the owner of each gate and protecting their week." S13 — Two go-lives, one engineering team, no 24-hour days	under Deliver Results

| S12 — Giving the skeptic the gate (Sephora agile conversion)

LPs: Hire and Develop the Best · Earn Trust · Strive to be Earth's Best Employer. **AIQB:** Hire and Develop 1, 3, 4 · Earth's Best Employer 6, 9 · Earn Trust 3.

Problem A VP sponsored converting a twenty-person product/engineering/QA group from a six-week to a four-week release. The QA lead — the most experienced person in the room — was the most skeptical, and QA was where a faster cycle would break first.

Goal Four-week releases without a quality regression; the mechanism survives my leaving.

Key metric Release cycle (6 → 4 weeks); critical defects per release; whether the gate was still in use after I rolled off.

Did it move? **Yes.** Four-week cycle reached in six weeks. Critical defects –50%. The QA-authored gate was adopted by neighboring teams and outlived my engagement.

Situation. "A VP sponsored moving a twenty-person product, engineering and QA group from six-week to four-week releases. The QA lead was the most skeptical person in the room — and QA was where a faster cycle breaks first.

Task. Four-week releases, no quality regression, and a mechanism that would survive me leaving. **Action.** Instead of managing around her, I gave her the load-bearing artifact: the definition of done with entry criteria, and the authority to hold a release that did not meet it. I coached her on writing a gate that can say no — numeric, published, owned by her — and on how to defend it to engineering without me in the room. Within two cycles she was the one defending it. **Result.** Four-week cycle in six weeks. Critical defects dropped by half, because the gate was stricter than what had existed at six weeks. Neighboring teams copied the model and the ceremony survived my leaving. Development that depends on me staying is not development."

Supply · Demand · Capacity. **Demand:** releases per quarter, up 50%. **Capacity:** the team's throughput at quality.

Supply: the gate — the mechanism that let throughput rise without quality falling. **What bound:** trust in the person who would hold the line.

The math

Release cadence	6 wk → 4 wk = +50% releases per year (8.7 → 13)
Critical defects	per release, trailing 3 releases: [4.0] → [2.0] = -50% (gate stricter than the old 6-week process)
Gate	= definition of done with numeric entry criteria (test pass %, open criticals = 0, perf within budget), owned by QA lead
Durability test	= gate still in use [N] quarters after I rolled off; copied by [2] neighboring teams

How it translated to a WBR

Outputs: releases on cadence; critical defects per release. **Inputs:** gate criteria pass rate; exceptions granted (with reason). **Red rule:** an exception without a documented reason code is a red on the release, not on QA.

Why the skeptic? Why not the person most bought in? The skeptic knew where it would break. Giving her the gate converted her objections into criteria, and her credibility made the gate real to engineering.

What specifically did you coach? How to write a criterion as a number, not an adjective; how to hold a release in a room full of engineers who want to ship; and when to let something through with a documented exception rather than block. She was good at the first two already; the third was the coaching.

What's the honest shape of your people leadership? Direct reports in the classic sense at MetricStream — onshore/offshore delivery teams, SOWs, the go-live chair. Everywhere since, matrix: this twenty-person conversion, UDC with 15+ tracks and 60+ stakeholders, about a hundred people across nine teams sequenced at Flexport. I do not dress matrix leadership up as a headcount I did not hold.

Did an investment in someone not work out? At Flexport I spent a quarter coaching an analyst toward owning the Monday review; he did not want the visibility and said so late. I should have asked earlier what he wanted rather than what I needed. I ask that question in the first month now.

At Amazon. The person who most doubts the capacity plan — in Ops, in Finance — is the person I would ask to write the gate the plan has to clear. Their objections become the criteria, and the plan is stronger for it.

The question bank — Hire and Develop the Best (10 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

HD-1. Helped a team member develop their career. How? Result?

S12. *Situation.* Sephora's QA lead was managed around because she said no; releases were six weeks apart. *Task.* Develop her into the owner of the release mechanism. *Action.* I gave her the definition of done and the authority to hold a release, then coached the form: turn objections into numeric criteria published in advance, and defend them in a room of engineers as readings, not opinions. Two cycles with me present and silent. *Result.* Four-week cadence, critical defects -50%, a gate copied by neighboring teams and still in use after I rolled off. Development that depends on me staying is not development.

Math. Critical defects per release, trailing 3: [4.0] → [2.0] ; cadence 6 → 4 wk ; durability = gate in use [N] quarters later with her name on it.

Probe "What did she get out of it?" — Visibility as the person who made four-week releases safe; a seat in design reviews she had stopped attending. **Probe** "How did you measure development?" — She defended the gate without me in the room by cycle two.

WBR / design: her name on the release scorecard as gate owner.

HD-2. Invested in an employee's development. What and why? Outcome? One that didn't work out?

S12 (worked) and the Flexport analyst (did not). *Situation.* Two investments. The QA lead, as above. And a Flexport analyst I coached for a quarter toward owning the Monday review. *Task.* Invest in what the person wants to own, not only what I need. *Action.* With the QA lead I invested in a strength and a form; it worked. With the analyst I invested in a role he did not want — he said so late, after a quarter. I helped him move to a role that fit. *Result.* Gate: -50% defects, copied. Analyst: a good outcome for him, a lesson for me. Since then I ask "what do you want to own in a year" in month one, not month four.

Math. The quarter I spent coaching toward the review $\approx [3 \text{ h/wk} \times 12]$; recovered by asking one question earlier.

Probe "What would you have done differently with him?" — Asked earlier; and separated the system problem — 60% accuracy — from his performance sooner. **Probe** "Honest shape of your people leadership?" — Direct reports at MetricStream; matrix since. I say so.

WBR / design: development goals written with the person, reviewed quarterly — same discipline as a metric.

HD-3. Feedback to develop the strengths of someone on your team. Impact on performance?

S12. Situation. Her strength was knowing where releases break; engineering's feedback was "blocker." **Task.** Develop the strength by changing the form, not the person. **Action.** I told her the strength was right and the delivery was wrong: write objections as criteria with numbers, published before the release, so a "no" is a reading. I reviewed her first two gates line by line. **Result.** The same objections became the gate; critical defects -50%; engineering stopped arguing with her and started arguing with the number — which is where the argument belongs.

Math. Gate entry criteria: test pass $\geq [X]\%$ · open criticals = 0 · perf within budget ; a hold = a criterion failing, not a judgment.

Probe "How did she take the feedback?" — Relieved; nobody had told her the instinct was right. **Probe** "Did you give feedback to engineering too?" — Yes: stop routing around QA; argue the criteria.

WBR / design: criteria as numbers on the scorecard; exceptions with reason codes.

HD-4. Coaching for a team member. What started it? Outcome?

S16. Situation. An EDI onboarding lead at Elementum was graduating supplier cohorts on relationships — "they're a good supplier" — rather than on valid-ASN rate. A cohort graduated at 88% and the dock found out. **Task.** Coach the gate, not the person's judgment of suppliers. **Action.** I coached on holding the 95% gate regardless of the relationship, and on using reason codes so a hold was a conversation about data rather than loyalty. We re-ran the cohort with the scorecard in front of the supplier. **Result.** Every later cohort graduated at $\geq 95\%$ valid ASNs; completeness 60 → 95%; the lead became the person suppliers asked for the list.

Math. Valid ASN rate = passing X12 validation + PO match ÷ shipments ; cohort gate $\geq 95\%$; a hold carries the failing supplier list with reason codes.

Probe "Was the supplier relationship damaged?" — Improved; the supplier got a scorecard instead of a vague complaint. **Probe** "What did the lead learn?" — A gate protects the relationship by making the no impersonal.

WBR / design: valid ASN rate by cohort as the graduation input.

HD-5. (Manager) Someone you hired who complemented your skills. How did you coach where you have a weakness?

MetricStream (direct reports). Situation. When I held headcount, I hired a delivery lead stronger than me in offshore team operations and hands-on configuration. **Task.** Coach where I was weak without pretending to be strong there. **Action.** I coached on what I was strong at — sponsor management, gate design, the go-live decision — and let him lead where he was stronger, including in front of the customer. In matrix roles since, I have given the gate to the person who knew more than me: the QA lead, the WMS specialists. **Result.** He ran the next program's delivery track; I ran the sponsor. Say plainly: "I hire into my weaknesses when I hold headcount, and in matrix roles I give the gate to the expert."

Math. Decision rights split: configuration and offshore cadence → him ; gates, sponsor, go/no-go → me ; reviewed weekly against the same RAID log.

Probe "How did you know it was a weakness?" — Mock-cutover questions I could not answer without him. **Probe** "How did you avoid becoming redundant?" — I owned the decision; he owned the knowledge; both were needed.

WBR / design: named owners per gate on the scorecard — the owner is whoever knows most, not whoever ranks highest.

HD-6. (Manager) Helped a remote team member develop their career across distance.

S13 (night-pass lead). Situation. The night-pass lead in another time zone executed what the day team decided and was never in the decision. **Task.** Develop a remote executor into a decision-maker. **Action.** Written decision rights for reversible actions at night; a weekly review of the night log — what he decided, what he escalated, what he would decide next time — without blame. **Result.** By week three he escalated less and decided well; the night pass became the model for the next program. Matrix; the development was in the decision rights, not a title.

Math. Escalations per night → decisions per night, week over week ; reversibility rule written in the log.

Probe "How did you build trust remotely?" — The log: every decision visible, reviewed without blame. **Probe** "What did he get?" — A decision role on his résumé and a reference I have given since.

WBR / design: night-pass decisions vs escalations as a weekly development metric.

HD-7. (Manager) Someone who struggled to get promoted; how did you facilitate their success?

S12. Situation. The QA lead's reputation was "blocker," which is not a promotion case. **Task.** Make her contribution visible as a measured mechanism. **Action.** The gate made her the owner of a visible artifact with a result — defects -50%, cadence +50%. I wrote that into the review language and made sure her name, not mine, was on the scorecard neighboring teams copied. **Result.** Her case became "owns the release mechanism two teams adopted." Honest framing: I influenced the case; I did not hold the promotion decision.

Math. Her scorecard: cadence 6 → 4 wk ; criticals -50% ; teams adopting = [2].

Probe "What was the barrier?" — Perception; the mechanism changed the evidence. **Probe** "What would you do with headcount?" — The same, with the decision too.

WBR / design: ownership visible on artifacts — the promotion case writes itself from the scorecard.

HD-8. (Manager) Most challenging talent review and promotion process. Factors? Bias tools? Perceptions of fairness? Learned?

MetricStream. Situation. Onshore and offshore delivery teams reviewed by a sponsor who saw one group daily and the other never. **Task.** Run a talent review that was fair to people the decision-maker never saw. **Action.** Factors: delivered outcomes against written criteria, not visibility. Bias tool: I scored everyone against the same written criteria before reading names, then calibrated; I had offshore leads present their own work to the sponsor. Perceptions: published criteria and a calibration conversation, not a private ranking. **Result.** Recognition and promotion distributed by contribution. Learned: visibility is a bias; mechanisms correct it. Since then my roles have been matrix; I apply the same criteria discipline to gate ownership and recognition.

Math. Score = $\sum$ criteria met (written in advance) ; names read after scoring ; offshore share of sponsor-facing presentations → from 0 to parity.

Probe "Did anyone feel it was unfair?" — One onshore lead, until the criteria were shown to be the same. **Probe** "What would you change?" — Calibrate with a peer manager, not alone.

WBR / design: written criteria first, names second — same as frozen metric definitions.

HD-9. (Manager) Constructed a team to accomplish a goal. Factors? Diversity? Work allocation? Effectiveness? Differently?

S1. Situation. Fifteen-plus workstreams and sixty-plus stakeholders for the omni-DC consolidation — a matrix team I constructed by track lead. **Task.** Build a team that could read its own gates without me. **Action.** Each track lead owned a gate they could read; I paired a skeptic with each optimist; vendor, ops and IT each had a seat on the issues log, not just in meetings. Diversity of function and of disposition — the people who thought it would fail wrote the criteria. Allocation by who could say no credibly. **Result.** One 836-row log with an owner and date per row, dated public gates, launched on date, +15% efficiency, zero incidents. Differently: name the zone-logic owner earlier.

Math. Gates: Nov 22 / Dec 20 / Jan 30 ; 836 rows, 100% with owner + date ; Oct-Dec change freeze.

Probe "How did you handle a weak track lead?" — Paired them with the gate's co-owner; moved the gate if it did not improve in two cycles. **Probe** "What made it work together?" — The log was the only place a date lived.

WBR / design: owner per gate per track on one page; rows past due as the team-health input.

HD-10. (Manager) A low-performing individual. How did you deliver feedback? Did performance improve?

MetricStream direct report. Situation. A configuration track lead missed internal dates repeatedly. **Task.** Deliver feedback that changed the performance, and know what came next if it did not. **Action.** Private, specific — the three dates, the impact on the go-live chain — and a question about what was in the way. The answer: an unclear spec and fear of asking. I fixed the spec and set a weekly check-in with a written "done." **Result.** Performance improved within a month; he led the next track. If it had not, the next conversation would have been about fit, with HR.

Math. Dates missed 3 → 0 over four weeks ; "done" = written exit per task.

Probe "How did you know it was the spec?" — I asked; and the dates he hit were the ones with clear specs. **Probe** "What if it had been effort?" — Then the written "done" would have shown it in two weeks, and the conversation changes.

WBR / design: written "done" per deliverable, reviewed weekly — the same instrument for a person as for a metric.

Insist on the Highest Standards

Relentlessly high standards; continually raise the bar; defects do not get sent down the line. They listen for: a standard you set, the moment it cost you to hold it, and whether you held it.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
MAIN	"I held wave three at 76% against an 80% gate in front of the VP. Standards without a stop-the-line are posters." MAIN — Shipment Visibility (Flexport)	Section 5
S1	"I published the 98% gate before the robots arrived, and throttled them to 30% when they hit 94%." S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived	below
S9	"Tolerance published before mock one: 0.5% inventory variance, or we do not cut over." S9 — Dual-DC WMS migration in eight months, tolerance published before mock one	under Earn Trust

S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived (Sephora UDC commingle · program-management story)

LPs: Insist on the Highest Standards · Customer Obsession · Bias for Action · Deliver Results. **AIQB:** Highest Standards 1, 4, 6 · Deliver Results 1, 2 · Bias for Action 1 · Customer Obsession 8.

Problem	Store replenishment and dotcom ran out of two warehouses: the same SKU stowed twice, two labor pools, two inbound docks for one vendor base, and no shared view of what was sellable to whom. Cost per unit above plan and a transfer every time one channel needed the other's stock.
Goal	One omnichannel DC on Blue Yonder with Locus robotics, live before peak, with zero customer-facing incidents and pick accuracy $\geq 98\%$ before the robots carried full volume.
Key metric	Processing efficiency (units per labor hour) vs the two-building baseline; pick accuracy by zone by hour vs 98%; customer-facing incidents.
Did it move?	Yes. Processing efficiency +15%. Pick accuracy 94% $\rightarrow \geq 98\%$ within one week of the throttle. Zero customer-facing incidents through peak. Launched on date.

Situation. "Sephora served stores and dotcom from two separate warehouses. Same SKUs stowed twice, two labor pools, one vendor base shipping to two docks, and no single answer to 'how many can we sell.' The decision was to collapse both into one omni DC on Blue Yonder with Locus robots — inside a window that ended at peak.

Task. I owned the program: fifteen-plus tracks, sixty-plus stakeholders, about a dozen systems — WMS, ESB and Kafka, OMS, finance, the data platform. I set the goal in numbers before we started: live before peak, zero customer-facing incidents, pick accuracy at or above 98% by zone before robots took full volume, processing efficiency above the two-building baseline.

Action. Three things that were not in the standard playbook. First, I published the pass/fail numbers before the robotics vendor was on site — 98% pick accuracy measured by zone by hour, not a building average — so the go/no-go would be read, not argued. Second, I designed the robot ramp as a dial, not a switch: the manual pick path stayed fully staffed and certified through the ramp, volume could move between bots and people within a shift, and the decision right for that dial sat with me. Third, one issues log — 836 rows over fourteen months, every row an owner and a date — behind dated public test gates, November 22, December 20, January 30, and an October-to-December change freeze so the building would survive peak. Week two after robots went live, pick accuracy printed 94%. I throttled robots to about 30% of volume the same day — only the zones that had cleared 98% in the debug sample — kept the launch green on the manual path, sat the zone-logic debug with the vendor, and restored full volume in week three when every zone cleared the bar.

Result. Processing efficiency finished plus fifteen percent against the two-building baseline. Pick accuracy from 94% to above 98% in a week. Zero customer-facing incidents through peak. Launched on the date. And structurally: one inventory, one dock, one labor pool — a unit can serve a store or a customer order without a transfer. The gate existed so the throttle needed no meeting."

Supply · Demand · Capacity. **Demand:** two profiles — store replenishment (lumpy, planned) and dotcom (daily, peaky) — now served from one pool. **Supply:** one vendor inbound stream to one dock instead of two. **Capacity:** two buildings' cube and hours collapsed into one, with robotics as the throughput multiplier. **What bound at launch:** pick throughput at *quality*, not cube — the building had the space; the question was whether the bots could pick at 98%. The throttle was a throughput-capacity decision made against a quality gate.

What was innovative. The usual approach is a cutover date plus a readiness checklist and a vendor who says "ready." I made the quality bar a number published before the technology existed in the building, built the ramp as a reversible dial with named decision rights, and kept a certified manual path as paid insurance. That turned a potential launch crisis into a routine Tuesday decision.

The math (program-management story — the math is throughput at quality)

Processing efficiency (UPH) baseline, two buildings:	= units processed ÷ direct labor hours, inbound + outbound, weekly [1.00M units ÷ 20,000 hrs] = 50.0 UPH	← bracketed inputs illustrative
after consolidation:	[1.05M units ÷ 18,260 hrs] = 57.5 UPH	← +15% is your number
where the 15% came from:	one stow instead of two (-dup handling), no inter-DC transfers, bots on dense zones	
Pick accuracy (zone, hour) week 2 building average 94% ;	= correct pick-confirms ÷ total picks by zone: dense multi-unit zones [88-91%], others [≥ 98%]	
Throttle share	= Σ volume of zones with accuracy ≥ 98% ÷ total volume	≈ 30% ← output, not target
Insurance price (manual path)	= 3 wks × [incremental hrs/wk] × [loaded rate]	vs
Cost of a 6-pt miss at peak	= [peak units/wk] × 6% × [cost per mis-ship: re-ship + concession +	

store 00S]
the second number is an order of magnitude larger → manual path is cheap insurance

Solution design / flow

vendor → ONE inbound dock → receive → stow (one inventory, one WMS)
└─ store replenishment wave → manual pick ──┐
└─ dotcom orders ──┐ Locus bots (zones ≥ 98%) ──┐ pack → ship
└─ manual pick (other zones) ──┘

DIAL (owned by program): zone list bots may carry; re-evaluated daily on zone × hour accuracy
GATES: Nov 22 / Dec 20 / Jan 30 test gates · Oct-Dec change freeze
· 836-row issues log (owner + date per row)

How it translated to a WBR

Outputs: processing efficiency vs baseline (weekly); customer-facing incidents (daily, target 0). **Inputs:** pick accuracy by zone by hour vs 98%; bot share of volume; issues-log rows past due; freeze exceptions. **Red rule:** any zone < 98% for two consecutive hours drops out of the bot list the same shift — no meeting.

SITUATION PROBES

Why was this important? What was the goal? Every unit stowed twice is cube paid for twice and a transfer waiting to happen. The business case was lower cost per unit and one view of sellable inventory; my program goal was the three numbers — zero incidents, ≥98% by zone, efficiency above baseline — because a consolidation that ships wrong units has destroyed the reason it was built.

What would have happened if nothing was done? Two buildings through another peak with inter-DC transfers to cover channel imbalances, duplicated safety stock, and a cost per unit that stayed above plan. The lease and labor math did not work for two buildings at our volume.

What was the initial scope? Did it change? WMS, robotics, and inventory merge. Scope grew to the integration layer when we found store and dotcom order flows hit the WMS through different interfaces with different unit-of-measure conventions. I added the track and re-baselined the gate dates once, publicly, in writing.

Biggest challenge? Time. The window ended at peak and peak was non-negotiable. Everything was designed to be safe to stop: the manual path, the freeze, the gates.

ACTION PROBES

What did you personally do versus the team? I wrote the gate definition and published it; I made the throttle call the day 94% printed; I owned the issues log and the change freeze; I decided which zones the bots kept. The vendor fixed the zone logic; ops ran the manual path; I decided when volume moved.

Why 30%? How did you arrive at it? It was an output, not a target. I let the bots keep only the zones that had already cleared 98% in the debug sample. Those zones carried about 30% of volume. Anything else would have been a guess dressed as a plan.

Why not delay the robotics go-live until the vendor was confident? Vendor confidence is not a metric. I had a measured path to run both and a number that said when to move volume. Delaying would have spent calendar we did not have and taught us nothing about the zone logic under real volume.

Why not push through — the business had bought robots? A gate that cannot stop the program is decoration. The robots were an input; the customer-facing output was accuracy. I will not ship a customer-facing miss to protect an automation narrative.

What did the manual path cost? Who paid? About three weeks of labor hours for a certified path we were planning to shrink. I priced it against one week of accuracy misses at peak — mis-ships, re-ships, concessions, store out-of-stocks — and against pulling the launch. It was the cheapest insurance in the program and I said so in the steering pack before the bots arrived.

How did you get sixty stakeholders to one log? I made the log the only place a date lived. If a dependency was not a row with an owner and a date, it did not exist for the program. People stopped sending emails because the log was where decisions happened.

What decision did you challenge? The original plan had robots at 100% from day one with a two-week "stabilization." I challenged the word stabilization — it had no number — and replaced it with the gate and the dial.

How did you handle the freeze? Someone always has an exception. One exception path, written down: a change in freeze needed a customer-facing defect it fixed, a rollback, and my signature. Three got through in ten weeks. The freeze held because the exception path was real and narrow.

RESULTS PROBES

How did you measure +15%? Units processed per direct labor hour, inbound and outbound, versus the two buildings' combined trailing baseline for the same weeks, normalized for the dotcom-versus-store mix. Weekly, from WMS and the labor system, in the steering pack.

How did you measure pick accuracy? WMS pick-confirm scan versus gold-standard location, by zone, by hour. A building average would have hidden the bad zones inside the good ones — which is exactly how 94% was first noticed.

Compared to your goal? Zero incidents: met. 98% by zone before full volume: met, one week late versus the vendor's own plan. Efficiency above baseline: exceeded.

What trade-off did you make? Three weeks of robotics ROI, deliberately, for zero customer risk through peak. I would make it again.

I'm concerned a 94% print means your testing was weak. Tell me more. Fair. The zone logic passed test volume and failed real volume mix — the test data under-represented multi-unit orders in the dense zones. The miss was in the test design. The mechanism caught it in week two instead of at peak, which is what the gate was for; and the fix I made since is below.

CHALLENGE

What would you do differently? Shadow the bots on one zone for a week at full real volume before they touched any customer order. I now put "one zone, real volume, a week" into every automation ramp I write.

Why this story for Highest Standards? Because the standard was set before it was tested, and it cost us something to hold it. Standards that are free are not standards.

At Amazon. A launch or retrofit ramp as a dial with zone- or path-level gates: the capacity plan counts a new building's or retrofit's throughput only for the process paths that have cleared the rate and quality bar at real volume. You never plan on capacity that has not earned it.

The question bank — Insist on the Highest Standards (7 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

HS-1. Refused to compromise your standards around quality or customer service. Who was the customer? Result?

S1; MAIN as backup. *Situation.* Sephora's stores and dotcom customers through peak; robots going live inside the peak window. The standard: 98% pick accuracy by zone before bots carried full volume, published before the bots arrived. *Task.* Hold it when it printed 94% in week two. *Action.* I throttled to the zones that had cleared 98% — about 30% of volume — the same day, kept the manual path certified and staffed, debugged zone logic with the vendor, and restored volume as zones cleared. At Flexport (MAIN) the same move: wave three held on a 76% ETA accuracy against an 80% gate. *Result.* Above 98% in a week; zero customer-facing incidents; +15% efficiency; launched on date. MAIN: boarded a week later with a kill switch; response 72 → 90%.

Math. Pick accuracy (zone, hour) = correct pick-confirms ÷ picks ; week 2 average 94% (dense zones 88–91%, others ≥ 98%) → throttle share ≈ 30%.

Probe "What did it cost?" — Three weeks of robotics ROI. **Probe** "Who pushed you to compromise?" — The robotics timeline; nobody by name, which is why the gate had to be written down.

WBR / design: accuracy by zone by hour vs the gate; bot share as the lever.

HS-2. Unsatisfied with the status quo. What did you do? Impact? Differently?

S3. Situation. A brochure number in the plan that everyone knew was soft, padded by feel. **Task.** Refuse the status quo without a mandate to change it. **Action.** I measured two hundred lanes, replaced the parameter with the empirical p99, got finance to sign the release rule, and built a drift flag so the status quo could not quietly return. **Result.** 99% accuracy held; ~\$1.5M released; the flag has reopened lanes since. Differently: a measurement audit in week one of any new domain — the standard I now start with.

Math. $\text{Planned LT} = \text{p99_lane}$; $\text{drift} = \text{rolling-90-day p99} > \text{plan} + 2 \text{ d} \rightarrow \text{planner review ; release where } \sigma_{\text{measured}} < \sigma_{\text{assumed}}.$

Probe "Why had nobody done it?" — The brochure was free and the pad was invisible; nobody owned the seam. **Probe** "What did the drift flag catch first?" — A lane whose carrier changed mid-year; p99 widened by five days.

WBR / design: % planning parameters measured vs inherited — the standing status-quo check.

HS-3. Improved the quality of something already getting good feedback. Why? How did customers react?

S8 (and S16). Situation. Flexport's Monday review got good feedback — on time, looked right. **Task.** Measure "looks right." **Action.** I recomputed each of seven metrics from source systems weekly and compared: 60% of metric-weeks matched within tolerance. I rebuilt the review as a Snowflake pipeline with the seven definitions frozen and a 1% parallel run before cutover. S16 similarly: 60% ASN completeness "worked" until we insisted on 95%. **Result.** 90% accuracy, zero manual hours, and a 4% ticket-sync drop caught in week one that the old review could not have seen. The VP's reaction went from surprise at the 60% to trust in the red.

Math. $\text{Accuracy} = \text{metric-weeks within tolerance of source recompute} \div \text{metric-weeks}$; $\text{parallel-run gate } |\text{manual} - \text{pipeline}| \div \text{manual} \leq 1\% \text{ for all 7 metrics, 3 consecutive weeks.}$

Probe "How did you know it needed improvement if nobody complained?" — Nobody had checked; I checked. **Probe** "Did the VP trust it less after learning 60%?" — Briefly; the parallel run rebuilt it.

WBR / design: review accuracy as a metric about the review itself.

HS-4. A decision between standards and delivery. Trade-offs? Outcome? Differently?

S9. Situation. Mock one printed 7% against a 0.5% reconciliation tolerance, with a hard go-live date. Delivery said "fix in hypercare." **Task.** Choose the standard without losing the date. **Action.** I kept the tolerance — it had been published before mock one for exactly this moment — burned the cause buckets (mapping first, then timing) through mocks two and three, and sequenced training late to protect the recon work. **Result.** Live at 0.3% on the date, under budget. Trade-off: training sign-off slipped 1.5 weeks. Differently: a what-if plan for the RF guns so the training slip never happened.

Math. $7\% \rightarrow \text{bucket by cause} \rightarrow \text{mapping fixes } (-5 \text{ pts}) \rightarrow \text{timing fixes } (-1.7 \text{ pts}) \rightarrow 0.3\%$; tolerance 0.5% published before mock 1.

Probe "What if mock three had been 0.8%?" — No-go, and I had said so before mock one; the sponsor had signed it.

Probe "Where did delivery win?" — Hypercare was four weeks, not eight; numeric exit criteria made it safe to compress.

WBR / design: tolerance published before rehearsal; go/no-go reads the scorecard.

HS-5. A goal you wish you had done better on. How could you have improved?

S10 (in-transit); S9 training as second. Situation. Goal: every stock state promiseable only under a measured rule. **Task.** Be honest about the one I missed. **Action.** I met it for every state but one — in-transit got "never," not a number — and dotcom paid for weeks before the divergence chart made me fix it through the policy. **Result.** Fixed, with a per-lane window; but I had the method from Elementum and did not apply it on day one. Better: measured default for every state in the first version, no exceptions. S9: training 1.5 weeks late on a what-if I should have written.

Math. $\text{Weeks of hidden sellable units} \times \text{units/week} - \text{unmeasured, and I say so}$; prevention: every default = measured number, day one.

Probe "Why did you default to never?" — It felt safe during a cutover; safety that is not priced is not safety. **Probe** "What tells you a default is wrong now?" — Two lines diverging; I chart them from day one.

WBR / design: in-transit units outside the lane window as an input from day one.

HS-6. Most significant continuous-improvement project you led. Catalyst? Approach? Outcome?

MAIN. Situation. Catalyst: \$0.66 a container at seven customers, sixty waiting, and an E-Team goal to replace human track-and-trace with automation. The pilot economics did not survive multiplication. **Task.** Run continuous improvement as a system, not a project. **Action.** A WBR with automation rate, coverage (automated vs with human inputs), tickets per active container and cost-to-serve as the spine; a backlog re-ranked every week on tickets × cost by carrier × terminal × milestone; six graduation criteria per wave measured at current load so each wave forced the fixes that made the next one cheaper; a public hold when a wave missed; and the review itself automated through Snowflake so it could outlive me. **Result.** \$0.03 a container at 68 customers (−95%); automation ~60 → 97%; accuracy 60 → 90%; response 72 → 90% at 5× volume; two months early.

Math. Each week: $\text{rank_c} = (\text{tickets_c} \times \$/\text{ticket} \times 52 + \text{interactions_c} \times \$0.10 \times 52) \div \text{eng-weeks_c}$; $\text{CTS_w} = (\text{T_w} \times \$/\text{ticket} + \text{platform}) \div \text{containers_w}$; gate check per wave.

Probe "What improved wave over wave?" — Tickets per customer, because bad sources were killed and ETA accuracy rose. **Probe** "What was the catalyst for the kill switch?" — Wave three's hold; I should have built it first.

WBR / design: CTS per container, tickets per customer, six criteria by wave.

HS-7. (Manager) Used feedback about your team to drive a change. How gathered? Outcome?

S4. Situation. Feedback about the alerting team's product arrived as dismissed alerts — planners voting with their mouse. **Task.** Turn informal feedback into a change mechanism. **Action.** Every dismissal categorized; top category fixed weekly in public; dismiss rate on the wall; the planner who flagged it named next to the fix. The team's design changed — the reason line came from this loop. **Result.** False alarms fell; the alert entered S&OP un-mandated; forecast accuracy 70 → 85%. Matrix: I ran the forum; the team owned the fixes.

Math. Dismiss rate = dismissed ÷ fired ; by cause (data / logic / timing / ignored) ; weekly burn-down of the top cause.

Probe "How did the team feel about public dismiss rates?" — Exposed for two weeks; proud by week four when the line fell. **Probe** "What else did you gather?" — S&OP attendance of our alert — whether it was on the agenda.

WBR / design: false-alarm rate by category as the team's quality input.

Think Big

Create and communicate a bold direction; think differently; look around corners. They listen for: the small ask, the big thing you saw in it, and how you sequenced it so it shipped.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S3	"The ask was a report. The product was lane-level planning as a system." S3 — Two hundred lanes: plan lead time at the empirical p99, release the inventory the brochure was holding	below
S7	"Think Big without sequencing is a slogan. The cheaper half first paid for the harder half." S7 — Flipping a socialized eighteen-month roadmap with a two-column cost-to-serve table	under Have Backbone; Disagree and Commit
MAIN	"The ask was fewer tickets. The product was every container in the client's book, forwarded by us or not." MAIN — Shipment Visibility (Flexport)	Section 5

S3 — Two hundred lanes: plan lead time at the empirical p99, release the inventory the brochure was holding (Elementum)

LPs: Think Big · Ownership · Dive Deep · Customer Obsession · Frugality · Invent and Simplify. **AIQB:** Think Big 1, 3, 6 · Ownership 3 · Dive Deep 5, 6 · Frugality 1 · Invent and Simplify 1, 6.

Problem	The inventory plan ran on carrier-brochure lead times — one optimistic number per lane, never measured. Safety stock on every lane was sized against a number nobody had checked; planners padded it by feel.
Goal	Planned lead times that hold 99% of the time, <i>and</i> less safety stock, not more — measured variability should buy inventory out, not in.
Key metric	Lead-time accuracy (arrivals inside planned lead time, by lane, by week) → 99%; safety stock units and dollars released, auditable by lane.
Did it move?	Yes. 99% lead-time accuracy held. 580,000 units and ~\$1.5M of safety stock released. Drift flag per lane made it permanent.

Situation. "A planner asked me for a lead-time variability report. When I looked underneath, the inventory plan ran on carrier-brochure lead times — one optimistic number per lane, never measured against an actual arrival. Planners knew the number was soft, so they padded safety stock by feel. Nobody owned the seam between the brochure and the plan.

Task. The ask was a report. The goal I set was different: planning lead times that hold ninety-nine percent of the time, with *less* safety stock — because if you measure variability honestly, it should release inventory on most lanes, not add it.

Action. I pulled the shipment history across two hundred lanes myself — ship date, arrival date, carrier, and the source of the transit number, brochure or scan. I built the empirical distribution per lane: mean, spread, tail. Then the move that mattered: I set every planning lead time to the lane's empirical p99 — the smallest number of days by which 99 of 100 shipments had already arrived — not the mean, and not a normal approximation, because right-skewed lanes have fatter tails than a normal distribution admits. The counter-intuitive result: on most lanes the measured p99 was tighter than brochure-plus-gut, so stock came out; on the minority where it was wider, stock went in, deliberately. I wrote the release rule with finance before any dollar moved: stock released only where measured spread was tighter than assumed. Then I productized it — a drift flag per lane that fires when the rolling distribution widens past a threshold, so widening triggers a review instead of a silent re-plan.

Result. Ninety-nine percent lead-time accuracy held. Five hundred eighty thousand units and about a million and a half dollars of safety stock released, auditable lane by lane. The ask was a report. The product was lane-level planning as a system — and the drift flag is why it was still true a year later."

Supply · Demand · Capacity. **Supply:** inbound lead time per lane — the variable I measured. **Demand:** replenishment requirements by SKU-location, unchanged. **Capacity:** safety stock is storage capacity and working capital consumed by supply uncertainty. **What bound:** a bad supply parameter was consuming cube and cash. Fixing the parameter released both without touching a building. This is the cheapest capacity there is: capacity hidden inside a wrong assumption.

What was innovative. The standard answer is a variability report and a safety-stock formula with a normal z-score. I planned to the empirical tail per lane, wrote a release rule that required the measurement to be *better* than the assumption before money moved, and built a drift detector so the plan re-opened itself. The system replaced the report.

The math

STEP 0 – TERMS (one lane = one origin → one destination; every number below is per lane)

d = average daily demand at the receiving location, units/day

σ_d = standard deviation of daily demand

LT_mean = measured mean transit time, days

σ_{LT} = measured std dev of transit time

LT_plan = the lead time the plan protects to

z = service factor: 99% → 2.33

n = shipments in the history window

t(k) = k-th smallest transit time

STEP 1 – MEASURE THE LANE (this is the "pull it yourself" step)

sort n transit times $t(1) \leq \dots \leq t(n)$; empirical p99 = $t([0.99 \ n])$

Lane A: n = 400 → p99 = $t(396)$ = 34 d ; LT_mean = 21 d ; σ_{LT} = 4 d ; right-skewed
brochure said 18 d. Count arrivals > 18 d: 240 of 400 = 60% "late" vs the brochure.

STEP 2 – WHY NOT THE MEAN, WHY NOT A NORMAL FIT

mean as plan: right-skewed lane → median < mean → $P(t > 21) \approx 0.40$ → 2 in 5 shipments late

normal p99: $21 + 2.33 \times 4 = 30.3$ d ; but actual arrivals > 30.3 d = 14 of 400 = 3.5%

→ a "99%" normal plan would have held 96.5% ; tail understated by ~4 days

brochure + pad: 18 d + "about a week" = 25 d ; arrivals > 25 d = 28 of 400 = 7% late → 93% held

empirical p99: 34 d ; arrivals > 34 d = 4 of 400 = 1% → 99% held, by construction

STEP 3 – SAFETY STOCK, TWO TERMS

$SS = d \times (LT_{plan} - LT_{mean}) + z \times \sigma_d \times \sqrt{LT_{mean}}$
└ lead-time cover: units ─┐ └ demand cover during the mean lead time ─┐
consumed if transit runs to (normal z is fine here: daily demand IS roughly
the p99 instead of the mean symmetric; it was lead time that was not)

The old ERP formula put lead-time risk inside a z-score with an ASSUMED σ_{LT} (system default, never measured) : $SS_{old} = z \times \sqrt{LT \times \sigma_d^2 + d^2 \times \sigma_{LT_assumed}^2}$ – and planners added a manual pad on top because the brochure was late 60% of the time.

STEP 4 – LANE A, BEFORE AND AFTER ($d = 500/\text{day}$, $\sigma_d = 150$, $\sigma_{LT_assumed} = [8 \text{ d}]$)

before: $ERP = 2.33 \times \sqrt{(18 \times 150^2 + 500^2 \times 8^2)}$
 $= 2.33 \times \sqrt{(405,000 + 16,000,000)} = 2.33 \times 4,050 = 9,437 \text{ units}$
pad = $[7 \text{ d}] \times 500 = 3,500 \text{ units}$
SS_before $\approx 12,900 \text{ units}$
after: LT cover = $500 \times (34 - 21) = 6,500 \text{ units}$
demand cover = $2.33 \times 150 \times \sqrt{21} = 2.33 \times 150 \times 4.58 = 1,601 \text{ units}$
pad = 0 (plan now holds 99%; the reason for the pad is gone)
SS_after $\approx 8,100 \text{ units}$
released on Lane A = $12,900 - 8,100 = 4,800 \text{ units} \times \$2.59 \approx \$12.4K$
release rule check: $\sigma_{LT} \text{ measured } 4 \text{ d} < \sigma_{LT} \text{ assumed } 8 \text{ d} \rightarrow \text{release allowed}$

STEP 5 – LANE B, WHERE STOCK WENT IN ($d = 200/\text{day}$, $\sigma_d = 80$, $LT_{mean} 30$, $\sigma_{LT} 11$, p99 58)

before: $2.33 \times \sqrt{(24 \times 80^2 + 200^2 \times 8^2)} = 2.33 \times \sqrt{2,713,600} = 3,838$; + pad 1,400 $\approx 5,240$
after: $200 \times (58 - 30) + 2.33 \times 80 \times \sqrt{30} = 5,600 + 1,021 \approx 6,620$
change: +1,380 units – stock ADDED, deliberately ; σ measured 11 > assumed 8 $\rightarrow$ no release
This lane is why the number is credible: the method is allowed to say "hold more."

STEP 6 – ADD IT UP ACROSS 200 LANES

~[150] lanes released, average [4,500] units = 675K units out
~[50] lanes took stock in, average [1,900] units = 95K units in
net $\Sigma (SS_{before} - SS_{after}) = 580K \text{ units}$
 $\times$ weighted average standard cost of the SKUs on those lanes [\$2.59] $\approx \$1.50M$ working capital
(P&L view, if asked: carrying cost [20%] $\times \$1.5M \approx \$300K/\text{yr}$; cube freed $\approx 580K \div [60/\text{case}]$
 $\approx 9,700$ cases of storage)

STEP 7 – PROVE IT HELD

LT accuracy = arrivals within planned LT $\div$ arrivals, by lane, by week $\rightarrow 99\%$
on Lane A that is ≤ 4 late arrivals per 400 ; a lane with > 8 late in a rolling 90 d is a red
validation before go-live: held-out quarter – 99.1% of arrivals $\leq$ p99_lane set on prior quarter

RULES THAT MADE IT PERMANENT

release rule (finance-signed): release only where $\sigma_{LT} \text{ measured} < \sigma_{LT} \text{ assumed}$, lane by lane,
written to a release ledger with the \$ and the signer
drift flag: fire when rolling-90-day p99 > $LT_{plan} + [2 \text{ d}]$ OR $\sigma_{90} > 1.25 \times \sigma_{baseline}$
 $\rightarrow$ planner review, not a silent re-plan
sensitivity: 1 day of planned LT on Lane A = $d = 500 \text{ units} \approx \$1,300$ – so the 4 days the normal
fit missed were 2,000 units of exposure on one lane

Solution design / flow

shipment events (ship scan, arrival scan, carrier, lane key, transit source)
└ lane history store $\rightarrow$ per-lane distribution (mean, σ , p50, p90, p99, n)
└ planning LT = p99 $\rightarrow$ ERP planning parameter (lane $\times$ SKU)
└ drift flag per lane $\rightarrow$ planner review
└ safety-stock recompute $\rightarrow$ release ledger
(finance-signed rule) $\rightarrow$ \$ by lane

How it translated to a WBR

Outputs: LT accuracy vs 99% (lane $\times$ week); safety stock \$ vs plan. **Inputs:** lanes with drift flag open; lanes with n below confidence floor; % planning LTs sourced from scan vs brochure. **Red rule:** drift flag open > 2 weeks without

a planner decision is a red on the planner's line, not the lane's.

SITUATION PROBES

Why important? What if nothing happened? Every lane carried either too much stock (cash and cube) or too little (stockouts), and nobody could say which. Doing nothing meant continuing to pay for an unmeasured number in inventory every week.

What was the initial scope and how did it change? One report for one planner. I expanded it to all two hundred lanes and to the planning parameter itself — I took that to leadership with mockups before I built it, because changing the planning parameter is a change to the plan, not to a report.

What was the goal before you set yours? There was none beyond "a report." That was the problem.

ACTION PROBES

Why p99 and not p95 or the mean? The mean is a coin flip — in a right-skewed lane, roughly two in five shipments arrive after it. p95 leaves one shipment in twenty late, which at our order frequency was a stockout a month on a busy lane. 99% was the service level the business had already committed to; I made the lead time honor it. On lanes where the cost of carrying justified less, the method allows a lower quantile per lane — the point is to choose it, not inherit it.

Why empirical quantile and not a normal distribution? Transit times are right-skewed — a shipment can be very late but not very early. A normal fit understates the tail, which is exactly where the stockouts live. I checked: on long ocean lanes the normal p99 was days short of the empirical p99.

How much history per lane? Seasonality? Enough shipments per lane to make a p99 meaningful — thin lanes were pooled with similar lanes (same origin region, mode, carrier) and flagged as lower confidence. Peak-season transits were kept as their own regime where the data supported it; the drift flag catches the seasonal widening on lanes where it did not.

What did you do versus the team? I pulled the data, built the distributions, chose the quantile rule, pitched leadership, wrote the release rule with finance, and designed the drift flag. Engineering productized the pipeline from my spec.

How did you get planners to trust a number that cut their safety stock? I showed each planner their own lanes: brochure number, measured p99, actual late arrivals under each. Where the data said "add stock," I added it first. Credibility came from the lanes where I gave inventory back.

What did you challenge? The assumption that the carrier's number was a fact. And the plan to ship a report — a report would have been read once.

RESULTS PROBES

How did you measure 99%? Actual arrivals inside the planned lead time, by lane, by week, from scan data — the same data the distributions came from, held out forward in time.

How did you compute \$1.5M? Safety-stock units before and after, by SKU-lane, valued at standard cost. Released only on lanes where the measured spread was tighter than the assumed spread — the finance-signed rule — so no lane's release depends on a judgment call.

Was the whole 580K units a release? Did any lanes get more stock? The 580K is net. A minority of lanes got more stock because their measured tail was wider than the brochure. I would rather not quote the lane count without the file in front of me.

Trade-off? Lanes with wider tails carried more inventory than before. That was the right trade: we had been running them short and calling it luck.

Compared to goal? 99% accuracy: met. Less safety stock, not more: met, net \$1.5M. Permanence: drift flag live.

CHALLENGE

What would you do differently? Start every new domain with a measurement audit in week one — "which planning parameters are measured, and which are inherited?" — instead of discovering the brochure problem mid-project. I have done that on every program since, including S/4HANA.

Why this story for Think Big? Because the opportunity was bigger than the ask, I took it, and I can show the gap: a report versus a planning system that still runs.

At Amazon. Every capacity planning parameter — dock-to-stow hours, process-path rates, vendor lead times, days of cover — is a distribution, not a number. I would ask on day one which of them are planned at a mean, plan the ones that bind at a chosen quantile, and put a drift flag on each so the plan re-opens when the world moves.

The question bank — Think Big (8 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

TB-1. Saw an opportunity to do something much bigger than the initial focus. Did you take it? Outcome?

S3. *Situation.* Asked for a lead-time variability report; I saw that the plan ran on an unmeasured parameter across two hundred lanes. *Task.* Take the bigger thing without dropping the smaller one. *Action.* I pitched leadership with mockups — brochure vs measured p99 vs actual late arrivals on three lanes — and delivered the report on time as proof of good faith. Then built the lane-level planning parameter with a finance-signed release rule and a drift flag. *Result.* 99% accuracy, ~\$1.5M released, a parameter that keeps itself true. The ask was a report; the product was lane-level planning as a system.

Math. 200 lanes × per-lane distribution (mean, σ , p50, p90, p99, n) ; $\Sigma(SS_{\text{before}} - SS_{\text{after}}) = 580K \text{ units} \approx \$1.5M$.

Probe "What would you have lost by not taking it?" — A report read once; the brochure in the plan for another year.

Probe "What did taking it cost?" — Engineering time to productize, and a quarter of my own.

WBR / design: lane LT pipeline → parameter → drift flag.

TB-2. Changed the direction or view of a function and helped them embrace new thinking. Why? Outcome?

S7. *Situation.* Product and engineering argued roadmaps in features and dates. *Task.* Change how the function decided, not just one decision. *Action.* I moved the argument to cost-to-serve with a two-column table — identical rows, only sequence differing — took it privately to the plan's owner, let engineering break it for two days, and let the Bellevue onsite ratify. After the flip the table became how the group compared options. *Result.* -66% effort to first value, ~\$5M avoided, and a function that priced sequences before debating them.

Math. Same rows: months to value · effort · CTS cut at horizon · cost avoided ; decision path: table → owner → pressure test → ratify → commit.

Probe "How do you know they embraced it?" — The next two roadmap decisions arrived at me as tables. **Probe** "Who resisted?" — Operators' advocates; the protected milestone kept them in.

WBR / design: CTS as the standing currency in roadmap review.

TB-3. Proposed a novel approach. Why novel? Successful?

S10 (returns plant, RTV as pick-reserve). *Situation.* Returns were a storage location; ungraded units leaked into promises and credit ran ahead of the floor. *Task.* Make "not promiseable until disposition" a system fact. *Action.* A separate returns plant in S/4HANA — receive, grade, then restock / scrap / RTV — and RTV stock-transport orders modeled as pick-reserve so finance could not credit until the pick. Novel because nobody treats returns as a plant; the heaviness was the point. *Result.* Zero returned units promised before grading; credit tied to the physical pick; the stock-state model has seven states and this one never argued again.

Math. Returns plant stock $\notin$ supply basis ; RTV STO = pick reserve (subtracted from ATS) ; credit memo waits for GI.

Probe "Why not a status flag?" — Flags get overridden; a plant does not. **Probe** "Cost of the heavier model?" — More STOs; cheaper than one damaged unit to a second customer.

WBR / design: returns-plant flow: receive → grade → restock / scrap / RTV; L&F and QI ageing as inputs.

TB-4. Drove adoption for your vision. How did you know it was adopted? How did you drive and track it? Differently?

S4. Situation. Vision: a supply-demand view planners trust enough to act on in S&OP. **Task.** Drive adoption without a mandate. **Action.** Earn it: reconciliation at zero before any alert, a reason line on every alert, a public false-alarm burn-down. Tracked adoption as % of alerts actioned, weekly. **Result.** The alert was on the S&OP agenda without anyone ordering it there; 70 → 85% forecast accuracy; ~\$2M released. Differently: the reason line in the first prototype, not the second.

Math. Adoption = alerts actioned in S&OP ÷ alerts fired, weekly ; output: stockouts without a prior alert → 0.

Probe "What was adoption at week four?" — Low; dismissals were the data that fixed it. **Probe** "How is it tracked now?" — % alerts actioned and stockouts without prior alert.

WBR / design: % alerts actioned as the adoption input.

TB-5. An idea adopted by global stakeholders. How did you drive and track adoption? Differently?

MAIN (and S16). **Situation.** A global 4PL visibility platform — customers across regions, data sources across hundreds of carriers and ports. **Task.** Get global stakeholders to adopt one admission rule and one WBR. **Action.** Published the six graduation criteria to sales and leadership; held wave three publicly; ran one Monday WBR for VP, core team and dependent EMs on the engineering org's conventions so it needed no translation; tracked adoption by scorecard per wave and cost-to-serve per container. S16: a thousand supplier and carrier routes on one 95% gate. **Result.** 68 customers, \$0.03 per container, a WBR still in the same format. Differently: the kill switch before wave one.

Math. Adoption = waves boarded on a green scorecard ÷ waves (target 100%) ; CTS_w trend ; S16: cohorts graduated ≥ 95% ÷ cohorts = 100%.

Probe "What differed by region?" — Data-source quality; waves were sequenced by it, not by geography. **Probe** "How did you know global stakeholders had adopted?" — Regional sales leads started quoting the criteria to customers.

WBR / design: graduation scorecard by wave, global.

TB-6. Thought differently to improve a process that was working. Assumptions questioned? How did you evaluate? Differently?

S8. Situation. The Monday review worked — on time, looked right. **Task.** Question the assumption that "looks right" is right. **Action.** Measured it: 60% of metric-weeks matched a recompute from source. Evaluated the alternative by a 1% parallel run, and by what the pipeline caught that the manual era could not. **Result.** 90% accuracy, zero manual hours, a 4% sync drop caught in week one. Differently: security lead time on the kickoff checklist.

Math. 60% → 90% metric-weeks within tolerance ; first catch = (records in source - records in pipeline) ÷ source = 4%.

Probe "Why fix what works?" — A VP steering on 60% is a decision-quality problem waiting for the wrong week. **Probe** "What did you not change?" — The seven definitions; changing them would have hidden the pipeline's accuracy.

WBR / design: review accuracy; frozen definitions; parallel-run delta.

TB-7. (Manager) Encouraged a team member or organization to take a big risk. How did you balance it? Outcome? Learned?

S1. Situation. I encouraged the organization to go live with robotics inside a peak window — a big risk. **Task.** Balance ambition with reversibility. **Action.** The gate (98% by zone, published), the dial (bot share of volume, re-evaluated daily), and the manual path (certified, funded) were built before the bots arrived. When 94% printed, the risk was handled as a Tuesday decision. **Result.** +15% efficiency, zero incidents, on date. Learned: the size of a risk you can take is set by the quality of the reversibility you build.

Math. Dial = $\sum$ volume in zones ≥ 98% ÷ total ; insurance cost (3 wk manual path) ≪ one week of a 6-pt miss at peak.

Probe "Who was most afraid?" — Store ops, of mis-shipped replenishment; the zone-level gate addressed their fear specifically. **Probe** "What risk did you refuse?" — Bots at 100% with a two-week "stabilization" that had no number.

WBR / design: bot share and zone accuracy — risk and its control on one page.

TB-8. (Manager) Established a vision where there wasn't one. Buy-in? Execution? Outcome? Differently?

S10. Situation. No vision for inventory truth across channels — each system had its own. **Task.** Establish one and get sixty-plus stakeholders behind it. **Action.** Vision: one definition per stock state, one owner, one promising rule, proven nightly. Buy-in through workshops where each group named the customer harmed by its own definition; signatures by a date. Execution as a program: RAID, weekly CIO/COO pack, 343-task runbook, rollback as two numbers. **Result.** Zero non-sellable promises; MCS retired; nightly OOB inside tolerance. Differently: in-transit from lane data on day one.

Math. $ATS = \text{basis} - \text{reserves} - \text{SNB} + \text{mech} + \text{L\&F}$; $OOB \leq 0.5\%$; rollback: $\text{sync} < 99\% / 4 \text{ h}$ or $\text{p95} > 2 \times \text{budget} / 1 \text{ h}$ — never fired.

Probe "How did you know it was a vision and not a config?" — Because the arguments were about words, not fields.

Probe "What did you say to get the first signature?" — "Here is the customer this state hurts if we promise it."

WBR / design: stock-state model + nightly ATS/OOB — vision as a table and a computation.

Bias for Action

Speed matters; many decisions are reversible and do not need extensive study; calculated risk taking. They listen for: the decision, how fast, what made it reversible, and what you did after.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S2	"Timestamped decision at 4 AM, then a post-mortem that made silence an alarm." S2 — Cyber Monday, 4 AM: rebalancing allocation across three DCs with 70% of the information	under Are Right, A Lot
S1	"Throttling the robots was a two-way door we had built. I turned it the same day." S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived	under Insist on the Highest Standards
S4	"I shipped the forecast alert at 70% accuracy with the error band shown, because a planner with an honest alert beats a planner with nothing." S4 — Forecast alerts that entered the weekly S&OP because planners trusted them	below

S4 — Forecast alerts that entered the weekly S&OP because planners trusted them (Elementum)

LPs: Invent and Simplify · Deliver Results · Earn Trust · Dive Deep. **AIQB:** Invent and Simplify 3, 5 · Earn Trust 1 · Dive Deep 6 · Customer Obsession 3, 4.

Problem	Planners learned about shortages when a production line stopped. Forecast accuracy ~70% at SKU-location-week. The SAP data any alert would run on was not trusted — the ERP's own reports disagreed with each other.
Goal	Alerts planners act on inside the weekly S&OP; forecast accuracy 70% → 85%; safety stock reduced — in that order, because the third is legitimate only after the first two.
Key metric	Forecast accuracy at SKU-location-week; daily reconciliation drift (MD04 vs MB52); alert false-alarm rate; safety stock \$ released.
Did it move?	Yes. 70% → 85%. Zero recon drift daily. ~\$2M safety stock released under a two-signal rule. Live in three months; adopted into S&OP without mandate.

Situation. "Planners found out about shortages when a line stopped. Forecast accuracy was about seventy percent at SKU-location-week, and the SAP data behind any alert was not trusted — the demand view and the stock view disagreed on any given day.

Task. Alerts planners would act on inside the weekly S&OP, forecast accuracy to eighty-five, and safety stock down. In that order.

Action. I refused to ship a threshold email on data nobody trusted. I led the SAP ingestion — inventory, production, POs, ASNs, forecasts — into one supply-and-demand view, with a daily MD04-to-MB52 reconciliation published and held at zero drift, and one rule: no alert fires on a day the recon is red. The alert itself was coverage math — projected supply against demand by week — with a thirty-second reason line: which input moved, by how much. Then a public false-alarm burn-down: every alert a planner dismissed was classified and the logic fixed in a weekly forum until the dismiss rate fell. The forecast-accuracy gain came mostly from the inputs the recon exposed — stale POs, duplicated demand, wrong lead times — not from a new model.

Result. Live in three months. Forecast accuracy from seventy to eighty-five percent at SKU-location-week. Recon at zero drift daily. About two million dollars of safety stock released — only where *both* forecast error and lead-time error had improved, a rule finance signed. The alert entered the weekly S&OP because planners trusted it, not because anyone mandated it."

Supply · Demand · Capacity. **Demand:** the forecast at SKU-location-week — the input I made 15 points more accurate. **Supply:** inventory, production, open POs, ASNs — reconciled into one view. **Capacity:** two kinds — the S&OP's capacity to act (alert lead time before a shortage), and safety stock as storage/working-capital consumed by forecast error. **What bound:** trust. Without a reconciled base, no amount of alerting moves a decision.

What was innovative. The standard approach is a dashboard with thresholds. I built the trust mechanism first — a daily reconciliation with a hard "no alert on a red day" rule — then alerts that explain themselves in one line, then a public burn-down of false alarms, and a two-signal release rule so no single improved number could justify a dollar. The alert was the least important part.

The math

```
Forecast accuracy (SKU-loc-week, lag-matched) =  $1 - \sum |F - A| \div \sum A$  70% → 85%
Recon drift (daily) =  $\sum_{\text{plant}} |\text{MD04 stock} - \text{MB52 stock}| \div \sum \text{MB52 stock}$  held at 0.0%
Coverage (weeks) = (on-hand + scheduled receipts inside LT)  $\div$  avg weekly demand
Alert fires when coverage < (LT weeks + safety weeks) AND recon is green today
Reason line = the input whose  $\Delta$  moved coverage most: "PO 4471 slipped 9 d → coverage 1.6 wk in wk 7"
False-alarm rate = alerts dismissed  $\div$  alerts fired, weekly, classified (data / logic / timing / ignored)
Two-signal release release SS on SKU-loc only if  $\Delta \text{FA} > 0$  AND  $\Delta \text{LT-accuracy} > 0$ 
SS_new =  $z \times \sqrt{(\text{LT} \times \sigma_d^2 + d^2 \times \sigma_{\text{LT}}^2)}$  with the improved  $\sigma_d$  and  $\sigma_{\text{LT}}$  ;  $\Sigma(\text{SS}_{\text{old}} - \text{SS}_{\text{new}}) \times \text{cost} \approx \$2\text{M}$ 
```

Solution design / flow

```
SAP ECC: inventory (MB52) · requirements (MD04) · POs · ASNs · production · forecast
└─▶ daily ingestion → reconciliation (MD04 vs MB52 = 0?) —red→ no alerts today; fix data
                        | green
                        ▼
supply-demand view by SKU-loc-week → coverage math → alert + reason line
                                   |
                                   └─▶ S&OP weekly
                                   └─▶ false-alarm burn-down (weekly, public) → logic fixes
                                       → dismiss rate ↓
```

How it translated to a WBR

Outputs: forecast accuracy at planning grain; stockouts with no prior alert (target 0); SS \$ released. **Inputs:** recon drift (daily); alert false-alarm rate; alert lead time before shortage; % alerts actioned in S&OP. **Red rule:** any stockout without an alert $\geq$ LT weeks earlier gets a root-cause row the same week.

SITUATION PROBES

Why important? Cost of nothing? A stopped line is the most expensive shortage there is, and planners were compensating with safety stock they could not justify. Doing nothing was paying for 30% forecast error in inventory and in expedites.

Who was the customer? The planners, and through them the plants. The S&OP meeting was where their decisions were made, so that was where the alert had to earn a seat.

ACTION PROBES

Why reconciliation first? Planners wanted alerts. An alert on untrusted data trains people to ignore alerts. I would rather be three weeks late with something they believe than a day early with something they dismiss. I told them that in the first meeting.

What's the MD04-to-MB52 reconciliation in plain terms? The planning view of stock and requirements versus the physical stock view, by plant, by material, by day. If they disagree, one of them is wrong, and any alert built on either is wrong. We held the difference at zero and published it.

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What's the MD04-to-MB52 reconciliation in plain terms? The planning view of stock and requirements versus the physical stock view, by plant, by material, by day. If they disagree, one of them is wrong, and any alert built on either is wrong. We held the difference at zero and published it.

Why a reason line? A planner has thirty seconds per alert. "Coverage falls below two weeks in week 7 because PO 4471 slipped nine days" is actionable. "Coverage alert" is noise.

How did you run the false-alarm burn-down? Every dismissed alert got a category — wrong data, wrong logic, right alert wrong timing, right alert ignored — and we fixed the top category weekly in a public forum. The dismiss rate was on the wall. Planners saw their feedback change the logic within a week; that is where trust came from.

How did forecast accuracy improve 15 points if you did not change the model? Garbage in. Stale POs were being counted as supply; duplicated demand from two systems inflated requirements; lead times were brochure numbers. The recon exposed each one. Fixing inputs moved the number more than a new model would have.

What did you personally own? The ingestion scope and the recon design, the "no alert on a red day" rule, the reason-line requirement, the burn-down forum, and the two-signal release rule with finance.

RESULTS PROBES

How did you measure forecast accuracy? $1 - |\text{forecast} - \text{actual}| / \text{actual}$ at SKU-location-week, lag-matched to the planning horizon, weekly, in the S&OP pack.

Why two signals before releasing stock? Forecast accuracy alone could be mix; lead-time accuracy alone could be a quiet quarter. Requiring both to improve on the same SKU-location is how I kept the \$2M honest. Finance signed the rule before the first dollar.

Compared to goal? Adoption: met, un-mandated. 85%: met. Stock release: \$2M, met under the stricter rule.

Trade-off? Three weeks of calendar on the recon before anyone saw an alert. Worth it; the alternative was alerts nobody read.

CHALLENGE

Differently? Put the reason line in the first prototype — I added it after planners dismissed the first week's alerts, and I lost a week of trust I had to earn back.

At Amazon. Any capacity signal I bring to S&OP or Finance carries its reconciliation state and a one-line reason — "FC-A storage utilization crosses 90% in week 9 because inbound for category X pulled forward two weeks." And a lever fires on two signals, not one: forecast and placement, or utilization and dock-to-stow.

The question bank — Bias for Action (8 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

BA-1. A calculated risk where speed was critical. Mitigation? Outcome? Differently?

S1. Situation. Robots went live into a window that ended at peak; week two, pick accuracy printed 94% against the 98% gate I had published before they arrived. **Task.** Act the same day, inside a risk I had chosen to take. **Action.** I throttled the bots to the zones that had cleared 98% — about 30% of volume — kept the launch green on the certified manual path, sat the zone-logic debug with the vendor, and restored full volume in week three. The mitigation was designed in advance: gate, dial, manual path. **Result.** +15% efficiency, zero customer-facing incidents, on date. Differently: one zone at real volume for a week before any customer order.

Math. Throttle share $\approx 30\% = \text{volume in zones} \geq 98\% \div \text{total}$; recovery: zones clearing 98% per day → full volume by week 3.

Probe "What was the risk you took?" — Going live with bots at all inside a peak window; the dial made it recoverable.

Probe "Who decided?" — Me; the decision right sat with the program.

WBR / design: bot share of volume as an input, pick accuracy by zone by hour as the gate; any zone below 98% two hours running drops out the same shift.

BA-2. Tight deadline, no time to consider all options. How much time? Approach? Learned?

S2. Situation. About an hour before the backorder pool doubled again, 4 AM on Cyber Monday. **Task.** Decide without the root cause. **Action.** Decide the reversible things now, learn the irreversible things in parallel: promise-at-risk ranking, manual node assignment with feasibility checks, a timestamped log; no promise downgrade, no emergency PO. **Result.**

99% fill held, ~\$8M protected, clean reconciliation when the scheduler restarted. Learned: a peak is its own regime — its own readiness checklist, including a scheduler heartbeat and a DST check.

Math. Pool doubles every ~40 min → waiting 1 h $\approx 2.8\times$ the pool ; node B room = 3 h $\times$ [4,000/h] = 12,000 units → assign in score order until full.

Probe "What option did you not consider that you should have?" — Pausing order intake for ten minutes to let the queue drain; I would weigh it next time. **Probe** "What did the log buy you?" — Clean reconciliation when the scheduler restarted; no double allocation.

WBR / design: allocation queue depth and scheduler age by hour; peak-regime checklist as a dated input.

BA-3. An important business decision without consulting your manager. How did it turn out? Differently?

S1. Situation. 94% printed on a Tuesday; the steering call was Thursday. **Task.** Decide under a decision right that had already been written. **Action.** The gate said what to do, and the decision right sat with the program in writing from the day the gate was published. I throttled, then told my sponsor afterward with the zone data. **Result.** Accuracy above 98% in a week, no customer impact. Where the line is: reversible and inside a published rule — act, then inform; irreversible or outside the rule — ask first.

Math. Decisions taken under gate authority, reported weekly ; the gate itself was signed by the sponsor before the bots arrived.

Probe "What if your sponsor had disagreed?" — The gate was theirs too — they had signed it. The conversation would have been about the gate, not the call. **Probe** "Where is the line?" — Reversible and inside a published rule: act, then inform. Irreversible or outside the rule: ask.

WBR / design: decision rights written next to each gate; "decisions taken under gate authority" reported weekly.

BA-4. Had to gather information and respond immediately. Outcome? Differently?

S2 (lead) or S11 (customer). Situation. S11: a couple of hundred orders mis-stated between OMS and WMS, ~\$2.5M, found that morning. **Task.** Gather exactly what was needed to act, and act the same day. **Action.** I pulled the complete affected list and the cause class — not the fix — argued for disclosure over a quiet fix, and had the customer re-rank the repair by their priorities that afternoon; we bulk-patched the class. **Result.** Trust kept, class never recurred, larger initiative won later. Differently: nothing on disclosure; faster on the class fix.

Math. Needed before acting: affected list complete (count reconciled both ways) + cause class ; not needed: the fix.

Probe "What information did you need before telling the customer?" — The complete list and the cause class; not the fix. **Probe** "Why not fix first?" — They would have re-ordered the fix anyway, and they would have learned we fix quietly.

WBR / design: OMS ↔ WMS mismatches per 10K lines published to the customer weekly — the disclosure built into the cadence.

BA-5. Decide between moving forward or gathering more information. What did you do? What information is necessary before acting?

S4. Situation. Planners wanted alerts now; the SAP data under them was not trusted. **Task.** Choose between shipping and gathering. **Action.** I gathered — three weeks on a daily MD04 ↔ MB52 reconciliation held at zero — because an alert on untrusted data trains people to ignore alerts. I published the drift number daily so the wait was visible progress. Then I moved fast: live in three months. **Result.** Forecast accuracy 70 → 85%, ~\$2M released, adopted into S&OP.

Necessary before acting: the data must reconcile and the action must be reversible. If both hold, act.

Math. Drift = $\sum |MD04 - MB52| \div \sum MB52$: 8–12% → 0 over 3 weeks ; rule: no alert on a red recon day.

Probe "How did you hold planners off for three weeks?" — I showed them the two SAP views disagreeing on the same SKU; they stopped asking. **Probe** "What did you do in the three weeks that was visible?" — Published the drift number daily; falling drift was the progress.

WBR / design: "no alert on a red recon day" rule; recon drift as the gating input metric.

BA-6. Saw an issue that would impact your team and took a proactive approach. Issue? Outcome? Learned?

S8 (and MAIN). *Situation.* I was the single point of failure for Flexport's Monday review — four systems into Excel, 30 hours a week across the people feeding two WBRs, 60% accuracy. Nobody had asked me to fix it. *Task.* Fix it before it failed on a Monday that mattered. *Action.* I wrote the BRD (Zendesk data in Snowflake, joined to container numbers from Crux), froze the seven metric definitions exactly as used — coverage automated, coverage with human inputs, tickets generated, tickets per active container, 4PL % within SLA, 3PL % within SLA, delta — priced the status quo at \$129,600/yr in our own rates, and gated cutover on a 1% parallel run. *Result.* Accuracy 60 → 90%, assembly 30 → 6 h/wk (\$103,680/yr), first catch a 4% sync drop. Learned: price the status quo in the builder's currency and the request prioritizes itself.

Math. $3 \times 5 \text{ h} \times 2 \text{ WBRs} \times \$90 \times 4 = \$10,800/\text{mo} = \$129,600/\text{yr} \rightarrow 6 \text{ h/wk} \times \$90 \times 48 = \$25,920/\text{yr} \rightarrow \text{saved } \$103,680/\text{yr}.$

Probe "What went wrong?" — Security review on ticket content: two weeks; owned upward the day I found it. **Probe** "Why not a better spreadsheet?" — Excel was the SPOF, not the formula.

WBR / design: the pipeline is the WBR's plumbing; frozen definitions and parallel-run delta are the design gates.

BA-7. (Manager) Your team was not moving to action quickly enough. What did you do? Outcome?

S13. *Situation.* Two go-lives on one engineering team; issues waited overnight because the same six people were on every call and exhausted. *Task.* Speed the team up without asking for more urgency. *Action.* I changed the structure: one RAID log, one 45-minute daily, and a two-pass war room — a day pass that triages, a night pass in another time zone that executes with written decision rights for reversible actions; nobody on both passes. *Result.* Issue age fell (p50 and p90 in the log, week over week); both go-lives landed; ~\$1M per DC protected; no 24-hour days. Matrix: I owned the structure, not the headcount.

Math. Issue age p50/p90 by pass ; people on both passes = 0 ; work moves 16–20 h/day on 8-h shifts.

Probe "Why was the team slow?" — Not motivation — a single queue with eight working hours a day. **Probe** "How did you know it worked?" — Issue age p50 and p90 in the log, week over week.

WBR / design: issue age by pass; people on both passes (target 0); decisions escalated to morning.

BA-8. (Manager) Removed a serious roadblock preventing progress. Barrier? How? Outcome? Differently?

S5 (sponsor's milestone); S8 (security review) as backup. *Situation.* The sponsor had promised a routing engine as milestone one; the team was being pushed to build on 55% data. The barrier was the commitment itself. *Task.* Move the milestone without making the sponsor defend it. *Action.* One page, privately: three lanes where the engine would have been wrong on current data. He moved the milestone himself. The team's weekly progress became the data-quality scorecard by defect class. *Result.* 99% by class; ~\$8M auditable; the engine built on a foundation that held. Differently: pull the data before the kickoff deck, not after.

Math. $3 \text{ lanes} \times (\text{engine recommendation on current data vs truth}) \rightarrow \text{wrong in both directions ; scorecard: \% by class, weekly, until } \geq 99\%.$

Probe "Why private?" — So he could own the change instead of defending the promise. **Probe** "What did the team see?" — The scorecard by defect class going up weekly; that was their progress metric.

WBR / design: data-quality scorecard by class as the team's weekly output until the engine milestone unlocked.

Frugality

Accomplish more with less; constraints breed resourcefulness, self-sufficiency and invention. They listen for: the constraint, the thing you did not buy or hire, and the unit cost.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
MAIN	"Cost-to-serve down 95% while the book grew ten times. Headcount would have scaled the waste." MAIN — Shipment Visibility (Flexport)	Section 5
S7	"Two columns of cost-to-serve beat eighteen months of roadmap. \$5M avoided, value in six months." S7 — Flipping a socialized eighteen-month roadmap with a two-column cost-to-serve table	under Have Backbone; Disagree and Commit
S15	"A scope cut that was priced, published and scored: 48 of 53 features, 11 partners, 10 countries, in a month." S15 — A scope cut that was priced, published, and scored	below
S8	"I priced the status quo at \$129,600 a year in the infra team's own rates and the fix at \$25,920." S8 — Four systems into one pipeline: the weekly review is only as good as the data under it	under Invent and Simplify

■ S15 — A scope cut that was priced, published, and scored (Flexport FF App export launch)

LPs: Deliver Results · Bias for Action · Customer Obsession · Frugality. **AIQB:** Deliver Results 1, 4 · Bias for Action 1 · Frugality 2, 4.

Problem	An export-side launch for forwarding partners with a hard date and a scope that could not all ship: Air and LCL, operator screens, and the partner path competed for the same engineers.
Goal	Ship the MVP on date; eliminate the most monthly operator interactions per engineering hour; migrate existing import partners without churn.
Key metric	Monthly operator interactions eliminated (the success metric set before scope); partners live; migration rate of import partners.
Did it move?	Yes. MVP April 2022; 11 partners, 10 countries; 48 of 53 import partners migrated within a month. Air/LCL and operator screens deferred — openly, with dates.

Situation. "The export launch was minimum-scope by design: partner path first; Air and LCL waited a year; operators kept their old screens two more quarters. Real cuts, real complaints. **Task.** Ship on date and make every cut defensible. **Action.** I set the success metric before scope — monthly operator interactions eliminated, not features shipped — so every proposed cut was scored against that number, in a published table, not in a hallway. Then a burn-up tripwire: any new work was estimated and swapped for something of equal size, never absorbed. **Result.** MVP landed April 2022 — eleven partners, ten countries — and forty-eight of fifty-three import partners migrated within a month. The people who lost scope could read why in the table."

Supply · Demand · Capacity. **Capacity:** engineering hours to the date — fixed. **Demand:** three constituencies' features. **Supply:** the scored scope. **What bound:** hours; the metric decided who got them.

The math

Success metric (set before scope) = monthly operator interactions eliminated
Scope score (feature) = interactions eliminated ÷ engineering weeks → rank; cut below the line

Migration	= $48 \div 53$ import partners in a month = 90.6%
Burn-up tripwire	= any added work estimated and swapped for equal size; absorbed work = 0 by rule

How it translated to a WBR

Outputs: operator interactions eliminated vs plan; partners live. **Inputs:** scope added vs swapped; partner migration rate. **Red rule:** scope absorbed without a swap is a red on the program, shown on the burn-up.

Why interactions eliminated and not revenue? Revenue lagged by quarters and was attributable to many things; operator interactions were measurable monthly and were the cost we were paid to remove.

How did you keep the door shut after launch? The burn-up chart was public; any line added had to show the line removed. Twice someone tried to absorb work; both times the chart made it visible within a week.

At Amazon. A capacity plan's scope is its lever list. Every lever added after the plan is published gets a cost and swaps for one of equal cost — or the plan is a wish list.

The question bank — Frugality (8 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

FR-1. Helped save costs or eliminate waste. Situation? Impact?

S3 (and MAIN). *Situation.* Safety stock on two hundred lanes sized against brochure lead times — paying in inventory for an unmeasured number. *Task.* Release what the bad number was holding, without creating stockouts. *Action.* Measured p99 per lane; released only where measured spread was tighter than assumed; added where it was wider; finance signed the rule. At Flexport (MAIN) the waste was operator hours — every ticket a container the machine could not track — and the fix was an ROI-ranked automation backlog. *Result.* 580K units and ~\$1.5M out at 99% accuracy; MAIN: cost-to-serve \$0.66 → \$0.03 while volume grew 5x. Also S4 ~\$2M, S5 ~\$8M.

Math. $\Sigma(SS_{\text{before}} - SS_{\text{after}}) \times \text{cost} \approx \1.5M ; $CTS = (\text{tickets} \times \$/\text{ticket} + \text{platform}) \div \text{containers}$; 1 pt automation at 10K/wk $\approx$ \$4K/yr.

Probe "Where did the units go?" — Not re-ordered; the release was in the planning parameter, so the next replenishment cycle simply bought less. **Probe** "Any lane got worse?" — A minority got more stock, deliberately; they had been running short.

WBR / design: safety stock \$ vs plan as output; lanes with open drift flags as input.

FR-2. Completed a project with limited resources. Approach? Impact? Differently?

S13. *Situation.* Two go-lives, one engineering team, no budget for a second. *Task.* Make the same hours cover more of the clock. *Action.* Two-pass war room across time zones, one RAID log, one daily, written night decision rights. *Result.* Both on date, ~\$1M per DC protected, no 24-hour days. Differently: write the night decision rights before the first night — I wrote them after the first escalation.

Math. Same headcount $\times$ 2 passes → 16–20 productive hours/day ; cost of delay $\approx$ \$200K/month/DC.

Probe "Why not ask for more people?" — New people on a compressed go-live are a cost for the first month; the structure was faster. **Probe** "What did the time zone give you?" — Sixteen to twenty working hours a day on eight-hour shifts.

WBR / design: issue age by pass; hours per person held to shift.

FR-3. Not enough resources for something important; found a creative way. Options? How did you decide? Outcome?

S8. *Situation.* No budget for a BI team; the Monday review depended on 30 hours a week of hand assembly across two WBRs, and me as the single point of failure. *Task.* Get it built without a budget line. *Action.* Options: keep stitching (SPOF), buy a tool (budget I did not have), or get the data-infra team to build the pipeline in the platform they already ran. I decided by pricing the status quo in their rates — \$129,600/yr — so the build was their saving, not my ask; the

BRD asked for a daily or hourly feed and an alert on failure, nothing exotic. *Result.* Snowflake pipeline with Looker SQL Runner on top; 60 → 90% accuracy; \$103,680/yr saved; zero manual assembly.

Math. \$129,600 → \$25,920 ; saved \$103,680/yr ; build cost ≈ [N] eng-weeks – payback inside a quarter.

Probe "How did you get engineering time?" — By competing on their number, with a frozen scope so the build was bounded. **Probe** "What did it cost?" — Two weeks of a security review I had not planned.

WBR / design: the pipeline itself; status-quo pricing as the business-case template.

FR-4. Trade-offs between quality and cost. How did you weigh them? Result? Differently?

S7. Situation. Partner-first was cheaper and faster; operator-first was "complete." *Task.* Weigh quality against cost in one currency. *Action.* Cost-to-serve as the currency, with the customer-side cost footnoted rather than priced. Partner-first won. *Result.* ~66% effort, ~\$5M avoided, value in six months; operators waited a year for screens — a date I protected. Differently: price the customer-side cost — speed, breach risk — as a third row, not a footnote.

Math. CTS cut ~50% vs ~25% ; months 6 vs 18 ; missing row: Σ operator hours lost × loaded rate + breach exposure.

Probe "What quality did you give up?" — Operator experience for four quarters; I protected a dated milestone so it was a delay, not a cut. **Probe** "Would partner-first have won with the third row?" — Yes, by less; the ratio was robust.

WBR / design: CTS vs plan and operator milestone on the same page so the trade stays visible.

FR-5. A creative solution without additional resources. Problem? Solution? How did you come up with it? Outcome?

S16 (or S2). Situation. A thousand routes, a routing guide people ignored, no budget to police it. *Task.* Solve it without resources. *Action.* Make overrides legitimate with a reason code — the skeptics were often right about specific lanes — and turn the codes into the lane-fix backlog. Came from asking why people overrode instead of forbidding it. *Result.* Completeness 60 → 95%; every cohort at ≥ 95% valid ASNs; zero enforcement headcount.

Math. Override reason-code coverage = 100% ; top codes → backlog ; residual 5% edge formats documented with owner (cost to fix > visibility gained).

Probe "Didn't that legitimize ignoring the guide?" — It legitimized telling us why; the guide got better because of it.

Probe "What was the top reason code?" — Lane-specific carrier capacity the guide did not know about.

WBR / design: override reason-code mix as the input that feeds the routing-guide backlog.

FR-6. Requested additional funding or budget. Why needed? Did you try another approach? Did you get it?

S1. Situation. I asked to fund a fully staffed manual pick path for three weeks through the robotics ramp — labor the plan had assumed we would shrink. *Task.* Justify spending more to risk less. *Action.* Alternative considered: delay the bots past peak — cost calendar, learned nothing. I priced the insurance against one week of a six-point accuracy miss at peak; an order of magnitude apart. *Result.* Funded. It paid for itself the day 94% printed and the manual path carried 70% of volume with zero incidents.

Math. Insurance = 3 wk × [incremental manual hrs/wk] × [rate] ; miss = [peak units/wk] × 6% × [cost per mis-ship] ≫ insurance.

Probe "What if you had not gotten it?" — Bots at full volume or bots after peak; I would have chosen after peak and said why. **Probe** "Did you use all three weeks?" — Just under; full volume restored in week three.

WBR / design: manual-path hours as a dated input with a sunset; bot share as the lever.

FR-7. (Manager) Decide whether to award or ask for additional resources. Criteria? Outcome? Differently?

MAIN. Situation. Data Services and support wanted to hire ahead of the 68-customer curve. *Task.* Decide whether to add people or fix the data layer. *Action.* Criteria: does the next wave's measured ticket rate justify it, and is there a cheaper fix in automation? I held hiring until each wave's scorecard showed tickets per active container; the ROI-ranked backlog and a kill switch for bad source classes cut tickets per customer wave over wave; we hired less than planned and let Data Services shrink by attrition. *Result.* CTS \$0.66 → \$0.03; response 72 → 90% at 5× volume. Differently: build the kill switch before wave one.

Math. Hire test: $\Delta \text{tickets_next wave} \times \$/\text{ticket}$ vs eng-weeks to automate the top carrier ; tickets/100 active containers $2.4 \rightarrow 0.6$.

Probe "How did support feel about being told no?" — Skeptical until wave two's ticket rate fell; then relieved they were not hiring into a mess. **Probe** "When do you say yes?" — When the scorecard shows the cost is structural, not a data defect.

WBR / design: tickets per customer by wave as the hiring trigger — a number, not a request.

FR-8. (Manager) Challenged your team to find a more efficient solution. What drove it? How did you help? Challenges? Results?

S4. Situation. The team's first design was a threshold-email alert. **Task.** Challenge it toward something cheaper in the only cost that mattered — planner attention. **Action.** An alert on untrusted data is waste — it trains people to ignore alerts. I specified the reconciliation and the reason line, and ran the false-alarm burn-down forum myself. Challenge: three weeks with nothing visible to planners. **Result.** $70 \rightarrow 85\%$ forecast accuracy, ~\$2M released, adopted into S&OP un-mandated.

Math. Planner cost per false alarm $\approx 20 \text{ min} \times [\text{rate}]$; false-alarm rate $\times$ alerts/week = the waste line the team burned down.

Probe "What did the team push back on?" — The reconciliation looked like scope creep; the drift number at 8–12% ended that. **Probe** "What did they invent?" — The reason-line logic — naming the input whose delta moved coverage most.

WBR / design: the alert flow with recon gate; false-alarm burn-down as the team's weekly metric.

Earn Trust

Listen attentively, speak candidly, treat others respectfully; be vocally self-critical; benchmark against the best. They listen for: a hard truth you told, to whom, and what it cost you.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S11	"Silent corrections are loans against trust." S11 — The \$2.5M discrepancy I refused to fix quietly	under Customer Obsession
S10	"Sixty stakeholders, one weekly pack, metrics frozen so nobody could re-litigate a red by changing the chart." S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback	under Ownership
S9	"I told the steering committee the migration was at 1.8% against a 0.5% tolerance, with the date it would hit it." S9 — Dual-DC WMS migration in eight months, tolerance published before mock one	below
MAIN	"Green and red were a published five-percent rule in the WBR's own instructions sheet. Nobody could argue a colour." MAIN — Shipment Visibility (Flexport)	Section 5

S9 — Dual-DC WMS migration in eight months, tolerance published before mock one (Sephora RP JDA · program-management story)

LPs: Learn and Be Curious · Deliver Results · Ownership · Earn Trust. **AIQB:** Learn and Be Curious 1, 2 · Deliver Results 1, 2 · Ownership 2 · Earn Trust 4.

Problem	Two DCs on a legacy ERP had to move to JDA WMS in eight months. A \$20M ROI pipeline of four programs was gated behind it. Nothing was documented, and I had never run a WMS migration.
Goal	Both DCs live on time and on the \$1.8M budget, with inventory reconciliation within 0.5% at cutover and no disruption to store or dotcom replenishment.
Key metric	Inventory reconciliation miss at each mock cutover vs the 0.5% tolerance; schedule; budget; hypercare exit criteria.
Did it move?	Yes. Mocks: 7% → 1.8% → 0.3% vs 0.5%. On time. ~3% under the \$1.8M budget. \$20M pipeline unblocked. One exit criterion slipped 1.5 weeks — owned, below.

Situation. "Sephora handed me a dual-DC migration from a legacy ERP to JDA WMS on a hard eight-month clock. Four programs worth about twenty million in ROI were gated behind it. Nothing was documented, and I had never run a WMS migration.

Task. Both DCs live on time and on a \$1.8 million budget, with inventory reconciled within half a percent at cutover, and stores and dotcom never noticing.

Action. I did not fake experience. I sat the sponsor for objectives, assumptions and success criteria in week one, and held discovery's written exit until every flow had a documented current state — that was the gate I could run without domain expertise, and it forced the expertise into the room. Then entrance and exit criteria for every phase. The decision I am proudest of: I published the 0.5% inventory reconciliation tolerance, agreed with finance, before mock one — so the first mock would be a measurement, not a verdict. Three mocks: seven percent, then 1.8, then 0.3. The seven went up unvarnished with my name on it. The go/no-go read the scorecard, not the mood.

Result. On time. About three percent under budget. 0.3% against a 0.5% tolerance at cutover. Four-week hypercare with numeric exit criteria. Twenty million of downstream ROI unblocked. One miss: training sign-off, a go-live exit criterion, slipped a week and a half — I had scored the RF-gun conversion as low effort, my own pre-training test caught the upgrade script failing on newer hardware, and COVID night shifts killed the recovery window. I told leadership it was mine. The program still landed on the date."

Supply · Demand · Capacity. **Demand:** daily store and dotcom replenishment that could not stop. **Supply:** inventory accuracy — the reconciliation tolerance is a supply-truth metric. **Capacity:** two DCs' ability to ship through a cutover weekend. **What bound:** the one-way door. Cutover is capacity you take offline on purpose; the mocks were how I bought it back with a number.

What was innovative. Publishing the tolerance before the first mock inverts the usual politics: a 7% first result is data, not failure, and the trend to 0.3% is the story. And running discovery as a gate I could hold without expertise turned my inexperience into a forcing function for documentation.

The math (program-management story — the math is tolerance and trend)

Recon miss (mock k) = $\sum_{\text{locations}} |\text{expected units} - \text{counted units}| \div \sum \text{expected units}$
mock 1: 7.0% (mapping 60% · timing 30% · physical 10%) → burn mapping first
mock 2: 1.8% (timing 70%) → freeze window + cutoff sequencing
mock 3: 0.3% < 0.5% tolerance (finance materiality) → GO
Budget: \$1.8M plan, ~3% under ≈ \$54K ; cost of a month's slip = 4 programs' ROI delay ≈ \$20M ÷ [36 mo] ≈ \$0.55M/mo
Training slip: 1.5 wks on a go-live exit criterion; program date held because it was not on the critical path

Solution design / flow

discovery (written exit: every flow documented) → design → build
→ mock 1 → mock 2 → mock 3 → cutover → hypercare (4 wk, numeric exit)
tolerance 0.5% published BEFORE mock 1 · each mock classified by cause
· go/no-go reads the scorecard

How it translated to a WBR

Outputs: recon miss vs 0.5%; schedule and budget variance. **Inputs:** RAID ageing; exit criteria met per phase; single-point dependencies with a what-if plan. **Red rule:** a single-point dependency in front of an exit criterion without a what-if plan is a red regardless of its size estimate.

SITUATION PROBES

Why important? Four programs and \$20M of ROI were waiting. Every month of slip was a month of four other teams idle.

What was the biggest risk? A cutover that reconciled badly and had to roll back — a week of two DCs shipping from a system of record nobody trusted.

ACTION PROBES

You'd never done this. Why not hand it to a specialist? I used specialists for the WMS. I owned the gates, the tolerance, and the go/no-go — which is the part that needs an owner, not an expert.

How did you learn fast enough? Discovery was my classroom. Holding the exit until every flow was documented meant the experts had to teach the program what the building did. By mock one I could read a reconciliation report.

Why publish the tolerance before mock one? So the first number could not be argued. If 7% arrives and the tolerance is set afterward, the tolerance becomes 7%.

Why 0.5%? Finance's materiality for inventory at the two buildings. I did not pick it; I got them to sign it.

What did you do about the 7%? Read it as a scorecard: classified every miss by cause — mapping, timing, physical — and burned the top class before mock two. 1.8% was mostly timing; 0.3% was noise.

What did you do versus the team? I set the tolerance and criteria, held the gates, made the go/no-go, and owned the slip. The integrator built; ops counted.

RESULTS PROBES

Compared to goal? On time: met. Budget: 3% under. Tolerance: 0.3 vs 0.5. Training exit: 1.5 weeks late. Three of four, and the fourth owned in writing.

Tell me more about the slip — it's a go-live exit criterion you missed. A single-point dependency — RF guns — in front of an exit criterion, scored low effort because it was "just a script." The script failed on newer hardware. The rule since: any single-point dependency in front of an exit criterion gets a what-if plan regardless of estimated size. It has caught two since.

CHALLENGE

Differently? The what-if rule above, from day one.

Why this story for Learn and Be Curious? Because the learning had a mechanism — discovery as a gate — and a measured outcome, not a book list.

At Amazon. Any one-way-door capacity change — a building cutover, a system migration, a network re-architecture — gets its tolerance published before the first rehearsal, and the rehearsals are read as a trend, not a verdict.

The question bank — Earn Trust (9 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

ET-1. Communicated a change in direction people would have concerns with. How did you understand and mitigate concerns? Changes along the way? Did they get comfortable?

S10. Situation. Telling finance they could no longer issue credit ahead of the physical unit, and telling the DC that returns would live in a separate plant. **Task.** Communicate a direction people had reasons to resist. **Action.** Stock-state workshops where each group named the customer harmed by its own definition; finance saw the duplicate-credit exposure, the DC saw returned units shipped to second customers. Changes along the way: my in-transit rule, corrected from lane data when it proved wrong — in the same forum, with my name on the error. **Result.** Signed before configuration; nightly OOB proved it held; they got comfortable because the policy was theirs and the proof was nightly.

Math. Duplicate-credit exposure = $\sum \text{RTV units credited before pick} \times \text{cost}$; OOB nightly $\leq 0.5\%$ as the proof the definition held.

Probe "What resistance did you not expect?" — Dotcom, who lost in-transit units to my conservative rule; they were right. **Probe** "How did you know they were comfortable?" — Nobody re-litigated a red in the weekly pack; metrics were

frozen so they could not.

WBR / design: frozen metrics in the weekly pack; OOB buckets with owners; policy changes through signature, not config.

ET-2. A tough or critical piece of feedback you received. What did you do about it?

S7 (and S4). *Situation.* After the roadmap flip, my manager: "You were right, and you nearly made it impossible for me to agree with you" — I had circulated an early draft of the table to a peer before bringing it to him. *Task.* Take the feedback as a rule, not a bruise. *Action.* I apologized, and the rule since is the principle itself: backbone is the private table first — every disagreement goes to the decision owner before anyone else. At Elementum (S4) planners said the first alerts were noise; the reason line and the burn-down shipped within a week. *Result.* Every roadmap disagreement since has gone to the owner first; the alerts were adopted.

Math. Order of operations: decision owner → pressure test → ratification → wider audience ; never the reverse.

Probe "How did it feel?" — Fair. He was right about the sequence and I had been right about the plan; both were true.

Probe "How do you get feedback now?" — A standing "what would you have done differently" at the end of every steering, including about me.

WBR / design: a "what we got wrong" line in every weekly pack — mine first.

ET-3. Influenced a peer with a differing opinion about a shared goal. What did you do? Outcome?

S12. *Situation.* The QA lead believed four-week releases meant lower quality. Shared goal — ship, safely — different path. *Task.* Influence without arguing. *Action.* I gave her the definition of done and the authority to hold a release, and coached her on writing a gate that says no with a number. Within two cycles she was defending it to engineering. *Result.* Critical defects -50%; four-week cadence; the gate outlived my engagement. Her opinion did not change; her instrument did.

Math. Cadence 6 → 4 wk and criticals/release -50% — the two numbers that made her case and mine the same case.

Probe "What if she had used the gate to block everything?" — Then the exception path with reason codes would have shown it, and we would have fixed the criteria, not her. **Probe** "Why her?" — She knew where it would break.

WBR / design: gate pass rate and exceptions with reason codes as inputs; defects per release as output.

ET-4. Not able to meet a commitment. Obstacles? Impact? Learned?

S9 (training, 1.5 wk) or S8 (security review, 2 wk). *Situation.* S9: the RF-gun conversion failed on newer hardware; COVID night shifts closed the recovery window. *Task.* Miss a commitment honestly. *Action.* I told leadership the day I knew, in the steering pack, with the new date and whose miss it was — mine, I had scored it. *Result.* A go-live exit criterion 1.5 weeks late; stores and dotcom saw nothing because it was off the critical path. Learned: what-if plans for single-point dependencies regardless of size.

Math. Float on the training path ≥ 2 wk → program date unaffected ; rule: dependency + downstream exit criterion → what-if.

Probe "Why did you not catch it earlier?" — I scored it as familiar; my own pre-training test is what caught it, just late.

Probe "How did peers react?" — Better than if they had found it themselves.

WBR / design: single-point dependencies in RAID with what-if status; exit-criteria slippage reported with owner.

ET-5. Your team's goals out of alignment with a team you relied on. How did you work with them? Did you achieve your goals?

S8 and MAIN. *Situation.* I needed the data-infra team to build a pipeline; their goal was platform work, not my Monday review. Sales's goal was boarding customers; mine was cost-to-serve. *Task.* Align without authority. *Action.* For data infra: priced the status quo in their own labor — \$129,600/yr — and froze the seven definitions so their build had a fixed target. For sales: the six-criteria scorecard as the shared instrument, with every hold carrying a re-test date. *Result.* Pipeline built, 60 → 90% accuracy, \$103,680/yr; 68 customers boarded on green scorecards, two months early.

Math. Their saving: \$129,600 - \$25,920 = \$103,680/yr ; sales: waves boarded on green ÷ waves =

100%.

Probe "What did you give them?" — A BRD with frozen definitions, so scope could not creep on them. **Probe** "What did they give you?" — Two weeks of pain on security review, and a pipeline that caught a 4% drop in week one.

WBR / design: status-quo cost in the builder's currency as the business case; frozen definitions as the contract.

ET-6. Uncovered a significant problem in your team. How did you communicate it? How did you address it? Manage the impact?

S5 (program) or S11 (customer). *Situation.* S5: the program I had joined to fix carriers could not measure lateness — a problem in our own data. *Task.* Communicate a problem that embarrassed the plan without embarrassing the people. *Action.* One page, three lanes, privately to the sponsor; then to the team as a scorecard they could move. Addressed: measurement to 99% by class before any routing. *Result.* The engine milestone moved; the team's weekly progress became the scorecard so nobody felt idle; ~\$8M auditable.

Math. True quality 55% at planning grain (reported 40%) → 99% by class ; team metric: % by class, weekly.

Probe "Did anyone get blamed?" — No; the defect classes were about fields, not people. **Probe** "How did you keep the team motivated on cleanup?" — Visible weekly movement by class, and the engine date tied to it.

WBR / design: defect-class scorecard as the team's weekly output during the pivot.

ET-7. Improved morale and productivity on your team. Underlying problems? How did you prevent recurrence?

S13. *Situation.* Same six people on every call, nights included; issues waiting for morning. *Task.* Fix morale by fixing the cause. *Action.* Two-pass war room with written night decision rights and a rule that nobody works both passes, enforced from the RAID log. *Result.* Issue age fell; both go-lives landed; ~\$1M per DC; no 24-hour days. Prevention: the structure is how I run every compressed program.

Math. Night calls per person per week → 0 for the day pass ; issue age p90 down week over week.

Probe "How did you measure morale?" — Hours per person per week in the log, and who showed up on both passes — zero after week one. **Probe** "What did the night pass decide alone?" — Reversible actions; irreversible waited for morning by design.

WBR / design: people on both passes (target 0) and issue age by pass as inputs.

ET-8. A team member struggling to keep up or fit in; you stepped in. Why? What did you do? Impact? Outcome?

S12, reframed. *Situation.* She was not struggling with skill — she was being managed around because she said no. *Task.* Step in without taking over. *Action.* Gave her the gate and coached her to write and defend criteria as numbers. *Result.* More of my time in the first two cycles, far less afterward; she owned the mechanism; defects -50%; neighbors copied it.

Math. My hours on releases: up for 2 cycles, then near zero ; her gate's durability [N] quarters.

Probe "Why did you think she was struggling?" — She had stopped speaking in design reviews; I asked her why over coffee. **Probe** "What did you learn?" — The person being routed around is usually the one who knows where it breaks.

WBR / design: gate ownership named on the release scorecard — her name, not mine.

ET-9. A team member not performing well and impacting your work. How did you handle it? Why? Outcome? Learned?

Flexport analyst (S8 context), matrix. *Situation.* An analyst feeding the Monday review delivered late and inconsistent numbers. *Task.* Separate the system problem from the person problem. *Action.* 60% accuracy was a four-system Excel problem, not his. For the rest: a weekly delivery time, a definition of "done" per metric, private review for three weeks, and the question of what he wanted to own. He did not want the visibility and said so late; he moved to a role that fit. *Result.* The review's accuracy problem was fixed by the pipeline; the person found the right seat. Learned: ask what someone wants in month one.

Math. Attribution: definition/system error vs assembly error, measured per metric-week before judging the person.

Probe "Was it fair to him?" — Fairer once I stopped blaming him for the system. **Probe** "Impact on your work?" — Two extra hours a week for a month; cheaper than a wrong number in front of a VP.

WBR / design: metric-level "done" definitions; delivery time as an input until the pipeline removed the dependency.

Dive Deep

Operate at all levels; stay connected to the details; audit frequently; be skeptical when metrics and anecdote differ. They listen for: the data pull, the grain, and what the detail changed.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S5	"I pull until the unit is a shipment, an order, a lane or a building — not a national average." S5 — Austria: I arrived certain it was carriers; the data said we could not measure lateness	below
MAIN	"Every red decomposed to carrier and ticket reason code before the review, with an owner and a date." MAIN — Shipment Visibility (Flexport)	Section 5
S8	"The first automated catch was a 4% ticket-sync drop the manual era could not see because it was built on top of it." S8 — Four systems into one pipeline: the weekly review is only as good as the data under it	under Invent and Simplify
S10	"The oversell was not a bug. It was five definitions of "available." I traced every one to the stock state that produced it." S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback	under Ownership

S5 — Austria: I arrived certain it was carriers; the data said we could not measure lateness (Elementum)

LPs: Dive Deep · Are Right, A Lot · Earn Trust · Have Backbone. **AIQB:** Are Right, A Lot 1, 6 · Dive Deep 1, 2, 4 · Earn Trust 6 · Have Backbone 2.

Problem	A 25-location European freight program: reports said carriers were late everywhere and freight cost was over plan. The sponsor had committed to a routing engine to move volume to better carriers.
Goal	Auditable freight savings through better routing — the sponsor's number — without building an engine on data that could not support a recommendation.
Key metric	Data quality at the planning grain (reported 40%; true ~55%) → 99% by defect class; freight savings \$ auditable to a shipment.
Did it move?	Yes. Data quality 99% by defect class before any routing ran. ~\$8M freight savings across 25 locations, every dollar auditable to a shipment under a finance-signed method.

Situation. "I walked into a 25-location European freight program in Austria convinced we had a carrier problem. Every report said carriers were late everywhere, freight cost was over plan, and the sponsor had promised a routing engine to move volume to better carriers.

Task. Deliver auditable freight savings. The sponsor wanted the engine first.

Action. Before the engine kickoff I pulled three years of shipment history myself — not an analyst, me. The data said something more embarrassing than late carriers: we could not measure lateness. True data quality at the planning grain was about fifty-five percent, against the forty the reports claimed — wrong in both directions because the reports were measuring different things. Missing rates, lanes keyed at the wrong grain, the same carrier under five spellings, transit times copied from brochures. So I changed the program's first milestone from engine to measurement, in front of a sponsor who had promised the engine, with one page: a routing recommendation on 55% data is a coin flip with a confident interface. We fixed quality by defect class to ninety-nine percent before any routing ran. And I had finance sign the savings method — baseline lane rate versus executed rate, per shipment — before anyone claimed a dollar.

Result. About eight million dollars in freight savings across 25 locations, every dollar traceable to a shipment. Data quality ninety-nine percent by defect class. And a changed mind, on the record: I had been hired to fix carriers, and I told the people who hired me that carriers were not the problem."

Supply · Demand · Capacity. **Supply:** carrier capacity and rates by lane. **Demand:** shipment volume by lane across 25 locations. **Capacity:** the program's capacity to decide — you cannot route what you cannot measure. **What bound:** measurement, not carriers. For a capacity planner the lesson is literal: a utilization or on-time number at the wrong grain is wrong in both directions, and spending a lever on it is a coin flip.

What was innovative. The usual response to "the data is bad" is to clean it on the way into the engine. I made measurement the deliverable, scored by defect class with a 99% bar, and signed the savings method with finance before the first claim — so the result could not be argued later. I also changed my own thesis in public, which is what made the sponsor willing to change theirs.

The math

```
Data quality (planning grain) = records passing ALL defect-class checks ÷ records
  defect classes: rate present · lane key at correct grain · carrier canonical · transit from
                  scan (not brochure)
  reports said 40% (field-present only, counted brochure as data) ; true = 55% ; after = 99% by
  class
Why a routing engine on 55% is a coin flip: recommendation needs rate AND lane AND transit
correct
  if each is ~0.82 complete independently →  $0.82^3 \approx 0.55$  → P(recommendation built on bad input)
  ≈ 45%
Savings method (finance-signed, per shipment):  $\Sigma_{\text{shipments}} (\text{baseline lane rate} - \text{executed rate})$ 
  × units
  25 locations, 3 years of baseline, no lane averages → ≈ $8M, each dollar joined to a shipment
  ID
```

Solution design / flow

```
3 yrs shipment history → profile by defect class → quality scorecard (by class, by location)
  | (my first pull: "we cannot measure lateness")
  ▼
MILESTONE 1 (new): fix measurement → 99% by class
  ↳ MILESTONE 2: routing engine (lane × carrier × mode)
    ↳ savings ledger per shipment (baseline - executed)
```

How it translated to a WBR

Outputs: freight \$ saved vs plan (auditable); on-time by lane (now measurable). **Inputs:** data quality by defect class; % shipments with scan-based transit; carrier canonicalization coverage. **Red rule:** savings claimed on any lane below 99% quality are held out of the ledger until the lane clears.

SITUATION PROBES

Why important? What if nothing happened? An engine on 55% data would have moved volume to carriers that looked good because their data was missing. We would have claimed savings nobody could audit and lost the sponsor's credibility with finance.

What was the initial scope? Routing engine, 25 locations, savings target. I kept the target and the locations and inserted a measurement phase in front.

ACTION PROBES

Why pull it yourself? Because I was the one who was going to tell the sponsor his first milestone was wrong. I needed to have seen the rows.

How can "40% reported" and "55% true" both be wrong? The reports counted a field as present if it had any value; some values were brochure numbers, so the reports overstated quality on those. Elsewhere they counted lanes as incomplete because of spelling variants that were actually resolvable, so they understated. Measured at the planning grain with defect classes, the true number was 55% — and the point is that neither report number could be used for anything.

How did you move a sponsor who had promised the engine? One page, privately first: here is what the engine would recommend on three real lanes using current data, and here is what actually happened on those lanes. Two of three recommendations were wrong. He took the page to his leadership himself.

What did you do versus the team? I pulled the history, defined the defect classes and the 99% bar, re-sequenced the program, and negotiated the savings method with finance. The team built the cleansing by class.

What decision did you challenge? My own first. Then the sponsor's milestone. Then the finance assumption that savings could be claimed on a lane average — we went per shipment.

RESULTS PROBES

How did you compute \$8M? Per shipment: baseline rate for that lane and mode before the program, minus executed rate after routing, summed across 25 locations, under the method finance signed. No lane averages.

Why 99% and not "good enough"? Because the engine's recommendations are only as reliable as the worst defect class. At 99% by class, a wrong recommendation was rare enough to catch by exception. At 90% it was a weekly argument.

Compared to goal? Savings target: met and auditable. The engine shipped later than the sponsor had promised. I owned that date change in writing.

CHALLENGE

Differently? Pull the data in week one, before the kickoff deck, not week four. It would have saved the sponsor an embarrassing re-plan.

Why this story for Are Right, A Lot? Because being right here meant changing my mind in public with data, which is the only version of that principle that is hard.

At Amazon. Before I plan to a utilization number, a design rate, or an on-time metric, I would ask what grain it is measured at and how it was last validated against actuals. A capacity ceiling that was set three years ago and never re-measured is a brochure number.

The question bank — Dive Deep (7 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

DD-1. Understanding a complex problem by digging into details. Who did you talk to, where did you look? How did you use it?

S5. Situation. Austrian freight program; the thesis was carriers. **Task.** Understand the problem at the row level. **Action.** Three years of shipment history pulled by me — not an analyst — plus the planners who keyed lanes and the finance team that would audit savings. The rows showed carriers under five spellings, lanes at the wrong grain, brochure transit times stored as actuals. I used it to flip the first milestone to measurement and to define the defect classes the 99% bar was scored on. **Result.** ~\$8M, auditable per shipment.

Math. Completeness by class ≈ 0.82 each $\rightarrow 0.82^3 \approx 0.55$ overall ; savings ledger = $\sum (\text{baseline} - \text{executed}) \times \text{units}$.

Probe "Why you and not an analyst?" — I was about to tell a sponsor his milestone was wrong; I needed to have seen the rows. **Probe** "What was the most valuable source?" — The field that said where the transit number came from — brochure or scan.

WBR / design: defect-class scorecard; "% transit from scan" as a standing input.

DD-2. Dug deep to a root cause. How did you know you were focusing on the right things? Outcome? Differently?

S2. Situation. Backorders climbing at 4 AM; the noise said capacity. **Task.** Find the root cause while acting on symptoms. **Action.** Stock by node (fine), DC hours (fine), allocation queue (growing with stock sitting still) — the only signal inconsistent with "capacity," which pointed at the engine. The trace confirmed a scheduler death at the DST change. **Result.** 99% fill; heartbeat and DST checklist since. Differently: the heartbeat before peak, not after.

Math. Queue depth $\uparrow$ while stock flat and hours available $\rightarrow$ engine, not capacity ; heartbeat alarm if no scheduler run in 15 min.

Probe "How long to the root cause?" — About an hour for the trace; ten minutes to rule out inventory and hours. **Probe** "What did the post-mortem change?" — Peak readiness checklist with system-level heartbeats, not just physical capacity checks.

WBR / design: "queue depth rising while stock flat" as a pre-wired red; scheduler age as an input.

DD-3. A problem requiring in-depth thought and analysis. How did you know you were focusing on the right things? Outcome?

S3. Situation. Two hundred lanes of transit history. **Task.** Choose the right statistic, not the convenient one. **Action.** Per-lane empirical distributions; the p99 over the mean and over a normal fit (the normal understated the tail by ~4 days on skewed lanes); a finance-signed release rule. I knew I was right because held-out arrivals landed inside the planned lead time 99% of the time, lane by lane. **Result.** 580K units, ~\$1.5M released; drift flag keeps it true. Differently: a measurement audit in week one.

Math. Validation: held-out arrivals $\leq$ p99_lane in 99% of cases ; normal p99 30.3 d vs empirical 34 d.

Probe "Why not a normal distribution?" — Right-skewed transits; normal p99 was ~4 days short on long lanes. **Probe** "Thin lanes?" — Pooled by origin region, mode and carrier, flagged lower confidence.

WBR / design: LT accuracy by lane by week; drift flags open; % parameters from scan data.

DD-4. A big problem in your organization you helped solve. How did you become aware? What was missing and how did you fill gaps? Reflection?

S10. Situation. Oversells and returned units shipped to second customers — and five groups each certain their "available" was right. **Task.** Find what was missing underneath. **Action.** Missing: a single definition of each stock state and a nightly proof that SAP and WMS agreed. I filled it with the signed stock-state model and the ATS/OOB reconciliation — supply basis minus reserves minus shipped-not-billed plus mech reserve and lost-and-found, reconciled to SAP supply, bucketed by cause with an owner. **Result.** Zero non-sellable promises; OOB in tolerance nightly. Reflection: conservatism has a price; my in-transit rule proved it.

Math. $\text{ATS} = 10,000 - 1,200 - 800 - 500 - 300 + 150 + 50 = 7,400$; $\text{OOB} = (10,420 - 120) - (10,000 + 300) = 0$.

Probe "What was the hardest gap?" — Shipped-not-billed: physically gone, still in SAP; treated differently for ATS and for the OOB. **Probe** "What did the reflection change?" — Measured defaults on day one, never "never."

WBR / design: OOB by bucket nightly; oversells as the output; a bucket repeating three nights becomes a defect.

DD-5. A specific metric you used to identify a need for change. Created or existing? How did it influence the change? Outcome?

S8 and MAIN. *Situation.* Nobody had asked how often the VP's Monday numbers were right. *Task.* Create the metric that would answer it. *Action.* Review accuracy — share of metric-weeks where the Monday number matched a recompute from source — 60%. It made the case for one pipeline with frozen definitions and a 1% parallel run. At MAIN, the created metric was tickets per active container: it turned Data Services' workload into a measured automation gap and ranked the engineering backlog. *Result.* 90% accuracy, zero manual hours, a 4% sync drop caught in week one; MAIN: automation ~60 → 97%, CTS ~95%.

Math. Accuracy = metric-weeks within tolerance ÷ metric-weeks ; tickets/active ctr = (S - A) ÷ N, rolled up to CTS and 4PL SLA.

Probe "Why was it 60%?" — Manual joins across four systems with different keys; some weeks the ticket export was truncated. **Probe** "Did anyone dispute the 60%?" — Infra did until they saw the recompute method; then they built the pipeline.

WBR / design: accuracy and source completeness as standing inputs under the review itself.

DD-6. Created a metric that identified a need for change. Why? How did it influence change? Outcome?

S4 (and S3). *Situation.* Planners were being asked to trust alerts built on two SAP views that disagreed. *Task.* Make the disagreement a number. *Action.* Recon drift: $\Sigma|MD04 - MB52| \div \Sigma MB52$, daily, by plant. Drift at 8–12% made the case to build reconciliation before alerts; drift at zero made the alerts credible. *Result.* Forecast accuracy 70 → 85%, ~\$2M released, alerts adopted un-mandated. S3: the per-lane p99 as the planning metric.

Math. Drift_plant,day = $\Sigma|MD04 - MB52| \div \Sigma MB52$; alert gate: drift = green today.

Probe "What caused the drift?" — Stale POs counted as supply; duplicated demand from two feeds; timing of goods movements. **Probe** "Is it still measured?" — Daily; "no alert on a red day" depends on it.

WBR / design: drift as the gating input; alert false-alarm rate as the quality input; stockouts-without-alert as the output.

DD-7. (Manager) Validated the assumptions underlying a report's project. How did you decide follow-up was necessary? Steps? Result?

S5 with the team's analyst, matrix. *Situation.* The routing-engine plan assumed the carrier data was good enough; the analyst's quality report said 40%. *Task.* Validate an assumption two documents disagreed on. *Action.* 40% and "good enough" could not both be true, so I pulled the rows myself and re-scored at the planning grain by defect class — true 55%, wrong in both directions (the report counted brochure values as data and missed grain errors). *Result.* The assumption was replaced with a measured scorecard; the program re-sequenced; the analyst's method became the standing one.

Math. Reported 40% (field-present only) vs true 55% (present AND correct grain AND canonical AND from scan) → neither supported building.

Probe "How did the analyst take it?" — Well; the defect classes became their scorecard and their work was visible weekly. **Probe** "What tells you to validate?" — A number nobody can say how it was computed.

WBR / design: every metric on page one has a computation note; metrics without one are flagged.

Have Backbone; Disagree and Commit

Respectfully challenge decisions when you disagree, even when uncomfortable; once decided, commit wholly. They listen for: who you disagreed with, how, what you conceded, and whether you committed.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S7	"Backbone is the private table. Theater is the public ambush." S7 — Flipping a socialized eighteen-month roadmap with a two-column cost-to-serve table	below
S11	"I disagreed with my manager about disclosure in the morning and committed to his repair sequence by noon." S11 — The \$2.5M discrepancy I refused to fix quietly	under Customer Obsession
MAIN	"I took "buy the Project44 data" off the table with a value-per-engineer-week stack, in writing — then committed to the lanes the E-Team picked." MAIN — Shipment Visibility (Flexport)	Section 5

S7 — Flipping a socialized eighteen-month roadmap with a two-column cost-to-serve table (Flexport)

LPs: Have Backbone; Disagree and Commit · Think Big · Frugality · Earn Trust. **AIQB:** Have Backbone 1, 2, 6, 8 · Think Big 2, 3 · Frugality 4 · Earn Trust 3.

Problem An inherited eighteen-month, operator-first launch sequence, already socialized with leadership. Engineering capacity was fixed; the sequence decided when value arrived and what it cost to serve.

Goal Value to customers sooner, at lower total cost-to-serve, with the same engineering headcount.

Key metric Engineering effort to first value; time to value (months); cost-to-serve reduction (%); cost avoided (\$).

Did it move? **Yes.** Engineering effort -66%. Time to value 6 months vs 18. Cost-to-serve cut ~50% vs ~25% under the original sequence. ~\$5M cost avoidance.

Situation. "I inherited an eighteen-month, operator-first launch plan that leadership had already seen and nodded at. Engineering capacity was fixed. The only real variable was sequence: which half of the problem we solved first.

Task. Get value to customers sooner at lower cost-to-serve, without new headcount — and do it without a public fight that would make my manager choose between me and the plan he had presented.

Action. I costed both sequences in the same currency — cost-to-serve, the number the business was actually paid on — rather than in features. Partner-first cut total cost-to-serve by about fifty percent and shipped value in six months; operator-first cut about twenty-five percent and shipped in eighteen. Two columns, same rows, same assumptions, every assumption footnoted. I did not flip it in a meeting. I took the table to my manager privately and asked him to break it. He pressure-tested it with engineering for two days; it held; and the Bellevue onsite ratified the new sequence. Then I committed completely — the operator work I had deprioritized, I personally protected a date for so the operators were not the losers.

Result. Sixty-six percent less engineering effort to first value. Value in six months instead of eighteen. About five million dollars of cost avoidance. I did not win the argument with seniority. I won it with a table — and the table is why it stayed won."

Supply · Demand · Capacity. **Capacity:** engineering hours — fixed, and the scarce resource. **Demand:** two customer populations, partners and operators, both wanting the platform. **Supply:** the sequence in which fixed capacity was allocated to demand. **What bound:** capacity allocation. The flip was a capacity-planning decision in a different costume: same hours, re-sequenced against value per hour.

What was innovative. Roadmaps are usually argued in features and dates. I converted the argument into the business's own P&L currency, made the two options structurally identical so only the sequence differed, and chose the private route so the decision-maker could change his mind without losing face. Backbone is the private table; theater is the public ambush.

The math

Two columns, same rows (cost-to-serve over the planning horizon):

	operator-first	partner-first	
months to first value	18	6	
eng effort to first value	[E]	[0.34 E]	→ -66%
CTS reduction at horizon	~25%	~50%	
cost avoided vs baseline	—	≈ \$5M	$= \sum \text{months} (\text{CTS}_{\text{baseline}} - \text{CTS}_{\text{partner-first}}) \times \text{volume}$
			$+ (E - 0.34 E) \times \text{loaded eng rate}$

Sensitivity: ratio held under engineering's two-day pressure test (effort ±20%, volume ±15%)

Solution design / flow

inherited: operators → screens → (18 mo) → partners

flipped: partners → partner path (6 mo, removes operator interactions at the source)
→ operators (protected date)

decision path: table → manager (private) → engineering pressure test (2 days)
→ Bellevue onsite ratifies → commit

How it translated to a WBR

Outputs: CTS vs plan; monthly operator interactions eliminated. **Inputs:** partner adoption by wave; engineering hours vs plan; operator-date milestone. **Red rule:** if partner adoption trails plan two months running, the table is re-run with actuals and re-presented — the decision stays data-owned.

SITUATION PROBES

Why important? Eighteen months of fixed engineering capacity is the most expensive thing the business owned. Spending it in the wrong order was a cost-to-serve decision nobody had priced.

Cost of nothing? Twelve extra months before customers saw value, and a cost-to-serve reduction half the size. The plan was not wrong; it was unpriced.

How socialized was it? Presented to leadership, dates in the planning system, teams staffed to it. Flipping it meant my manager going back to the people he had presented to.

ACTION PROBES

Why not raise it in the planning meeting? Because a public challenge forces the person who owns the plan to defend it, and then the argument is about him, not the number. Privately, he could attack the table and own the change. He did both.

What if he had said no? I would have asked what number would change his mind, gone to get it, and committed to operator-first in the meantime without relitigating. Disagree once, with data; then commit.

Why not do both in parallel? We did not have the engineers. Parallel would have shipped neither in six months. Sequencing was the whole bet.

How confident were you in the ~50% vs ~25%? Confident in the ratio, less in the absolutes — I said so in the footnotes. Engineering's two-day pressure test moved a few assumptions and the ratio held.

What did you do versus the team? I built the table, chose the currency, chose the private route, and protected the operators' date afterward. Engineering validated the effort estimates.

What decision did you challenge? The plan itself, and the implicit assumption that a socialized plan is a finished plan.

RESULTS PROBES

How did you compute \$5M? Difference in projected cost-to-serve between the two sequences over the planning horizon, plus the engineering effort not spent on the operator-first path to first value. Conservative assumptions, footnoted, reviewed by engineering and finance.

What did the operators lose? Twelve months of waiting for their screens. I owned that trade and protected a date for it. They got a worse year and a better platform.

Compared to goal? Sooner: 6 vs 18. Cheaper: -66% effort, ~50% CTS cut. Same headcount: yes.

CHALLENGE

Differently? Price the customer-side cost explicitly — speed, breach risk — not only cost-to-serve. The table was right; it would have been more right with a third row.

Why this story for Backbone? Because it was a real disagreement with the person who wrote my review, about a plan with his name on it, and I can show the artifact that resolved it.

At Amazon. When Finance says a building is not justified, I do not argue — I price both columns in the same currency: annualized cost of the build against overflow, transfers, premium transportation, speed downgrades and the risk-adjusted cost of a peak miss, with the break-even forecast shown. The six-page narrative this JD asks for is that table with sentences around it.

The question bank — Have Backbone; Disagree and Commit (8 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

HB-1. Strongly disagreed with your manager or peer on something important. How did you handle it? Differently?

S7. Situation. My manager had socialized an eighteen-month operator-first plan; I disagreed on sequence. **Task.** Disagree with the person who wrote my review, about a plan with his name on it, and keep the relationship. **Action.** I built both options in cost-to-serve — identical rows, footnoted assumptions — took the table to him privately and asked him to break it. He pressure-tested it with engineering for two days; the Bellevue onsite ratified partner-first. Then I committed completely and protected a dated milestone for the operators who lost the sequence. **Result.** -66% effort to first value, ~\$5M avoided, value in six months. Differently: nothing on the method; I would have priced the customer-side cost as a row.

Math. Two columns, same rows ; ratio held under $\pm 20\%$ effort / $\pm 15\%$ volume ; operator milestone dated and tracked in the WBR.

Probe "What if he'd said no?" — Ask what number would change his mind, go get it, commit to his plan meanwhile.

Probe "Why not in the meeting?" — Public challenge makes it about him, not the number.

WBR / design: the two-column table — the artifact that makes a disagreement data-owned.

HB-2. An unpopular stance in a meeting with peers and your leader. Why strongly? What did you do? Outcome?

S5. Situation. Kickoff for a routing engine the sponsor had promised; I said the first milestone should be measurement. Unpopular with the sponsor and with the team that wanted to build. **Task.** Hold the stance in the room. **Action.** I felt strongly because a recommendation on 55% data is a coin flip with a confident interface. I had already shown the sponsor privately; in the room I showed the three lanes where the engine would have been wrong. **Result.** Re-sequenced; 99% by class; ~\$8M auditable. The sponsor owned the change because he had seen it first.

Math. $P(\text{bad input}) \approx 45\%$; 3 lanes: engine pick vs truth on current data.

Probe "Did you lose anything?" — Some credibility with the build team for a month, until the scorecard moved. **Probe** "Would you have committed if overruled?" — Yes, with the savings-method signature as my condition so claims stayed auditable.

WBR / design: quality by defect class as a gate; savings ledger per shipment.

HB-3. Committed to a group decision you disagreed with. What led you to commit? Same decision now?

S15. Situation. I argued for a thinner FF App MVP — partner path only, no operator screens for a year. The group kept a minimal operator path in. **Task.** Commit to a decision I had argued against. **Action.** I committed because the success metric — operator interactions eliminated — had been set before scope, and the group's version scored acceptably against it; my disagreement was taste, not data. **Result.** MVP landed April 2022; 48 of 53 import partners migrated in a month. Same decision now: yes; the metric was the right arbiter.

Math. Scope score = interactions eliminated ÷ engineering weeks ; both versions above the line → group's version stands.

Probe "How did you commit visibly?" — I owned the operator-path milestone personally. **Probe** "When would you not commit?" — When the decision contradicts the agreed metric and nobody will change the metric.

WBR / design: success metric set before scope; burn-up tripwire so the compromise did not grow.

HB-4. Felt strongly but the team went another direction. How hard did you press? How did you approach the project afterward?

S10 (in-transit), reversed. Situation. I pressed for "never promise against in-transit"; the team's instinct for a window was right. **Task.** Press hard, then be corrected by the data I had asked for. **Action.** I pressed hard enough to get it signed. When the data showed dotcom starved, I brought the divergence chart to the same policy forum and asked for the window from lane data. **Result.** Same process, measured default; sellable units recovered. The lesson is the one I now press for first: a measured number on day one.

Math. Window_lane = p90 transit from scans ; paired chart sellable-by-channel vs in-transit-by-lane as the standing check.

Probe "Give one where you lost and were right." — S15's operator path: I thought it was waste; it scored acceptably and I committed. The metric decides, not me. **Probe** "How hard is too hard?" — Past the point where the other side has heard the number.

WBR / design: sellable-by-channel vs in-transit-by-lane as the divergence chart that reopens a policy.

HB-5. Pushed back against a decision that negatively impacted your team. Issue? Outcome? Differently?

MAIN (wave three). Situation. Leadership's inclination was to board the onboarding pool on sales's timeline, which would have dumped tickets on Data Services and bad ETAs on sixty customers. **Task.** Push back with something better than an opinion. **Action.** The scorecard: POD-ETA accuracy 76% vs the 80% gate, one source class causing 70% of the misses, fix in a week, kill switch built. Public, with the re-test date. **Result.** Boarded a week later; ticket response rose to 90% at 5× volume. Differently: kill switch before wave one.

Math. Cost of boarding anyway = $\Delta \text{tickets} \times \$/\text{ticket} \times \text{cohort weeks} + \text{ETA misses propagated into customer plans}$; one week of delay $\ll$ that.

Probe "What if leadership had overruled?" — Board with a signed exception naming who absorbs the tickets, and the metric would show it. **Probe** "Did it hurt you?" — With Sales, for a week.

WBR / design: gate scorecard visible to Sales; exceptions signed, not implied.

HB-6. Submitted a great idea your manager did not support. Idea? How did you handle it?

S3. Situation. My idea: replace the planning lead-time parameter itself, not just report on variability. My manager: scope — deliver the report. **Task.** Move him without going around him. **Action.** I built the smallest proof — three lanes, brochure vs measured p99 vs actual late arrivals — and pitched leadership with mockups while delivering the report on time. The proof moved him. **Result.** 200 lanes, 99%, ~\$1.5M released, drift flag.

Math. 3 lanes: brochure 18 d / plan 25 d / p99 34 d / late arrivals under plan = 7% → proof of the unmeasured number.

Probe "Did you go around him?" — No; the leadership pitch had him in the room. **Probe** "What if the proof had failed?" — Then the report was right and I would have shipped it.

WBR / design: three-lane proof → planning parameter → drift flag — proof before scope.

HB-7. Supported a business initiative you didn't agree with. How did you handle it? How did you deliver the message to your team?

S15. Situation. Air and LCL deferred a year, operators kept old screens two quarters — cuts I had argued against in part. **Task.** Support it and deliver the message. **Action.** Once scored against the pre-set metric, I owned the cuts. To the team: the published table — every cut next to its score — and "the metric decided; here is how we re-open it," with the burn-up tripwire as the door. Nobody was asked to pretend the cuts did not hurt. **Result.** MVP on date; cuts re-opened through the door, not around it.

Math. Burn-up tripwire: any added work estimated and swapped for equal size ; absorbed work = 0 by rule.

Probe "How did the operators take it?" — Badly, then better when they saw their date held. **Probe** "Did anyone re-open it?" — Twice; both times the swap rule made the cost visible and the request withdrew.

WBR / design: scope table with scores; burn-up with swaps — the message and the mechanism are the same page.

HB-8. The business gained because you persisted for a length of time. Why determined? How did it turn out?

S10 (and S3). Situation. One definition of "available" signed by five groups, and a returns plant nobody initially wanted. **Task.** Persist for months without a mandate. **Action.** Workshops, failure modes, signatures by a date; the plant designed and defended three times. Determined because every oversell was a cancelled order and every ungraded return was a damaged unit to a second customer. **Result.** Zero non-sellable promises; nightly OOB in tolerance; legacy system off. S3: the drift flag took a quarter of persistence after the release.

Math. Oversells $\rightarrow 0$; OOB $\leq 0.5\%$ nightly ; drift flag live one quarter after release.

Probe "When did you nearly stop?" — When finance asked for a credit exception; the duplicate-credit exposure number ended it. **Probe** "What kept the pressure productive?" — A signature date; silence counted as consent in writing.

WBR / design: stock-state model signed; ATS/OOB nightly — persistence encoded as a mechanism.

Deliver Results

Focus on the key inputs and deliver with the right quality and on time; rise to the occasion despite setbacks. They listen for: the goal as a number, the obstacle, and the number at the end.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S9	"Results with a rollback are not results. The gates are why it counted." S9 — Dual-DC WMS migration in eight months, tolerance published before mock one	under Earn Trust
S10	"343-task runbook, rollback as a number, never fired, legacy system off, zero customer incidents." S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback	under Ownership
S13	"Two go-lives, one engineering team, no 24-hour days. Both dates held." S13 — Two go-lives, one engineering team, no 24-hour days	below
MAIN	"Seven to sixty-eight customers, 66 cents to 3. The metric was published weekly, so the result is a chart, not a claim." MAIN — Shipment Visibility (Flexport)	Section 5

S13 — Two go-lives, one engineering team, no 24-hour days (Sephora)

LPs: Strive to be Earth's Best Employer · Deliver Results · Invent and Simplify. **AIQB:** Earth's Best Employer 3, 10 · Deliver Results 1, 7.

Problem	Two programs compressed onto one engineering team with overlapping go-lives; roughly \$200K per month per DC at stake in delay, and a team that would otherwise work around the clock.
Goal	Both go-lives on date, with issue resolution moving around the clock and no individual working around the clock.
Key metric	Go-live dates met; issue age in the RAID log; hours worked per person per week (held to normal shifts).
Did it move?	Yes. Both landed on date; ~\$1M per DC protected; no 24-hour days.

Situation. "Two programs compressed onto one engineering team, with roughly two hundred thousand dollars a month per DC at stake in any slip. The default was heroics — the same six people on every call, day and night. **Task.** Both go-lives on date without burning the team. **Action.** One RAID log across both programs, so an issue existed once and had one owner. One 45-minute daily forum, not two. And a two-pass war room: a day shift that triaged and assigned, a night shift in another time zone that executed and handed back — so issues moved around the clock without anyone working around the clock. I set the rule publicly: no one on both passes. **Result.** Both landed on date, about a million dollars per DC protected, and the issue-age metric in the log fell because work never waited for morning. Heroics are a process miss."

Supply · Demand · Capacity. **Demand:** two programs' issues. **Capacity:** one team's hours — fixed. **Supply:** the two-pass structure that made the same hours cover twice the clock. **What bound:** hours per day, not people.

The math

Cost of delay	≈ \$200K / month / DC ; single-pass plan projected [~5] months of slip → ≈ \$1M per DC protected
Issue age (RAID)	= hours from open to resolved ; two-pass war room: work moves 16–20 h/day with 8-h shifts → age ↓
Rule	= no person on both passes (log shows who touched what, when) ; night decision rights = reversible actions only

How it translated to a WBR

Outputs: both go-live dates; issue age p50/p90. **Inputs:** issues per pass; people appearing on both passes (target 0); decisions escalated to morning. **Red rule:** anyone on both passes in a week is a red on me, not on them.

How did you enforce "no one on both passes"? The RAID log recorded who touched what and when. Anyone appearing on both passes got a conversation with me, not a thank-you.

How did you measure the \$1M per DC? Monthly cost of delay per DC — roughly \$200K — times the months of slip the original single-pass plan projected. Conservative: it ignores the morale cost of the alternative.

What if the night pass hit something only the day pass could decide? Published escalation: one named person per night with decision rights for anything reversible; anything irreversible waited for morning by design. That boundary was written down before the first night.

At Amazon. Peak readiness is the same shape: a 24-hour bridge staffed in passes with written decision rights, so the 4 AM call has an owner who is awake and a boundary they do not cross alone.

The question bank — Deliver Results (7 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

DR-1. Delivered an important project under a tight deadline. Sacrifices? Impact on the deliverable? Outcome?

S9. Situation. Dual-DC WMS migration in eight months, \$20M of ROI behind it, nothing documented, first time for me.

Task. Deliver on date without cutting the standard. **Action.** I held discovery's exit until every flow was documented — cost early weeks, bought every later one; published the 0.5% tolerance before mock one; ran three mocks; sequenced training late. **Result.** 0.3% recon vs 0.5% tolerance, on time, ~3% under \$1.8M. **Sacrifices:** my comfort and 1.5 weeks of training sign-off, not the standard.

Math. Budget \$1.8M, ~3% under $\approx$ \$54K ; cost of a month's slip $\approx$ $\$20\text{M} \div [36 \text{ mo}] \approx \$0.55\text{M}/\text{mo}$ of ROI delay.

Probe "What did you cut?" — Nothing from the tolerance or the mocks; I compressed hypercare to four weeks with numeric exit criteria. **Probe** "What would you not do again?" — Score a dependency as small because it was familiar.

WBR / design: mock trend 7 → 1.8 → 0.3 against a pre-published tolerance; exit criteria per phase.

DR-2. Significant unanticipated obstacles to a key goal. Successful? Differently?

S1. Situation. 94% pick accuracy in week two of robotics, inside a peak window. **Task.** Clear an obstacle the plan had not anticipated — but the mechanism had. **Action.** Throttle to cleared zones, manual path carries the rest, zone-logic debug, full volume in week three. **Result.** +15%, zero incidents, on date. Differently: one zone at real volume for a week before any customer order.

Math. UPH: $[1.00\text{M} \div 20,000 \text{ h}] = 50 \rightarrow [1.05\text{M} \div 18,260 \text{ h}] = 57.5 = +15\%$; accuracy 94 → $\geq 98\%$ in 1 week.

Probe "Why did testing miss it?" — Test mix under-represented multi-unit orders in dense zones; the miss was in test design. **Probe** "What if the bots had not cleared in week three?" — Manual path through peak; bots after. The plan said so.

WBR / design: zone × hour accuracy as the gate; test-mix coverage as a readiness input next time.

DR-3. Not only met a goal but considerably exceeded it. How? Challenges?

MAIN. Situation. Goal: 68 customers by date; MAU 3×; volume 2×. **Task.** Meet it — and I added cost down and accuracy up. **Action.** Onboarding gated on six published criteria at current load so each wave made the next cheaper; the human-in-the-loop measured as tickets per active container and burned down through an ROI-ranked automation backlog; the WBR run by hand, then automated. **Result.** 68 two months early; volume 5× (vs 2×); cost-to-serve -95%; accuracy 60 → 90; response 72 → 90 at 5× volume. The goal was the number; the exceeding was the direction of the cost curve. Challenge: holding wave three in front of sales.

Math. Volume 2,000 → 10,000/wk ; CTS \$0.66 → \$0.03 ; automation ~60 → 97% ; WBR cost - \$103,680/yr.

Probe "Isn't ~95% mostly denominator?" — Partly; 10× containers at flat cost gives \$0.066. Cost also fell ~55% because tickets per customer fell. **Probe** "What did you hire?" — Less than planned; I hired after each wave proved the next one's cost.

WBR / design: CTS per container and tickets per customer by wave; six criteria as inputs.

DR-4. More than halfway to a goal when you realized it may not be the right goal. What did you do? Outcome?

S5 (and S3). *Situation.* Four weeks into a routing-engine program, the data said the goal — "fix carriers" — was wrong; the problem was measurement. *Task.* Change the goal more than halfway in. *Action.* Re-sequenced: measurement to 99% by class first, engine second, savings method signed with finance before any claim. S3: the goal was a report; halfway through I changed it to a planning parameter. *Result.* ~\$8M auditable; S3 ~\$1.5M at 99%. Differently: measurement audit in week one.

Math. Milestone 1 (new) = 99% by class ; milestone 2 = engine ; ledger per shipment.

Probe "How did you tell the sponsor?" — One page, three lanes, privately. **Probe** "What was the cost of the pivot?" — Engine shipped later than promised; I owned the date in writing.

WBR / design: data-quality scorecard replaced the engine milestone as the first output.

DR-5. A goal you didn't think was achievable. How did you help the team try? Successful? Differently?

S9. *Situation.* Eight months, two DCs, no documentation, no WMS experience on my side — not achievable as handed to me. *Task.* Make it achievable by making it measurable. *Action.* Discovery exit, phase criteria, a tolerance published before mock one so a 7% first result was data, not failure; three mocks; one RAID log. *Result.* 0.3%, on time, under budget. Differently: the what-if rule for single-point dependencies from day one.

Math. Mocks: 7% → ~2% → 0.3% vs 0.5% ; each mock classified by cause before the next.

Probe "What did you tell the team at 7%?" — "This is the scorecard; here are the three cause buckets; we burn the biggest one first." **Probe** "When did you believe it?" — At 1.8%; the trend was the evidence.

WBR / design: recon miss by cause bucket per mock — the trend chart was the team's morale.

DR-6. Did not effectively manage your projects and something was late. Impact? How do you focus on the right deliverables with competing priorities?

S8 (security review). *Situation.* I did not plan the data-security review for ticket content; two weeks late. *Task.* Own it and show how I keep the right things on top. *Action.* Told my manager the day I found it with the new date; the parallel run kept Monday alive on the manual path. How I focus: one RAID log across programs, every item with an owner and date; critical path published; dependencies in front of exit criteria get what-if plans; approval lead times are a kickoff question. *Result.* No business impact; credibility repaired by owning it; pipeline landed at 60 → 90%.

Math. Slip = 2 wk ; business cost $\approx 2 \times 30 \text{ h} \times \$90 = \$5,400$ of manual assembly ; critical-path items with what-if = 100% since.

Probe "Was it really no business impact?" — The pipeline's first catch — a 4% sync drop — arrived two weeks later than it could have. **Probe** "What else did you miss?" — S9's RF guns; same lesson from a different angle.

WBR / design: approvals and their lead times as RAID rows; critical-path items flagged.

DR-7. (Manager) Setting goals that are challenging yet achievable. How did you approach it? Outcome?

MAIN. *Situation.* The 68 and the date came from sales. *Task.* Make the goal hard in a way the team could win. *Action.* I added two conditions — cost-to-serve down, ticket response up — so the goal could not be met by burning Data Services. Each wave's criteria were achievable at the current load and challenging because the load grew; the WBR made progress visible weekly. Matrix team across support, data ops and engineering. *Result.* All three met, two months early.

Math. Goal set: 68 by date AND CTS ↓ AND response ↑ ; gate at current load each wave ; WBR weekly as the progress instrument.

Probe "How did you know 90% data quality was the right bar?" — It was the level below which tickets per customer rose sharply in the pilot. **Probe** "Did anyone think it was impossible?" — Support did, at \$0.66; the first two waves' cost curve changed their mind.

WBR / design: per-wave scorecard — goals set as gates the team can see itself clearing.

Strive to Be Earth's Best Employer

Work to create a safer, more productive, higher-performing, more diverse and more just work environment; lead with empathy. They listen for: what you did for the people doing the work, and what it cost the plan.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
S13	"Heroics are a process miss. I re-cut the plan so nobody worked a 24-hour day." S13 — Two go-lives, one engineering team, no 24-hour days	under Deliver Results
S12	"I gave the skeptic the gate and the credit." S12 — Giving the skeptic the gate	under Hire and Develop the Best
S1	"Zero incidents through peak was a safety number before it was a customer number." S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived	under Insist on the Highest Standards

The question bank — Strive to Be Earth's Best Employer (11 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

BE-1. Helped build a more inclusive working environment. Problems? What did you do? Reactions? How did it help?

S13. *Situation.* The offshore night team executed and the onshore day team decided — the night team was never in the room. *Task.* Make inclusion structural. *Action.* Written decision rights for reversible actions on the night pass; weekly review of their decisions without blame. *Result.* Escalations fell by week three; the night lead started proposing fixes, not just applying them; issue age fell. The people doing the work were making the calls.

Math. Night decisions ÷ night escalations, week over week ; issue age p90 down.

Probe "What problem were you solving — inclusion or speed?" — Both; they were the same problem. **Probe** "How did you make the rights real?" — The log recorded night decisions as decisions, reviewed the same way as day ones.

WBR / design: decisions by pass as a visible input; escalations to morning trending down.

BE-2. Advocated for someone with less influence or seniority. Why? Outcome?

S12. *Situation.* The QA lead was being routed around. *Task.* Advocate for someone with less influence. *Action.* Gave her the load-bearing artifact and put her name on it; told engineering to argue the criteria, not around her. Also MetricStream: offshore leads presenting their own work to the sponsor. *Result.* She owned the gate; defects -50%; neighbors copied her model.

Math. Her name on the scorecard [2] teams adopted ; defects -50%.

Probe "What did it cost you?" — Two cycles of friction with engineering. **Probe** "Why her and not just fix the process?" — The process was her; fixing it without her would have confirmed the exclusion.

WBR / design: owner names on scorecards — the advocacy is structural.

BE-3. Implemented a change to improve the work experience of yourself or coworkers. Goal? Why necessary? Outcome?

S13 (and S8). *Situation.* The same six people on every call, around the clock. *Task.* Change the work experience, not the people. *Action.* Two-pass war room, written night rights, nobody on both passes. S8: removing 30 hours a week of hand assembly from the people feeding the review. *Result.* Both go-lives on date, ~\$1M per DC, zero 24-hour days; S8: \$103,680/yr of people's time returned.

Math. 24-hour days = 0 ; assembly hours 30 → 6 per week.

Probe "How did you enforce it?" — The log showed who touched what and when; anyone on both passes heard from me. **Probe** "Did anyone resist resting?" — Yes; the rule made resting a compliance item, which oddly helped.

WBR / design: hours per person and double-pass count as inputs.

BE-4. Promoted diversity, equity or inclusion. What forms of diversity? Approach?

S1, S13, MetricStream. *Situation.* Teams where the under-heard were the skeptics, the offshore team, the night shift. *Task.* Practice inclusion structurally. *Action.* Skeptics write the criteria (S1); offshore teams get decision rights and the floor with the sponsor (S13, MetricStream). I do not ask people to be inclusive; I put the under-heard person's name on the artifact and their decision in the log. *Result.* Decisions and recognition distributed by contribution, visible in the log. Say plainly this is how you have practiced it; do not claim programs you did not run.

Math. Share of gates owned by non-leads ; share of sponsor-facing presentations by offshore leads: 0 → parity.

Probe "A specific moment?" — The night lead's first unescalated decision being reviewed as a good call in front of the day team. **Probe** "What did you learn?" — Inclusion that is not written into decision rights evaporates under pressure.

WBR / design: decision rights and owners written down — inclusion as a mechanism.

BE-5. Showed compassion during a stressful situation. How did you recognize it mattered? What did you do? Outcomes?

S9 (RF guns). *Situation.* The track lead whose script failed was convinced he had sunk the program; he stopped speaking in the daily. *Task.* Recognize it and act. *Action.* I told leadership the miss was mine — I had scored it — and told him the recovery plan was his to run. *Result.* He ran it; training closed 1.5 weeks late; the program landed. S13: the two-pass rule was compassion with a schedule.

Math. Attribution in the steering pack: owner of the estimate (me), owner of the recovery (him), date.

Probe "Was it really your miss?" — The scoring was; the script was his. Owning the part that was mine let him own his.

Probe "What did he do next?" — Led device readiness on the next DC.

WBR / design: misses attributed to the decision, not the person, in the RAID.

BE-6. Supported or empowered someone to achieve a goal. Your role? Impact on you?

S12. *Situation.* A QA lead with the right instinct and the wrong form. *Task.* Empower rather than direct. *Action.* I gave the authority and coached the form; she did the work. S13's night lead: decision rights and a weekly review turned an executor into a decision-maker. *Result.* Two cycles of more time for me, then much less, and a mechanism that outlived me.

Math. My hours per release cycle ↓ after cycle 2 ; gate durability [N] quarters.

Probe "How did you know when to step back?" — When she defended the gate to engineering without looking at me.

Probe "What did you get?" — A release process that did not need me.

WBR / design: gate owner named; my name absent.

BE-7. Organized or participated in an initiative to foster an enjoyable work environment. Reactions?

S1 and S13. *Situation.* Sixty stakeholders, three vendors, an 836-row issues log. *Task.* Make the grind visible and a little enjoyable. *Action.* Every Friday, rows closed that week were read out with the owner's name. S13: hand-off notes between passes ended with one thing that went well. *Result.* People started closing rows on Thursday; the night team kept the hand-off habit after the program. Say only what you actually did; do not inflate this.

Math. Rows closed Thu–Fri as share of weekly closes ↑ ; small, real.

Probe "Why does it matter?" — Fourteen months is a long program; a weekly moment of visible progress kept the log honest and the people in it. **Probe** "Reaction?" — Vendors asked to be included.

WBR / design: rows closed by owner, read weekly — recognition as part of the cadence.

BE-8. Noticed someone being excluded or treated unfairly. Could you help?

S12 and MetricStream. *Situation.* A QA lead routed around because she said no; offshore leads reviewed by a sponsor who never saw them. *Task.* Help structurally. *Action.* Gave her the gate; had offshore leads present their own work and scored everyone against written criteria before reading names. *Result.* Both helped; both were structural; both outlived me.

Math. Criteria written before names read ; presentations by offshore leads 0 → parity.

Probe "How did you notice?" — She stopped speaking; they were never on the agenda. **Probe** "What did you not do?" — Make it a conversation about fairness; I made it a change to who held what.

WBR / design: criteria before names; owners on artifacts.

BE-9. (Manager) Created an environment where people felt comfortable speaking up with concerns, ideas or mistakes. How do you know it worked?

S9, S10. Situation. Programs where bad news had to travel fast. **Task.** Make speaking up safe by going first. **Action.** Mock one at 7% went up unvarnished with my name on it (S9); my own wrong in-transit rule was corrected through the public policy forum (S10); every weekly pack has a "what we got wrong" line, mine first. **Result.** I know it worked because the RF-gun failure and the dotcom in-transit challenge both came to me early and directly. Matrix.

Math. "What we got wrong" line present in 100% of weekly packs ; time from issue known → reported = same day.

Probe "A time it did not work?" — The Flexport analyst who said late that he did not want the role; I had not asked.

Probe "How do you measure it?" — Time from a miss occurring to it appearing in the log.

WBR / design: "what we got wrong" as a standing pack section; miss-to-log latency.

BE-10. (Manager) Improved your team's work environment. Why needed? How did you decide steps? Impact?

S13. Situation. The RAID log showed six names on most rows and night calls on all of them. **Task.** Improve the environment from the evidence. **Action.** Steps decided from the log: one log, one daily, two passes, written night rights, no double passes. **Result.** Issue age down; both go-lives on date; no 24-hour days.

Math. Names per row ↓ ; night calls per person ↓ ; issue age p90 ↓ — all from the same log.

Probe "What did the team ask for?" — More people; the structure was faster and I said why. **Probe** "What did you measure?" — Hours per person, rows per owner, issue age by pass.

WBR / design: load per owner as a standing input.

BE-11. (Manager) Influenced a change that fostered DEI on your team or org. Indicators? Measures?

MetricStream and S13. Situation. Offshore leads absent from sponsor reviews; night team's decisions recorded as escalations. **Task.** Change it with measures, not slogans. **Action.** Written criteria scored before names; offshore leads presenting their own work; night decision rights in writing with weekly review. **Result.** Decisions and recognition distributed by contribution, visible in the log. Honest framing: my teams' practices, not an org-wide program.

Math. Indicators: sponsor-facing share by location ; decisions vs escalations by pass.

Probe "How did you know the indicators were real?" — The agenda and the log; both were countable. **Probe** "What would you scale?" — Criteria-before-names into every review I influence.

WBR / design: who decides and who presents, counted.

Success and Scale Bring Broad Responsibility

Be humble and thoughtful about the secondary effects of your actions; leave things better than you found them. They listen for: a second-order effect you saw, and who you protected from it.

STORY	HOOK SENTENCE — SAY THIS FIRST, THEN THE STORY	FULL BLOCK
MAIN	"We scaled ten times and ticket response improved. The cost of scale was paid in gates, not in someone else's backlog." MAIN — Shipment Visibility (Flexport)	Section 5
S1	"A gate I published before the robots arrived protected the floor team from a narrative." S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived	under Insist on the Highest Standards
S10	"Returns became first-class stock so a customer's credit was never ahead of the physical unit." S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback	under Ownership

The question bank — Success and Scale Bring Broad Responsibility (9 questions, in the bank's order)

Format: question → **story** and STAR answer → **Math** → **Probes** → **WBR / design**. Where the story is MAIN, the full math and the WBR sheet are in Section 5.

SS-1. A project with impact beyond your immediate customer or organization. How did you approach downstream impact? When did you start thinking about it? Risks?

MAIN. *Situation.* Scaling a visibility platform 10× affects every customer's downstream planners, every carrier whose data we score, and the Data Services team whose work I was shrinking. *Task.* Design for downstream impact before wave one. *Action.* The gates were designed so growth did not dump cost on Data Services or bad data on customers; the public site created no tickets so it could not bury the human loop; Data Services were told on day one that the loop would shrink and were moved into other roles by attrition. *Risk:* a wave boarded on bad data would have propagated wrong ETAs into sixty customers' plans. *Result.* Response 72 → 90% at 5× volume — the proof the downstream was protected; CTS −95%.

Math. Downstream exposure of a bad wave $\approx$ customers $\times$ containers $\times$ P(wrong ETA) $\times$ [planner cost per wrong ETA] ; the 80% gate priced against that.

Probe "Who outside Flexport felt it?" — Carriers, who got source-quality scorecards and in some cases a kill switch.

Probe "What did you not foresee?" — The need for the kill switch itself.

WBR / design: per-source quality and kill-switch activations — the downstream made visible.

SS-2. Changed a decision because of unintended consequences for customers. Impact? Trade-offs? How did the new solution serve better?

S10 (in-transit). *Situation.* Unintended: dotcom customers saw fewer units for sale than physically existed. *Task.* Change my own decision. *Action.* "Never" became a per-lane window from measured transit. Trade-off: a small, priced risk of promising a unit on a late truck. *Result.* Sellable units recovered without re-opening the policy's integrity.

Math. Expected late promises = $\sum \text{lanes units} \times (1 - P(\text{arrive inside window}))$ — small by construction at p90.

Probe "How did you find out?" — Dotcom's lead and two diverging lines. **Probe** "How did you bound the new risk?" — Window set from measured lane transit, same method as the 200 lanes.

WBR / design: in-transit outside window as an input; sellable by channel as the output.

SS-3. Changed course because of negative impact to a third party.

S16 (and S7). *Situation.* The original plan cut off suppliers whose ASNs failed validation — stopping their shipments and their cash. *Task.* Change course for a third party. *Action.* Cohort gates with supplier scorecards and reason codes, so a failing supplier got a specific list, not a cutoff. S7: operators' date protected in a flip that cost them. *Result.* Completeness 60 → 95% without a supplier losing a route.

Math. Suppliers cut off = 0 ; cohorts graduated $\geq 95\% = 100\%$.

Probe "Did it slow you down?" — A little per cohort; faster overall because suppliers fixed their own data. **Probe** "Who was the third party in S7?" — Operators, inside the company but outside the decision.

WBR / design: supplier scorecard by cohort.

SS-4. Thought carefully about social responsibility in decision-making. Why? How did it affect the decision?

S5 (or S10). *Situation.* The routing engine's cheapest answer on some lanes was air expedite. *Task.* Weigh it. *Action.* Measured properly, the data also showed which lanes could consolidate to ground without a service miss; I weighted the engine toward consolidation where service held, so fewer expedited miles was a scored outcome. Say this only to the extent you did it; otherwise S10: the returns plant's grade step means restock before scrap. *Result.* Savings and a lower expedite share, both on the ledger.

Math. Expedite share = expedited shipments ÷ shipments, by lane ; constraint: service $\geq$ target.

Probe "Did it cost savings?" — Marginally on a few lanes; the \$8M held. **Probe** "How did you measure it?" — Expedite share by lane, before and after.

WBR / design: expedite share and restock-vs-scrap ratio as inputs.

SS-5. Considered everyone affected by your work, not just direct customers.

S10. *Situation.* Direct customers: dotcom shoppers and stores. Also affected: finance's auditors, the returns team, vendors, the DC night shift. *Task.* Give each a seat. *Action.* Each had a row in the stock-state workshops and the runbook; the rollback written as a number was for the night shift specifically — so a tired bridge did not carry a judgment call. *Result.* Zero incidents on the cutover weekend; rollback never fired.

Math. Rollback triggers as numbers ; runbook 343 tasks with owner per task.

Probe "Who did you almost miss?" — Vendors on RTV timing; the pick-reserve model changed when they got credit notices. **Probe** "How did you include them?" — A vendor-facing row in the runbook and a communication before the weekend.

WBR / design: stakeholders as rows in the runbook with owners.

SS-6. Balanced growth or progress with environmental or societal impacts. Impacts? Steps?

S1. *Situation.* Consolidating two warehouses into one removed inter-DC transfers and duplicated stock. *Task.* Be honest about scope. *Action.* I did not run it as an environmental program; I kept transfer elimination as a tracked outcome rather than a side effect. *Result.* Fewer truck miles, measured; +15% efficiency. Also S5 expedite share.

Math. Inter-DC transfer legs before vs after ; miles $\times$ legs.

Probe "Did you measure emissions?" — No; I measured transfers and miles, which are the input. **Probe** "What would you do at Amazon?" — Placement decisions already carry a distance term; I would want it on the cost page, not a separate one.

WBR / design: transfer volume and miles as inputs alongside cost.

SS-7. Realized a project would have unintended consequences in the future; and one where you didn't realize until complete.

MAIN (foresaw) and S10 (did not). *Situation.* Two programs, two kinds of consequence. *Task.* Compare them honestly. *Action.* MAIN: scaling without gates would have buried Data Services and shipped wrong ETAs; I built the gates and the no-ticket rule for public tracking before wave one. S10: the in-transit rule's effect on dotcom only showed in production. *Result.* Both now shape how I write defaults: every rule gets a measured number and a divergence chart from day one.

Math. MAIN: gate at current load ; S10: paired chart sellable vs in-transit as the early-warning line.

Probe "What did S10 teach about foresight?" — Conservative defaults are decisions too; price them. **Probe** "What else did you miss on MAIN?" — Kill switch.

WBR / design: divergence charts on every new policy for its first quarter.

SS-8. Left something better than you found it. How did your actions influence things for the better?

S3, S12, S10, MAIN. *Situation.* Four places I have left. *Task.* Leave mechanisms, not dependencies. *Action.* S3: a planning parameter that measures itself and a drift flag planning still uses. S12: a release gate owned by QA that neighbors copied. S10: a stock-state model and a nightly reconciliation that run without me. MAIN: a WBR with frozen definitions and a Snowflake pipeline that does not need me on Monday. *Result.* The test I apply: is the mechanism still firing a year after I left, with my name nowhere on it?

Math. $\text{Durability} = \text{mechanisms in use [N] quarters after roll-off} \div \text{mechanisms built}.$

Probe "How do you know they're still running?" — Former colleagues; the drift flag and the gate were both referenced in later programs. **Probe** "What did you leave worse?" — Two weeks of security-review scar tissue on the infrastructure team; I fixed the checklist.

WBR / design: mechanisms with owners that are not me.

SS-9. (*Manager*) Influenced an organizational change toward social awareness and responsibility. Activities? Outcome?

Honest answer, with S16, S7, S10. *Situation.* I have not run a social-responsibility program. *Task.* Answer the question as asked. *Action.* What I have done is make downstream parties visible in decisions — suppliers as scorecards rather than cutoffs (S16), operators' dates protected in a flip that cost them (S7), vendors and night shifts as rows in a cutover runbook (S10). The activity was structural: a row, an owner, a metric. *Result.* Decisions that counted who they affected. Say that, and stop.

Math. $\text{Parties with a row and an owner in the decision artifact} \div \text{parties affected} \rightarrow 100\%.$

Probe "Is that social responsibility?" — It is the version of it a program manager controls. **Probe** "What would you add at Amazon's scale?" — The downstream row on every capacity decision: delivery stations, DSPs, communities around a new building.

WBR / design: a "who else is affected" row in every decision document.

7A. How Amazon manages capacity — the unit, the flow, the money, the review

This is the depth section. You will not recite it. It exists so that when the interviewer says "walk me through how you would think about capacity in a region," every step, metric and dollar you name is attached to a unit moving through a building. **Three kinds of statements appear here.** (1) Structure that Amazon has described publicly — FC types, inbound and outbound process paths, sort centers and delivery stations, CPT. Use freely. (2) Standard capacity-planning economics — utilization ceilings, fixed-cost dilution, peak factors, labor math. Use freely; this is the job. (3) Numbers. Every number below is **ILLUSTRATIVE**, chosen so the arithmetic can be checked on paper. Say so if you use one, and say what replaces it on day one: the S&OP demand file, Ops Engineering design rates, Finance's cost-per-unit ledger.

7A.0 Supply, demand, capacity — three words, four equations

Say these three words in this order, with these meanings, and the interviewer knows you are speaking the team's language. **Demand** is units customers will order, by week, by region, by building type, by speed tier — the forecast, with its error. **Supply** is units arriving to be stowed — vendor inbound, FBA inbound, transfers, returns — and the inventory already on hand. **Capacity** is what a building can do with those units: **store** them (cube) and **move** them (hours through each process path). Demand and supply are flows; inventory is the stock between them; capacity is the pipe and the tank. The plan is the statement that for every building and every week the four equations below hold.

- (1) INVENTORY BALANCE
$$I(t) = I(t-1) + \text{Supply_in}(t) - \text{Demand_out}(t) - \text{Shrink}(t)$$

what you stow this week minus what you ship is the change in what you hold
- (2) STORAGE CONSTRAINT (Ceiling_storage $\approx$ 0.85–0.90)
$$I(t) \times \text{cube_per_unit}(\text{mix}) \leq \text{UsableCube} \times \text{Ceiling_storage}$$

what you hold must fit — at the ceiling, not at 100%
- (3) INBOUND THROUGHPUT (steps: unload, receive, decant, stow)
$$\text{Supply_in}(t) \leq \min \text{ over steps } s \text{ of } [\text{Hours}_s(t) \times \text{Rate}_s]$$

you can only stow what every inbound step can push in the hours you have
- (4) OUTBOUND THROUGHPUT (steps: pick, rebin, pack, SLAM, ship-sort)
$$\text{Demand_out}(t, \text{CPT}) \leq \min \text{ over steps } s \text{ of } [\text{Hours}_s(\text{window before CPT}) \times \text{Rate}_s]$$

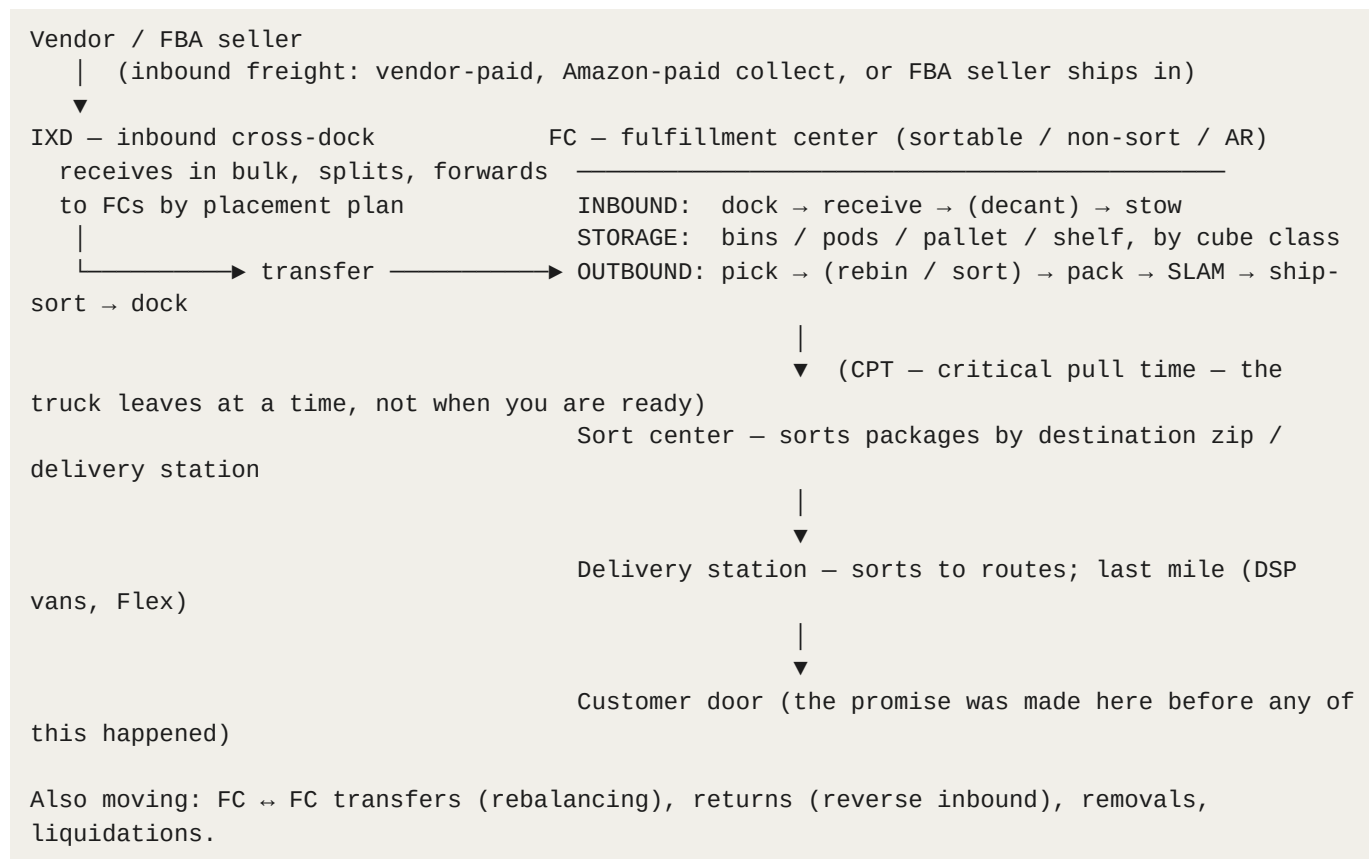
you can only ship what every outbound step can push BEFORE EACH TRUCK LEAVES
- Rate_s is not a constant:
$$\text{Rate}_s = \text{DesignRate}_s \times f(\text{utilization}) \times g(\text{new-hire share}) \times h(\text{flow evenness})$$

f falls past ~85–90% storage utilization (walk, hunt, congestion)
g $\approx$ 0.6 $\rightarrow$ 1.0 over ~2 weeks of ramp; h < 1 when work arrives in bursts

How to read a gap with these. If (2) fails, you have a *storage* gap: lower $I(t)$ (days of cover, slow movers out, inbound pushed back) or raise UsableCube (overflow, retrofit, new build). If (3) fails, *inbound* gap: hours, appointment shaping, IXD absorbs. If (4) fails, *outbound* gap: hours, placement to buildings with slack, CPT structure. If (1) is drifting up week over week with (2) approaching the ceiling, you have a *placement or forecast* problem wearing a capacity costume — supply is being sent to a building whose demand does not justify it. Name the equation that failed before you name a lever. That is the whole discipline.

The one sentence: "Demand tells me how many units must leave each building each week; supply tells me how many must arrive; inventory is the difference; capacity is whether the difference fits in the cube and whether the flows fit in the hours — before each truck leaves."

7A.1 The network one unit travels



The sentence that organizes everything below: capacity is planned in two currencies — **cube** (can the unit be stored) and **hours** (can the unit be moved through each step before its truck leaves) — and the plan is honest only if both are checked at the step where each binds. A building can have empty shelves and still miss CPT because pack is short of hours. A building can have hours and still refuse inbound because it is 97% full.

7A.2 Inbound — step by step, what the unit does, how it is measured, what it costs

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Inbound transportation	Trailer or container travels vendor → IXD/FC. Appointment booked; trailer arrives at a dock door.	Trailers/day vs appointments; on-time arrival; units per trailer (cube fill); dwell in yard (hours).	Dock doors × turns/day; yard spots; appointment slots.	Variable: linehaul \$/mile or \$/trailer, drayage, fuel; semi-fixed: yard jockeys, gate staff. Paid by Amazon on collect freight; embedded in cost of goods on prepaid.
Dock / unload	Pallets or fluid-loaded cartons come off. Cartons scanned to a receive queue.	Cartons unloaded/hour; door dwell (hours door is occupied per trailer); unload backlog (trailers waiting).	Doors; unload labor hours; forklift/equipment.	Variable labor: unload hours × rate. Fixed: doors, levelers, forklifts (capex / lease).

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Receive	Carton opened, ASIN identified, quantity counted, condition checked, unit now exists in inventory as "received, not yet stowed."	Units received/hour per associate (receive rate); receive backlog (units or hours of work waiting); vendor compliance defects (mislabeled, wrong qty) per 1,000 cartons.	Receive hours; problem-solve queue for exceptions.	Variable labor: receive hours. Exceptions cost 5–10× a clean carton (illustrative) — the compliance metric is a cost metric.
Decant / prep	Units pulled from vendor cartons into totes or prepped (poly-bag, label) so they can be stowed.	Units/hour; prep required %; tote fill.	Prep hours; consumables.	Variable: labor + packaging consumables.
Stow	Unit placed into a bin (shelf, pod in AR, pallet location). Inventory becomes available to promise at this scan.	Units stowed/hour (stow rate); dock-to-stow hours (the headline inbound flow metric: carton arrival to unit available); stow backlog; bin-fullness at stow.	Stow hours; empty cube — if bins are full, stow slows or stops; in AR, pod availability and robot drive-unit count.	Variable labor: stow hours. Stow rate falls as the building fills (walk time, hunting for empty bins) — this is the first place the utilization curve costs money.

Why dock-to-stow is the inbound metric that matters. Until the unit is stowed, the customer cannot buy it against this building's promise, the vendor's trailer is still at your door or in your yard, and the carton is occupying floor space that is not a bin. A dock-to-stow target (say ≤ 24 hours, illustrative) converts inbound capacity into a flow commitment. Miss it and you have an inbound backlog, which shows up as refused appointments, which shows up as inventory stuck upstream, which shows up as a speed miss three weeks later.

7A.3 Storage — the cube currency

QUANTITY	HOW TO COMPUTE IT (ILLUSTRATIVE)	WHAT IT TELLS YOU
Gross storage cube	Bins $\times$ cubic feet per bin, by bin type (shelf, pod, pallet, bulk floor).	The building's theoretical ceiling. Never plan to it.
Usable cube	Gross $\times$ bin-fill efficiency (air inside a bin because units are odd shapes; ~60–75% illustrative) $\times$ operating ceiling (85–90%).	The number you actually plan storage to.
Units on hand	Daily outbound units $\times$ days of cover (inventory policy, by ASIN velocity class).	Required storage. Slow movers are the cube problem; fast movers are the hours problem.
Storage utilization	Cube occupied $\div$ usable cube, weekly.	Above the ceiling: stow rate falls, inbound gets refused, pick paths lengthen. This is the capacity-plan red light.
Inventory health	% of cube held by units with $> N$ weeks of cover; aged inventory; unsellable (damaged, quality hold).	Cube you are paying for that cannot ship. Your S/4HANA story: quality-hold counted as available overstates capacity.

Illustrative storage math. A building with 2.0M usable cubic feet at a 90% ceiling has 1.8M plannable. If its mix averages 0.25 cu ft per unit and the inventory policy is 21 days of cover on 120K daily outbound units, it needs $2.52\text{M units} \times 0.25 = 630\text{K cu ft}$ — 35% utilized, lots of room. Push the same building to 20 days of cover on a 300K/day peak outbound and it needs 1.5M cu ft — 83%, at the edge. The same shelves went from comfortable to critical because *days of cover* and *daily outbound* moved, not the building. That is why capacity planning owns the inventory policy conversation, not just the real-estate one.

7A.4 Outbound — step by step

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Order drop / batching	Customer order is assigned to this building (placement + promise decision). Units are batched into pick work by CPT.	Units dropped per hour by CPT window; work-in-queue hours ahead of each CPT.	Planning logic, not labor. Over-drop and pack drowns; under-drop and pick idles.	No direct cost; it decides every downstream cost. Wrong-building placement adds transportation and transfer cost.
Pick	Associate (or robot brings pod to associate) retrieves unit into a tote.	Units picked/hour (pick rate); picks per tote; pick backlog vs CPT; mispicks per 1,000.	Pick hours; in AR, drive-unit and pod availability; pick-station count.	Variable labor: largest single labor line in most buildings. Rate drops as storage fills (path length) and at peak with new hires on a learning curve.
Rebin / sort (multi-unit orders)	Totes from many pickers are consolidated so one order's units land together.	Units sorted/hour; wall utilization; orders waiting on last unit.	Sort-wall slots; hours.	Variable labor; fixed sort walls / sorters.
Pack	Units go into a box or mailer; box size chosen; dunnage added.	Units or packages packed/hour; packages per hour per station; box-size accuracy (cube out — shipping air is a transportation cost).	Pack stations and hours. Pack is frequently the outbound bottleneck at peak.	Variable labor; variable packaging (corrugate, mailers, tape, dunnage — \$/package, mix-dependent).
SLAM	Scan, Label, Apply, Manifest — package weighed, label printed and applied, manifested to a carrier and a truck.	Packages/hour; weight/dimension mismatches; label errors.	SLAM lines (machines) — a fixed-capacity step.	Fixed equipment; small variable labor and labels.
Ship sort / dock	Package routed to the right outbound door for its carrier, sort center, and CPT; loaded into a trailer.	Packages sorted/hour; CPT attainment (% of planned packages on the truck before it leaves — the headline outbound flow metric); trailer cube fill; late trailer departures.	Sorter capacity; outbound doors; trailer availability.	Fixed sorter; variable load labor; variable transportation — a half-full trailer costs the same as a full one.

Why CPT attainment is the outbound metric that matters. A critical pull time is the moment the promise stops being a plan and becomes a truck. A package that misses its CPT either rolls to the next truck (speed downgrade, possibly a missed promise) or goes on a more expensive mode (premium transportation). Every upstream step — pick, pack, SLAM — is measured against hours-ahead-of-CPT, not against the shift. That is what "flow" means inside a building: work arriving at each step early enough that the last step finishes before the door closes.

7A.5 Downstream — sort center, delivery station, last mile

- **Sort center.** Receives trailers from many FCs, sorts packages by destination delivery station or carrier injection point. Capacity in packages/hour through the sorter and outbound doors by destination. Metrics: sort rate, missort rate, trailer utilization, sort-center CPT attainment to delivery stations.
- **Delivery station.** Receives sorted packages overnight, sorts to routes, dispatches to delivery partners. Capacity in packages/day by route count × packages/route. Metrics: packages per route, dispatch on time, delivery success rate.
- **Why NACF capacity planning cares.** FC outbound capacity is only useful if there is downstream sort and delivery capacity on the same lanes on the same night. A region can have FC slack and still miss speed because the delivery stations near the demand are full. Say "FC capacity and downstream capacity have to be planned on the same page" — it is the regional version of "storage and throughput both."

7A.6 A worked example — demand meets capacity in one region (ILLUSTRATIVE)

Three FCs serve a region. Numbers invented so you can check them.

BUILDING TYPE		USABLE STORAGE (M CU FT, AT CEILING)	OUTBOUND DESIGN RATE (K UNITS/WK)	PLANNABLE OUTBOUND AT 90% (K/WK)	AVG WEEK DEMAND PLACED (K/WK)	PEAK WEEK (×1.6)
FC-A	Sortable AR	1.8	1,400	1,260	1,000	1,600
FC-B	Sortable (manual)	1.2	900	810	700	1,120
FC-C	Non-sort	2.5	500	450	300	480
Region		5.5	2,800	2,520	2,000	3,200

Read it. Average week: 2,000 against 2,520 plannable — 79% utilized, fine. Peak week: 3,200 against 2,520 — a **680K-unit gap**, and it is not evenly distributed: FC-A is 340K over, FC-B 310K over, FC-C is 30K over. "The region is 27% short" is the wrong sentence. "Two sortable buildings are each about 35% short in peak week; non-sort is fine" is the right one.

Check the other currency. FC-A peak inventory: $1.6\text{M/wk} \div 7 = 229\text{K units/day} \times 18 \text{ days of cover} \times 0.25 \text{ cu ft} = 1.03\text{M cu ft}$ against 1.8M usable — 57%, storage is not the problem. FC-B: $160\text{K/day} \times 18 \times 0.30 = 0.86\text{M}$ against 1.2M — 72%, getting tight but not binding. So this is a **throughput** gap in the two sortable buildings, concentrated in pick and pack hours. The lever stack is a labor and flow conversation first, a steel conversation never (for this peak).

Lever stack with illustrative costs for the 650K sortable gap, cheapest and most reversible first:

LEVER	CAPACITY IT BUYS	ILLUSTRATIVE COST	TRIGGER AND OWNER
1. Placement shift — move sortable-eligible demand to FC-C's slack and to the next region's sortable buildings	~150K	Extra outbound transportation ~\$0.40/unit on shifted volume (longer zone) → ~\$60K/wk	Fires when week-8 forecast still shows >10% gap. Owner: placement / SCOT partner.
2. Hours — add a shift, extend to 6 days, overtime	~250K (FC-A 150K, FC-B 100K)	Overtime premium 1.5× on ~180K incremental hours-equivalent... illustratively +\$1.2M for the peak period	Labor plan locked at week-10. Owner: FC Ops + regional labor planning.

LEVER	CAPACITY IT BUYS	ILLUSTRATIVE COST	TRIGGER AND OWNER
3. Peak hiring with learning-curve haircut	~200K	New hires at ~60% rate for 2 weeks; training and attrition cost; ~\$2M for the period	Requisitions open week-12; ramp complete week-2. Owner: FC Ops / HR.
4. Inbound shaping — pull fast-mover inbound forward to pre-peak weeks, push slow-mover inbound to post-peak	Frees pick/stow hours in peak week (~50K equivalent)	Carrying cost on earlier inventory (small); vendor negotiation effort	Inbound appointment caps by week published week-10. Owner: inbound planning.
5. Overflow site / 3P	Residual	\$1.50–3.00/unit all-in	Only for what 1–4 do not cover. Contract trigger at week-8.
6. Retrofit / automation	+15–30% building throughput	Capex, 2–4 quarters	Not this peak. Goes into the OP cycle for next year.
7. Demand shaping — speed tier, promise	Last resort	Conversion cost (unpriced trust)	Only when a customer-facing miss is the alternative.

What you say at the end of the example. "The plan closes 650K with placement, hours, hiring and inbound shaping, at roughly three to four million dollars of incremental cost for the peak period, and leaves overflow as a priced option with a trigger date. Every lever has an owner and a week it fires. The two things I will not do: spread a pad across all three buildings, which hides that FC-C is fine; and plan either sortable building above 90% of design rate, because congestion arrives before the ceiling does."

7A.7 The labor model — how hours are computed

Required hours (step, week) = units through the step ÷ planned rate (units/hour) for that step

Planned rate = engineered standard × building-specific factor × learning-curve factor × congestion factor

Headcount (step) = required hours ÷ (hours per shift × shifts per week × attendance %)

Total direct hours (week) = Σ over steps (receive, stow, pick, rebin, pack, ship-sort, dock, problem-solve)

Indirect hours = supervision, training, quality, ICQA (inventory counts), safety, maintenance labor

ILLUSTRATIVE FC-A peak week: 1.6M units

pick	$1.6M \div 120/hr = 13,300 \text{ hrs}$	stow	$1.5M \text{ inbound} \div 110/hr = 13,600 \text{ hrs}$
pack	$1.6M \div 90/hr = 17,800 \text{ hrs}$	receive	$1.5M \div 150/hr = 10,000 \text{ hrs}$
rebin	$0.8M \div 250/hr = 3,200 \text{ hrs}$	ship/dock	$\approx 4,000 \text{ hrs}$

direct $\approx 62,000 \text{ hrs}$; at 36 productive hrs/associate/week and 92% attendance $\rightarrow \sim 1,870$ associates

average week $\approx 1,170 \rightarrow$ peak requires ~ 700 incremental associates, ~ 2 weeks of ramp

Two things the interviewer wants to hear you know: **rates are not constants** — they fall when storage is full, when the building is on new hires, and when flow is uneven (idle pack then flooded pack) — and **pack, not pick, is often the peak bottleneck**, because pack rate per unit is lower and pack stations are fixed.

7A.8 The cost model — every dollar a unit picks up

Think of cost per unit as the sum of what the unit touched, divided by how many units shared the building that week. Fixed costs make the denominator matter; variable costs make the process paths matter; congestion makes both worse at the same time.

COST FAMILY	LINES	FIXED / VARIABLE	HOW IT BEHAVES WITH VOLUME AND UTILIZATION
Facility	Lease or depreciation on the building; property tax; insurance; base utilities; security	Fixed	Per-unit cost falls as volume rises — <i>fixed-cost dilution</i> . This is why under-utilized buildings are expensive per unit and why Finance resists new steel until the existing floor is used.
Equipment	Conveyors, sorters, SLAM lines, AR drive units and pods, racking, forklifts, IT/scanners	Fixed (capex, depreciated) + maintenance (semi-variable)	Same dilution. Automation shifts cost from variable labor to fixed equipment — good at high utilization, bad at low.
Direct labor	Receive, stow, pick, rebin, pack, ship-sort, dock, problem-solve	Variable (hours × wage), with step-ups	Roughly linear until the building is full or on new hires; then rate falls and cost per unit rises. Overtime at 1.5×; peak new-hire premium; shift differentials.
Indirect labor	Area managers, process assistants, learning, ICQA, safety, HR, facilities	Semi-fixed (steps with shifts and headcount bands)	Moves in steps when you add a shift; otherwise flat.
Packaging	Corrugate, mailers, tape, dunnage, labels	Variable per package	Mix-driven. Box-size accuracy also drives outbound transportation (dimensional weight).
Inbound transportation	Linehaul vendor → IXD/FC; IXD → FC transfers; drayage	Variable per trailer / per unit	Driven by placement: a unit stowed far from its vendor and far from its demand pays twice. Trailer cube fill is the efficiency metric.
Outbound transportation	FC → sort center linehaul; sort center → delivery station; last mile; carrier injection	Variable per package, zone and weight dependent	Usually the largest cost family after labor. Driven by placement (distance to customer), speed tier (air vs ground), package cube, and trailer fill. A placement shift to close a capacity gap is paid here.
Transfers and rebalancing	FC ↔ FC moves to fix placement after the fact	Variable	A tax on bad placement or bad forecasting. Track it as its own line so it cannot hide in inbound.
Inventory carrying	Cost of capital on units on hand; shrink; damage; obsolescence/markdown	Variable with inventory level	Rises with days of cover. Inbound shaping (pull-forward) spends this to save labor in peak week — price the trade.
Quality and defects	Mispicks, mis-ships, damages, concessions, returns processing	Variable per defect	Rises when rates are pushed past standard. A building running hot pays here two weeks later.
Overflow / 3P	Short-term space and labor contracts	Variable (all-in per unit) with minimums	Expensive per unit but no fixed commitment — the right lever for a short peak, the wrong one for a structural gap.

Cost per unit, illustrated at three utilizations (same building, one week, invented numbers):

	60% utilized	85% utilized	97% utilized ("full")	
Units shipped	840K	1,190K	1,360K	
Fixed (facility+equipment)	\$1.70M → \$2.02/u	\$1.70M → \$1.43/u	\$1.70M → \$1.25/u	
Direct labor	\$2.0M → \$2.38/u	\$2.7M → \$2.27/u	\$3.6M → \$2.65/u	← rates
fall, OT rises				
Indirect labor	\$0.45M → \$0.54/u	\$0.50M → \$0.42/u	\$0.55M → \$0.40/u	
Packaging	\$0.55/u	\$0.55/u	\$0.55/u	
Outbound transport	\$3.20/u	\$3.20/u	\$3.45/u	← CPT
misses → premium mode				
Defects / concessions	\$0.06/u	\$0.08/u	\$0.18/u	
Total	≈ \$8.75/u	≈ \$7.95/u	≈ \$8.48/u	

Read it. Cost per unit is U-shaped in utilization. Too empty and fixed cost is undiluted; too full and labor, transport and defects spike. The economic optimum sits near the operating ceiling, not at 100%. That is the whole argument for planning to a ceiling, and it is the argument you make to Finance in their own currency: "the last 7% of utilization costs more per unit than the first 85% saved."

Inbound vs outbound cost, said plainly. Inbound cost per unit (freight + receive + stow) is small per unit but front-loaded and placement-sensitive. Outbound cost per unit (pick + pack + packaging + transportation) is larger and speed-sensitive. Placement is the one decision that moves both: put the unit near its demand and you pay more inbound and less outbound; the capacity plan has to know which side of that trade a region is on.

7A.9 The WBR this role would run

Outputs are what the business feels. Inputs are levers a named owner can move this week. Every line has a grain, a target, an owner, and a red that arrives pre-decomposed: which input moved, whose lever, what correction, verified by the same number next week. Illustrative targets.

	METRIC	GRAIN	ILLUSTRATIVE TARGET	OWNER / WHY IT IS ON PAGE ONE
Outputs				
O1	Capacity delivered vs plan (units/week the network could process)	Region × building type × week	≥ 100% of committed plan	You. The headline: did the capacity exist when the plan said it would.
O2	Throughput utilization vs ceiling	Building × week (peak: × day)	75–90% band	You with FC Ops. Below band = cost; above = speed and defects.
O3	Storage utilization vs ceiling	Building × week	≤ 90%	You with inventory planning. The second currency.
O4	Customer speed attainment (promised speed met)	Region × speed tier × week	≥ target by tier	Shared with S&OP / transportation. The customer-facing reason the plan exists.
O5	Cost per unit vs plan (fully loaded, and by family)	Building × week; region × month	≤ plan	Finance partner. The second half of the charter.
O6	CPT attainment	Building × CPT × day	≥ 98%	FC Ops. Outbound flow health; leading indicator of O4.
O7	Dock-to-stow hours	Building × day	≤ 24 h	FC Ops / inbound. Inbound flow health; leading indicator of refused appointments.

	METRIC	GRAIN	ILLUSTRATIVE TARGET	OWNER / WHY IT IS ON PAGE ONE
Inputs				
I1	Forecast accuracy at planning grain (and bias)	Building type × region × week, 4- and 8-week horizons	MAPE within band; bias ≈ 0	S&OP / SCOT. A biased forecast is a capacity error wearing a demand costume.
I2	Placement compliance (units landed in the planned building)	Region × week	≥ 95%	Placement partner. Off-plan placement is the silent capacity thief.
I3	Labor plan attainment (hours staffed ÷ hours required, by step)	Building × step × week	≥ 97%	FC Ops / labor planning.
I4	Rate attainment vs planned rate, by step	Building × step × week	≥ 95% of plan	FC Ops. Where congestion and learning curve show up first.
I5	Launch and retrofit milestones on dated gates	Project × gate	Green/red vs date	Launch / Engineering / Real Estate. New capacity that is late is a gap you planned on.
I6	Inbound appointment adherence and vendor compliance defects	Building × week	≥ 95%; defects ≤ x per 1,000	Inbound planning / vendor management.
I7	Inventory health: days of cover by velocity class; % cube in aged / unsellable	Building × week	Within policy; aged ≤ x%	Inventory planning. Cube you cannot sell is capacity you do not have.
I8	Trailer cube fill, inbound and outbound	Lane × week	≥ target	Transportation. The cheapest transportation dollar.
I9	Transfers and rebalancing volume and cost	Region × week	Trending down	You. The tax on bad placement, kept visible.
I10	Lever status: which planned levers have fired, on which trigger, at what cost	Lever × week	All triggered levers fired on time	You. The plan is a list of triggers; this line proves it is alive.

Page one vs annex. Page one: O1–O7, I1–I10, RAG with week-over-week deltas, five callouts, and the action log with owner and date. Annex: forecast histograms, rate curves by building, placement heatmaps, cost bridges, launch Gantt. **Correction of error** is written when O1 misses plan by more than a published threshold, when O2 or O3 exceeds ceiling for two consecutive weeks, or when a lever that should have fired did not. It ends in a mechanism change — a trigger moved earlier, a ceiling re-measured, an owner named — never in "try harder."

7A.10 How you would model it

Four layers, each feeding the next. Build the simplest version of each first; add fidelity where the plan is actually sensitive.

LAYER 1 – DEMAND	units by week × region × building type × speed tier, from S&OP peak factors by week; a range (P10 / P50 / P90), not a point – the plan is sized to a quantile you choose on purpose
LAYER 2 – PLACEMENT	which building absorbs which demand: minimize inbound + outbound transport + transfer cost, subject to each building's storage and throughput ceilings → units/week per building. The network LP lives here; SCOT owns the engine, you own the constraints and whether the answer is plannable

LAYER 3 – BUILDING	per building per week: required hours by step (7A.7) and required cube (7A.3) vs available; gap by currency by step; rate curves as a function of utilization and new-hire share; CPT structure (how many trucks, when) – flow inside the week, not just the weekly total
LAYER 4 – COST	per building per week: fixed + labor + packaging + transport + carrying + defects (7A.8); cost per unit; incremental cost of each lever; the U-curve
OUTPUTS	gap by week × building × currency; ranked lever stack with cost and lead time; triggers; sensitivity – which forecast error, rate miss or launch slip breaks the plan
VALIDATION	backtest on last year's weeks: did the model's "available" match what the buildings shipped? Where it missed, which layer was wrong – demand, placement, rate, or ceiling?

Three modeling judgments to say out loud. (1) Plan to a quantile, not a point: if you size to P50 you will be short half the time. Decide with Finance which quantile the cost of shortage justifies. (2) Rates are curves, not constants: a model with fixed rates will tell you a 97%-utilized building is fine. It is not. (3) Weekly totals hide CPT misses: a building can hit its weekly number and miss every 6 PM truck. Model at least the daily CPT structure for peak weeks.

What you would ask on day one. What is the current ceiling by building type, who set it, and when was it last re-measured against actuals? Does placement see the throughput ceiling or only storage? Which lever fired last peak that was not in the plan? Where does forecast bias live by region? And: which of these numbers does Finance accept, and which does it dispute?

7A.11 Whiteboard math — the ten computations you must be able to do cold

Each one: the formula, why it is the right formula, a worked number (ILLUSTRATIVE), and what it tells a capacity planner. Do them on paper once before Friday. In the room, say the formula first, then the numbers, then the implication.

1. Peak sizing from a forecast with error — plan to a quantile, not a point

Required capacity at service level α = Forecast_P50 × (1 + z_{α} × CV)
 CV = forecast error ($\sigma \div$ mean) at the planning grain and horizon
 $z_{0.80} \approx 0.84$ $z_{0.90} \approx 1.28$ $z_{0.95} \approx 1.65$

Worked: peak-week P50 = 1.6M units at FC-A; 8-week-ahead CV = 12%
 $P80 = 1.6M \times (1 + 0.84 \times 0.12) = 1.76M$
 $P90 = 1.6M \times (1 + 1.28 \times 0.12) = 1.85M$
 $P95 = 1.6M \times (1 + 1.65 \times 0.12) = 1.92M$

Implication. Sizing to P50 is short half the time. The quantile is a *business decision*: the cost of being short (premium transport, speed misses, overflow at spot rates) vs the cost of being long (undiluted fixed cost, idle hours). Choose it with Finance, write it down, and size every lever to it. If CV is 12% at 8 weeks and 6% at 2 weeks, the plan can hold reversible levers (hours, placement) at P90 and commit irreversible ones (overflow contracts) at P80, then upgrade as the forecast tightens.

2. Storage required and utilization — the cube currency

Inventory units = DailyOutbound × DaysOfCover
 Cube needed = Inventory units × cube_per_unit(mix) ÷ BinFillEfficiency
 Utilization = Cube needed ÷ (GrossCube × Ceiling)

Worked (FC-A, same as 7A.6):
 Inventory = 229K units/day × 18 days = 4.12M units; × 0.25 cu ft = 1.03M cu ft of product

Usable = 2.86M gross bin cu ft × 0.70 bin fill × 0.90 ceiling = 1.80M cu ft
 Utilization = 1.03 ÷ 1.80 = 57%
 Sensitivity: +1 day of cover = 229K × 0.25 = 57K cu ft ≈ 3.2 points; +10 days ≈ 32 points → 89%,
 at the ceiling

Implication. Days of cover is the biggest storage lever and it is an inventory-policy decision, not a real-estate one. Three days of cover is ten points of utilization — the difference between comfortable and congested. The capacity planner must own the days-of-cover conversation with inventory planning.

3. Throughput required vs available — the hours currency

Required hours_s = Units_s ÷ Rate_s
 Available hours_s = Heads_s × ShiftHrs × Shifts/wk × Attendance
 Gap_s = Required - Available ; binding step = largest positive gap

 Worked (FC-A peak week, 1.6M outbound):
 pack: 1.6M ÷ 90/hr = 17,800 hrs needed; staffed 420 heads × 8 h × 5 days × 0.92 = 15,460 hrs
 gap = 2,340 hrs ≈ 210K units at pack → pack binds even though pick (13,300 hrs needed) is
 staffed to 14,000
 Fixes priced in hours:
 +1 shift-day/wk on pack = 420 × 8 × 0.92 = +3,090 hrs at 1.0× wage
 OT on existing heads = 2,340 hrs at 1.5× wage
 move 210K units to FC-C = transport Δ (7A.6: ~\$0.40/unit)

Implication. A building's throughput is its slowest step *at the hour the truck leaves*. Always compute the gap per step; the binding step is where the money goes. Pack binds more often than pick because its rate is lower and stations are fixed.

4. Little's Law — turning a flow metric into a backlog and back

WIP = Throughput × CycleTime ⇔ CycleTime = WIP ÷ Throughput

 Inbound: 215K units/day arriving, dock-to-stow target 24 h
 → WIP allowed on the inbound floor = 215K × 1.0 day = 215K units
 if the floor holds 430K units, dock-to-stow is running 48 h, whatever the dashboard says
 Outbound: a CPT with 180K units due in 6 h needs 30K units/h through pack
 at 90 units/h that is 333 pack heads working that CPT

Implication. You can measure cycle time by counting what is on the floor. Dock-to-stow hours, backlog units and receive rate are one relationship, not three metrics; if two are green and one is red, one of the greens is wrong.

5. Congestion — why cost per unit turns up past the ceiling

Queueing approximation: Wait ∝ $\rho \div (1 - \rho)$, ρ = utilization
 $\rho = 0.70 \rightarrow 2.3$ $0.85 \rightarrow 5.7$ $0.90 \rightarrow 9.0$ $0.95 \rightarrow 19$ $0.97 \rightarrow 32$ (relative wait)

 Storage analog — stow rate falls as empty bins get scarce (illustrative curve):
 Rate(ρ) = DesignRate × $[1 - k \times \max(0, \rho - 0.85)^2 \div 0.0225]$, $k \approx 0.3$
 $\rho = 0.85 \rightarrow 100\%$ of design $0.90 \rightarrow 97\%$ $0.95 \rightarrow 87\%$ $1.00 \rightarrow 70\%$

Implication. Going from 90% to 95% utilization doubles waiting; going to 97% triples it again. That is the arithmetic behind "plan to a ceiling, not to 100%." It is also why the last five points of utilization cost more per unit than they save in fixed-cost dilution (7A.8).

6. Fixed-cost dilution vs variable-cost inflation — the U-curve, derived

CPU(ρ) = Fixed ÷ Units(ρ) + VarLabor(ρ) + Packaging + Transport(ρ) + Defects(ρ)
 Fixed ÷ Units falls like 1/ ρ
 VarLabor, Defects rise past the ceiling (Rate(ρ) falls, queueing grows)
 Transport rises when CPT misses force premium modes

Worked (7A.8 numbers): CPU at 60% = \$8.75 at 85% = \$7.95 at 97% = \$8.48
 60 → 85 dilution: fixed/unit \$2.02 → \$1.43 (-\$0.59); labor/unit \$2.38 → \$2.27 (-\$0.11)
 85 → 97 inflation: fixed/unit \$1.43 → \$1.25 (-\$0.18); labor +\$0.38, transport +\$0.25, defects +\$0.10 (+\$0.73 total) → net +\$0.53 per unit

Implication. The minimum of the curve is the economic ceiling, and it is *below* the physical ceiling. When Finance asks why you will not plan to 100%, this is the table you show: the last 12 points cost 53 cents a unit.

7. Lever cost per unit of capacity — ranking the stack

Lever cost (\$ per unit of gap closed) = Incremental cost ÷ Units of gap it closes
 Rank ascending, subject to lead time ≤ weeks remaining to peak

Worked (7A.6 lever stack, 650K/wk gap over a 4-week peak):

placement shift	\$60K/wk × 4 ÷ (150K × 4) ≈ \$0.40/unit	lead 1-2 wks	reversible
inbound shaping	carrying cost on pulled-forward stock ≈ \$0.30/unit	lead 6-8 wks	
reversible			
OT / extra shift	\$1.2M ÷ (250K × 4) ≈ \$1.20/unit	lead 4-6 wks	reversible
peak hiring	\$2.0M ÷ (200K × 4) ≈ \$2.50/unit	lead 10-12 wks	semi-reversible
overflow / 3P	\$1.50-3.00/unit all-in	lead ~8 wks	contractual
demand shaping	conversion loss — unpriced, so last		

Implication. Flow before hours, hours before hiring, hiring before overflow, overflow before the promise — but the ranking is an output of this arithmetic, not a slogan. If placement shift costs \$0.40 and the next region has no slack, it drops off the list and hours move up.

8. Build vs no-build — the break-even forecast

Build (annualized) = Capex ÷ Life + Lease/fixed opex + Fixed labor ← paid regardless of volume
 No-build (annual) = $\sum_{\text{weeks}} [\text{Overflow\$} + \text{Transfers\$} + \text{PremiumTransport\$} + \text{SpeedDowngrade} \times \text{ConversionCost} + P(\text{peak miss}) \times \text{PeakMiss\$}]$
 Break-even demand D* = the demand at which the two are equal; report it as a forecast quantile

Worked (illustrative): build = \$28M/yr all-in

no-build cost at P50 demand = \$19M/yr; at P75 = \$31M/yr; at P90 = \$44M/yr → D* ≈ P68

sizing to P80: the build pays. sizing to P50: it does not, and overflow is the plan.

Implication. Do not argue "we need the building." Show D*, show where the planning quantile sits, and let the decision fall out. Also show the middle path — retrofit for +20% throughput at a third of the capex — on the same axis.

9. Learning curve and attrition — why peak heads are not peak hours

Effective hours (wk) = $\sum_{\text{cohorts}} \text{Heads}_c \times \text{ShiftHrs} \times \text{Attendance} \times \text{Productivity}_c(\text{weeks since hire})$

Productivity ≈ 0.60 (wk 1) 0.80 (wk 2) 0.95 (wk 3) 1.00 (wk 4+) [illustrative]

Ramp attrition ≈ 10-15% of a cohort → hire ~1.15× the heads you need

Worked: need +700 effective associates in peak week 0

hire 300 in wk -3 and 500 in wk -2 (800 gross)

effective in wk 0 ≈ 300 × 0.95 + 500 × 0.80 = 685 → 15 short, borderline → add 50 to the wk -3 cohort

Implication. A labor plan in heads is wrong by the learning curve. Plan in effective hours by cohort, and the hiring trigger moves earlier than the heads math suggests.

10. Speed — the CPT cliff and what a late unit costs

A unit that misses its CPT by 1 minute loses a full truck cycle — often a calendar day of promise.
Speed loss (units/day) = $\sum_{\text{CPT}} \max(0, \text{Units due at CPT} - \text{Pack capacity in the window before CPT})$
Cost per missed unit $\approx$ premium-transport uplift (if upgraded)
or $P(\text{cancel} \mid \text{late}) \times \text{margin} + \text{trust (unpriced)}$

Worked: 18:00 CPT has 180K due; pack capacity 14:00–18:00 = 4 h $\times$ 36K/h = 144K $\rightarrow$ 36K miss,
daily
60% upgraded to premium at +\$2.50 $\rightarrow$ 21.6K $\times$ \$2.50 = \$54K/day in transport alone, before cancels

Implication. Weekly utilization can be 80% and the 18:00 CPT can miss every day. Capacity must be planned at the CPT window grain for peak weeks. This is Case C in Section 7, in numbers.

7A.12 Lines for this section

- "Capacity is two currencies — cube and hours — and the plan is honest only if both are checked at the step where each binds."
- "Dock-to-stow is inbound flow. CPT attainment is outbound flow. Everything else is a component."
- "Cost per unit is U-shaped in utilization. The optimum is near the ceiling, not at a hundred percent."
- "Placement is the one decision that moves inbound and outbound cost at the same time."
- "A biased forecast is a capacity error wearing a demand costume."
- "Inbound shaping spends carrying cost to buy peak-week labor. Price the trade; don't just do it."
- "Pack, not pick, is usually where peak breaks."

7. The functional block — capacity planning

You will get at least one of these. Format they expect: clarify out loud (60–90 seconds) $\rightarrow$ structure (60 seconds) $\rightarrow$ a math checkpoint $\rightarrow$ recommendation with owners, triggers and risks. Say "both constraints — storage and throughput" early in every answer. Label every number you make up as illustrative, and say what you would replace it with on day one.

Primer: the shape of a capacity plan

Demand side	units by week × building type × speed tier, with a peak factor (forecast owner: S&OP / SCOT; your job is to interrogate its grain and mix)
Required capacity needed	storage: units on hand ≈ daily outbound units × days of cover → cube throughput: units/week each process step must push (inbound, stow, pick, pack, ship) both, by building type; the binding one is the one you plan to
Available capacity before 100%)	buildings × design rate × a utilization ceiling (congestion starts before 100%) minus planned downtime, retrofits, launches still ramping
Gap by week	required – available, by region, by type; where and when it binds
Levers, cheapest and fastest first	
1. flow: move units – placement / allocation to buildings with slack (days to weeks)	
2. hours: shifts, overtime, labor plan (weeks)	
3. inventory policy: pull inbound forward or push it back; trim slow movers (weeks)	
4. overflow: 3P / temporary space (weeks to months)	
5. retrofit / automation: more throughput in the same cube (quarters)	
6. expansion / new build: more cube (OP cycle; years)	
7. demand shaping: speed tier, promise, selection (the lever nobody wants to admit)	
Plan	named owner per lever, trigger that fires it, review cadence, and what you will not do

Case A — "Build next year's capacity plan for a region."

Clarify (say these out loud). Which building types are in scope? Is the demand forecast handed to me or do I own it? What is the peak factor and what week does it peak? What are the policy ceilings on utilization? Which launches and retrofits are already committed? What is the planning grain — week, building, type?

Structure. The primer, top to bottom. Spend the most time on the gap-by-week and the lever ranking.

Math checkpoint (illustrative). "If a building ships 1.0M units a week at average and the peak factor is 1.6, peak week is 1.6M. If its throughput design rate is 1.4M and the ceiling I am allowed to plan to is 90%, usable is 1.26M. The gap is 340K units in peak week. Flow first: how much of that can neighboring buildings absorb inside their own ceilings? Then hours. Overflow only for what is left. I would replace every one of those numbers with the S&OP demand file and the engineering design rates on day one."

Recommendation. A one-page plan: gap by week by region, the lever stack with owners, the triggers that promote a lever (e.g., "if the week-8 forecast is still 5% above plan, overflow contracts sign"), and the two things you will not do — plan to 100% of design rate, or pad every building's forecast to hide a placement problem.

Risk: forecast mix shifts between types; you plan by type, not by total.

CASE A — THE PROBES THEY WILL RUN INSIDE THE CASE, AND THE ANSWERS

"Where does your demand number come from, and how much do you trust it?" From S&OP/SCOT at week × region × building type × speed tier. I trust it as a distribution, not a point: I ask for the error (CV) at the 8-week and 2-week horizons and the bias by region, and I size reversible levers at P90 and irreversible ones at P80 until the forecast tightens (7A.11 #1).

"Which constraint binds first in your example — storage or throughput?" Throughput, in the two sortable buildings, specifically pack hours before the evening CPTs. Storage is at 57% and 72%. I know that because I ran equations (2) and (4) separately per building; the network total would have hidden it.

"Why 90% and not 95%?" Queueing: waiting roughly doubles between 90 and 95 (7A.11 #5), stow and pick rates fall as bins fill, and the cost-per-unit curve turns up past the economic ceiling (7A.11 #6). If the building's own measured rate curve says the knee is at 93%, I plan to 93%. The 90 is a placeholder for a measured number.

"Your placement shift costs transport. How much, and who pays?" Illustratively \$0.40 a unit on shifted volume — longer zone. Transportation's budget pays; the capacity plan's cost line shows it. I would not hide a capacity fix inside someone else's cost center; the WBR line I9 (transfers and rebalancing) exists so this stays visible.

"What if the forecast is wrong by 15% the other way — you over-hired?" The levers are sequenced by reversibility for exactly this: placement and OT unwind in a week; hiring is the first semi-irreversible lever and it fires at week -12 only if the P80 still shows the gap. If demand comes in low, the cost is undiluted fixed cost and some idle hours — I would show that number next to the cost of being short, and that comparison is how we chose the quantile in the first place.

"How do you know the lever actually fired and worked?" WBR line I10: each lever has a trigger date, an owner, and a 'fired / not fired / effect measured' state. Effect is measured in the same currency as the gap — units of pack capacity in the week it was needed — not in 'we added a shift.'

"What would you not do?" Spread a pad across all three buildings — it hides that FC-C is fine and spends everywhere. Plan either sortable building above its ceiling. Pull the promise before the priced levers are exhausted. And I would not present a plan without the quantile written on it.

"How would you present this to Finance?" Two columns, same rows (S7): cost of the plan by lever versus cost of the shortfall it prevents — overflow at spot, premium transport, speed downgrades at conversion cost, risk-adjusted peak miss. Break-even demand marked on the forecast distribution (7A.11 #8). Six pages, the table on page one.

"Walk me through the data pull." Demand: forecast file at week × building type × region × speed tier with error by horizon. Capacity: design rates by process path by building from Ops Engineering; measured rates by week for the last 52 weeks to fit the utilization curve; usable cube and bin fill by building. Labor: heads, attendance, cohort start dates. CPT structure: trucks and due units by hour. Costs: wage rates, OT premium, lease and depreciation by building, transport \$/unit by lane, overflow quotes. Join everything at building × week; keep CPT × day for peak weeks.

If they ask for it as a STAR ("tell me about a time you balanced capacity and demand"): S2 is the lead — demand exceeded the apparent capacity of the network at 4 AM; the task was 99% fill; the actions were promise-at-risk ranking and rebalancing to nodes with hours; the result was 99% held and \$8M protected — and the capacity turned out to be a planning-system failure, which is the lesson. S1 is the backup: consolidation as a capacity decision, throttled against a quality gate.

Case B — "Peak forecast just came in 10% above capacity. Go."

1. **Verify the number before you spend on it.** Grain and mix: is the 10% real demand, or a type-mix shift that makes one building type bind while another has slack? Same cohort weights as last forecast?
2. **Find where it binds.** Storage or throughput? Inbound dock, stow, pick, pack, or outbound? One region or network? "A network 10% is usually three buildings at 25% and the rest fine."
3. **Cheapest lever first, with a trigger.** Placement and allocation shifts this week. Labor plan next. Pull slow-moving inbound back; pull fast-moving inbound forward. Overflow for the residual. Speed tier only if the customer promise is the last protection.
4. **Publish the plan with a date on every lever** and a freeze on non-essential change into peak — the UDC October-to-December freeze.
5. **What you will not do.** Spread a national pad across every building. That hides the cell that is actually breaking and spends cost everywhere.

Story to attach: S2 — rank by promise-at-risk, rebalance across nodes, timestamped log, then a mechanism so you never discover the regime in November again.

CASE B — PROBES AND ANSWERS

"You have one hour. What's the first number you pull?" The gap by building by step, not the network 10%. Equation (4) per building: units due at each CPT vs pack-and-ship hours in the window. Then (2): any building above 85% storage. Ten percent on the network is usually three buildings at 25–35% and the rest fine (7A.6).

"Is it real demand?" Three checks before spending: same cohort weights and promotions as last forecast; mix shift between building types (a 10% rise in non-sort with sortable flat is a different plan); and bias — if this region's 8-week forecast has run 6% high for a year, the 10% is 4%.

"What fires this week?" Placement to buildings with slack, with the transport delta priced (19). OT authorization on the binding step in the binding buildings. Inbound appointment caps on slow movers into those buildings, pulling fast movers forward. All reversible.

"What fires only on a trigger?" Hiring at week –12 if P80 still shows the gap. Overflow contract at week –8 for the residual above what hours and placement close. The promise lever only when the priced levers are exhausted — and I would write the dollar value of a one-day speed downgrade next to it so nobody pulls it casually.

"How do you communicate it?" One page by Friday: gap by building by week by currency, the lever stack with cost per unit closed, triggers with dates, owners, what I will not do, and the number I will report next week to show whether the first levers worked.

Case C — "Utilization looks fine but speed is missing."

Utilization is a weekly average. Speed misses live at the hour, the dock door, the sort type, the cutoff. Change the grain: by building by hour by process step. Check mix: did fast-promise demand land in buildings far from the customer? Check cutoff cliffs: units arriving an hour late lose a calendar day. **Story:** S5 — the reports were wrong in both directions because they measured the wrong grain. **Line to say:** "If I am still talking about network utilization after two follow-ups, I am failing the question."

CASE C — PROBES AND ANSWERS

"Give me the math." 7A.11 #10: a CPT with 180K due and 144K of pack capacity in its window misses 36K units every day while weekly utilization reads 80%. Speed is lost at the CPT window, not the week.

"What else could it be?" Placement — fast-promise demand landing in buildings far from the customer, so the unit ships on time and still arrives late; downstream — sort center or delivery station capacity on that lane; or a dock-to-stow backlog making units unavailable to promise at order time (Little's Law, 7A.11 #4).

"Which one first?" CPT attainment by building by CPT by day (O6) splits it: if CPT attainment is high and speed is low, the problem is placement or downstream; if CPT attainment is low, it is inside the building, and the step with the largest hours gap in the window is the cause.

Case D — "Finance says the new building is not justified."

Do not argue. Price both columns. Cost of the build, annualized. Cost of not having it: overflow contracts, transfers and split shipments, speed downgrades and their conversion cost, labor premiums, the risk-adjusted cost of a peak miss. Show the sensitivity: at what forecast does the build break even? Offer the middle path — retrofit or flow redesign — priced in the same table. **Story:** S7 — the two-column table that flipped an eighteen-month plan; Finance attacked it and it survived. **Line:** "I would rather lose this argument to a better number than win it with a louder voice."

CASE D — PROBES AND ANSWERS

"What's in the no-build column?" Overflow at contract or spot rates for the shortfall weeks; transfers and rebalancing; premium transportation for CPT misses; speed downgrade at conversion cost; labor premium from

running existing buildings past the economic ceiling (7A.11 #6); and a risk-adjusted cost of a peak miss — probability × cost — which I state as an assumption and let Finance change.

"Where's the break-even?" 7A.11 #8: the demand level D^* where annualized build cost equals annual no-build cost, reported as a quantile of the forecast — "the build pays at P68 and above." Then the question is not "is the building justified" but "what quantile are we planning to" — a decision Finance owns with us.

"What if they still say no?" Commit. Then the no-build plan becomes the plan: overflow contracts signed by week -8, the transport and labor premiums budgeted, and the speed exposure written down. I bring the realized no-build cost back to the OP cycle next year as data, not as an argument.

Case E — "Define the metrics for a network capacity program."

Outputs (what the business feels): capacity delivered vs plan by week; utilization vs ceiling by type; cost per unit; customer speed attainment. Inputs (what a named owner can move this week): launch and retrofit milestones on dated gates; placement compliance; forecast accuracy at the planning grain; labor plan attainment; inbound ASN accuracy. Mechanism: a weekly review where reds arrive pre-decomposed — which input moved, whose lever, what correction, verified by the same number next week — and a correction-of-error write-up for any miss above a published threshold that ends in a mechanism change, not "try harder." **Line:** "Inputs without outputs are a hobby. Outputs without inputs are weather."

CASE E — PROBES AND ANSWERS

"Pick the one output you'd be judged on." Capacity delivered vs plan (O1), because it is the promise this team makes to everyone else. Cost per unit (O5) is the constraint on how I deliver it.

"Pick the one input you'd watch hardest." Forecast bias by region (I1), because a biased forecast makes every other metric lie in the same direction, and placement compliance (I2) a close second, because off-plan placement is the silent thief.

"How do you stop the WBR from becoming a status meeting?" Reds arrive pre-decomposed — which input moved, whose lever, what correction, verified by the same number next week. Definitions frozen (S8). A correction of error above a published threshold that ends in a mechanism change. And a line (I10) that shows whether triggered levers actually fired.

"What's the full list?" 7A.9: seven outputs, ten inputs, each with grain, target and owner.

Fermi, if they throw one

- State assumptions. Compute out loud. Label it illustrative. Say what you would replace it with.
- *"How many units does a large sortable FC ship in a week?"* Structure: shifts × hours × headcount × units per hour, cross-checked against outbound dock capacity. Do not quote an Amazon number you have not seen. "I would take the design rate from Ops Engineering on day one."
- *"How much storage for a week of demand?"* Daily outbound × days of cover ÷ units per cubic foot ÷ bin utilization. Say the structure, pick round numbers, flag the two you are least sure of.
- *"Cost of a 4-point speed miss at peak?"* Volume × delta × cost per late unit, plus the unpriced trust cost. Name the grain. Never say "huge."

Lines that read as judgment, not worship

- "I plan to a ceiling, not to design rate. Congestion starts before the building is full."
- "A network average hides the three buildings that are actually breaking."
- "Flow is the cheapest capacity in the building. Steel is the most expensive. I spend them in that order."
- "The forecast is an input I interrogate, not a fact I plan to."
- "Every lever gets a trigger and an owner before peak, or it is a wish."

8. The hard questions

"You have never built an Amazon FC capacity plan. Why you?"

"Correct, and I will not pretend otherwise. An internal transfer brings Amazon scar tissue. I bring the three things a capacity plan is made of, each proven: allocating demand across a network when the plan did not fit — 99% fill through peak; consolidating buildings and running a launch behind gates — plus fifteen percent, zero incidents; and turning measured data into planning parameters — two hundred lanes, a million and a half released. I also bring the review grammar this team already runs — input levers, output metrics, pre-wired reds. Days one to thirty I learn your numbers, your building types, and the five decisions last peak that nobody wants to talk about. I will not arrive trying to rebuild it. I will arrive with the mechanism."

"This role is in Bellevue."

Decide before Friday and say it in one sentence. Do not improvise a relocation answer in the room. [Your line here.]

"Tell me about a failure."

Pick one, own it in the first sentence, name the mechanism it built, stop. Three ready:

- **UAT that crashed under load (Elementum).** Five users, ten scenarios; I sent screenshots instead of digging. Concurrent load crashed it; sign-off missed; sponsor escalation. The implicit requirement — performance at the customer's shape — was not in the entrance gate. Every critical UAT since has numeric load criteria; Flexport's launch proof was 500 containers in 20 minutes at 99.6% connectivity before a customer touched it. "I was the mechanism that failed; I wrote the mechanism that replaced me."
- **RF devices, 1.5-week training slip (Sephora JDA).** In S9. Single-point dependency in front of an exit criterion now gets a what-if plan regardless of size.
- **Security review, two weeks (Flexport Snowflake).** In S8. "Whose approval does this data need, and what is its lead time" is now a kickoff question.

"Tell me about a time you had to influence without authority, two levels up."

S7. Then one sentence: "I set the instrument, not the person — a named line, a numeric target, a published grain, a date. Next Monday's number is the appeal."

"How do you set goals for people who don't report to you?"

"I set the instrument, not the person. A named input line, a numeric target, a published grain, and a date the number gets checked. On UDC the issues log made it unemotional: an owner's row either moved or it did not. I have never won a goal argument with a title; I have won them with a line on the page everyone reads."

"What would you do in your first 90 days?"

"Thirty days: learn the plan as it actually exists — the demand file, the design rates, the ceilings, which lever fired last peak and why. Sixty: take over the weekly review's output block and pre-wire the reds. Ninety: one output metric moved by a mechanism the review can audit — a trigger that fires a lever before the week it is needed, or a placement change that bought capacity without steel. Not a new dashboard."

"Are you an L6 or an L7?"

If asked, do not oversell. "L6 is running the plan and the review well, independently, on complex problems with real trade-offs. That is what I have done. I have also flipped a socialized roadmap with a table, published

tolerances that stopped go-lives, and throttled my own launch. I have not held a large direct-report org, and I will not pretend that is my differentiator."

9. Your questions — pick two

1. "When the capacity plan was wrong last peak, how long did it take to know which input was wrong — the forecast, the design rate, or the placement?" (*Shows you think in decomposition. Whatever they name, you have the matching story.*)
2. "Where does the storage-versus-throughput call actually get made today — in the model, in a planning table, or in a negotiation between S&OP and Ops?" (*The answer is almost always "it's complicated," which is your opening.*)
3. "What does PPT own relative to this team, and where do the two of you most often disagree?"
4. "Which lever is the team least willing to pull — flow, hours, overflow, or the promise — and why?"
5. "What does a good first ninety days look like to you, in one output metric?"

10. Close — fifteen seconds

"I want this work. I have not built an Amazon capacity plan, and I have said so. I have allocated a network through peak, consolidated buildings behind gates, and turned measured data into the parameters a plan runs on. In ninety days I will move one output with a mechanism your review can audit. If that is the job, I am here for it."

11. Numbers worth memorizing

All on your resume or in your war book. If you cannot say what sits under one, leave it out.

NUMBER	WHAT IT IS
3 mega-DCs, ~100 stores, 99% fill, ~\$8M, ~20% stockouts	Manhattan allocation and promising through peak; Cyber Monday 4 AM
200 lanes, p99, 99% LT accuracy, 580K units, ~\$1.5M	Elementum lead-time distributions replacing brochure numbers
70% → 85%, ~\$2M, zero recon drift	Elementum forecast accuracy and safety stock; MD04/MB52 daily recon
40% / 55% / 99%, ~\$8M, 25 locations	Austria: reported vs true data quality; freight savings
~1,000 routes, ≥95% valid ASN, 60% → 95%	EDI onboarding gate; shipment completeness
7 → 68 MAC, 5× volume (~2,000 → ~10,000/wk), 60% → 90% system accuracy, \$0.66 → \$0.03/container, automation rate ~60% → 97%, 17-line WBR, wave 3 held at 76% vs 80%	MAIN — Flexport Shipment Visibility: human-powered track-and-trace to automated product; the WBR the E-Team asked for; six graduation criteria
$\$0.66 \approx 800 \text{ tickets} \times \$1.50 \div 2,000 + \$0.06 \text{ platform}; \$0.03 \approx 300 \times \$0.80 \div 10,000 + \$0.015$	Cost-to-serve decomposition (illustrative inputs, real endpoints)
~50% vs ~25%, 66%, \$5M, 6 vs 18 months	Roadmap flip on cost-to-serve

NUMBER	WHAT IT IS
4 → 1, 60% → 90%, 30 h/wk → 6, \$129,600 → \$25,920/yr (\$103,680 saved), 1% parallel run, 4% first catch	Snowflake WBR pipeline (BRD TSR-852: Zendesk → Snowflake on container number, 7 frozen metrics, Looker SQL Runner)
8 months, 7% → 1.8% → 0.3% vs 0.5%, ~3% under \$1.8M, \$20M	Dual-DC JDA migration; mocks; pipeline unblocked
15+ tracks, 60+ stakeholders, 836 rows, Nov 22 / Dec 20 / Jan 30	UDC omni DC program; test gates; Oct–Dec freeze
343 tasks; sync < 99% × 4 h or p95 > 2× × 1 h; never fired; MCS off; 0 incidents	S/4HANA ↔ ISM cutover runbook and rollback triggers; legacy inventory system retired
Returns plant; RTV STOs as pick-reserve; receive → grade → restock / scrap / RTV	S/4HANA stock-state model: nothing promiseable until disposition; credit never ahead of the physical unit
94% vs 98%, ~30%, week 3, +15%, zero incidents	Locus robots throttled; processing efficiency
6-week → 4-week in 6 weeks, defects –50%	Agile conversion; QA owned the gate
~\$2.5M, a couple hundred orders	OMS–WMS discrepancy disclosed, not hidden
~\$200K/month per DC, ~\$1M per DC	Two compressed go-lives on one team
\$25 → \$0.40, 98%, 100% citations	LLM triage agent (side build)
11 partners, 10 countries, 48 of 53 in a month	FF App export MVP, April 2022

Leave "\$27M+ cost savings" off the table unless you can show the addition live.

12. Traps

Do not quote an Amazon internal number. You do not know their design rates, ceilings or peak factors. Label every number illustrative and say what replaces it on day one.

Do not say "we" for a decision. "I published the gate. I made the throttle call. I wrote the tolerance."

Do not stack stories. One question, one story, one number. Offer depth; do not volunteer it.

Do not let Situation eat the clock. Fifteen seconds of context. Seventy percent of the story is what you did and what it measured.

Do not open with SAP governance. This room is network and capacity. S/4HANA (S10) is a strong answer to inventory-flow, returns, cutover, and being-wrong questions, and a diluted answer to "tell me about yourself." Use it when they ask about inventory; do not lead with it when they ask about capacity.

Do not invent a headcount. Matrix leadership, said plainly, is fine at L6. A fake org chart is not.

Do not solve the case before clarifying. Sixty seconds of questions out loud is scored, not tolerated.

"What would you do differently" needs a real answer. You have five: shadow one zone first; scheduler heartbeat; measurement audit in week one; dual-path devices; security lead time on the kickoff checklist.

Decide the Bellevue answer before Friday.

13. The morning-of card

1. Read "tell me about yourself" and "why Amazon" (section 3) out loud once. End on the gap sentence.
2. Tell S1 and S2 to a wall in STAR — problem, goal, metric, S, T, A, R, did it move — in under two minutes each. Then say the supply/demand/capacity box for each.
3. Walk Case A on paper: demand → two currencies (cube, hours) → gap by week by building → lever stack with cost → owners and triggers. Re-read the 7A.6 worked example and the 7A.8 U-curve.
4. Do whiteboard math #1, #2, #3 and #6 (7A.11) on paper. Say the five judgment lines in section 7 once.
5. Pick your two questions. Pick your Bellevue sentence.